Adaptive Subgradient Methods for Online Deep Learning

Master Thesis

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Eduard Bopp

from Tschinas, Uzbekistan

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of the Faculty of Mathematics and Natural Sciences
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Supervisor: Prof. Dr. Florian Jarre
Second Supervisor: Prof. Dr. Christiane Helzel

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Introduction

In supervised machine learning, the objective is to construct a model that captures the underlying relationships or correlations present in a given dataset. Formally, we are provided with a collection of labeled data points

$$\{(x^{(j)}, y_j)\}_{j=1}^{N} \subseteq \mathcal{X} \times \mathcal{Y},$$

where $\mathcal{X}$ and $\mathcal{Y}$ denote the input and output spaces, respectively. These spaces are often considered as vector spaces to accommodate general structural and continuity properties of the data. In many practical applications, the input space is chosen as a Euclidean space, such as $\mathcal{X} = \mathbb{R}^d$ for d-dimensional feature vectors or $\mathcal{X} = \mathbb{R}^{d_1 \times d_2}$ for matrix-valued inputs, where $d, d_1, d_2 \in \mathbb{N}$. The output space is typically taken as $\mathcal{Y} = \mathbb{R}$, for instance in regression problems.

A model is represented by a parametric function

$$m_w : \mathcal{X} \to \mathcal{Y},$$

where $w \in \mathbb{R}^n$ denotes the vector of model parameters. To evaluate the quality of a given parameter choice, an empirical risk function is defined as

$$\mathcal{L}(w) = \frac{1}{N} \sum_{j=1}^{N} \ell(m_w(x^{(j)}), y_j),$$

where $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$ is a suitable loss function that quantifies the discrepancy between prediction and observation, for example $\ell(a, b) = (a - b)^\top (a - b)$ in the case of least squares regression. The corresponding optimization problem is

$$w^* = \arg\min_{w \in \mathbb{R}^n} \mathcal{L}(w),$$

which aims to determine the parameters minimizing the empirical loss. This formulation provides the foundation for many methods in supervised learning and allows the application of classical optimization algorithms, such as gradient descent and its numerous variants, to iteratively update the parameters w.

This setting corresponds to the *offline learning* paradigm, in which the entire dataset is available prior to training. Once the model has been fitted, it is subsequently applied to unseen data. However, many modern applications, such as financial forecasting, adaptive control, or personalized recommendation systems, operate in a fundamentally different regime. In these contexts, data points arrive sequentially and may reflect a time-varying or non-stationary environment. Retraining the model from scratch after each new observation would be computationally infeasible and conceptually inefficient.

The *online learning* paradigm addresses this challenge by formulating learning as a sequential decision-making process. The model receives data points one at a time and must continuously adapt its parameters in real time, typically under constraints of limited memory and computation. This introduces several distinctive challenges: the learner must make predictions before observing the corresponding outcomes, and the full data distribution is generally unknown and may even evolve over time.

A mathematically rigorous and widely studied abstraction of online learning is provided by the framework of *Online Convex Optimization* (OCO) [1]. This framework offers a powerful and general foundation for analyzing online learning algorithms and forms the central focus of this thesis.

Chapter 1 provides the necessary mathematical background. We begin with fundamental definitions and concepts from convex optimization, as well as selected notions from analysis and linear algebra that will be required in later chapters. We also discuss projection operators and their properties, since they play an essential role in many optimization algorithms. The chapter concludes with a short overview of several classical optimization algorithms in the setting of *offline learning*.

Chapter 2 introduces the framework of *Online Convex Optimization* (OCO) based on [1]. We formalize the online learning setting and present key algorithms, including *Follow-The-Regularized-Leader* (FTRL), *Online Mirror Descent* (OMD), and *Adaptive Gradient method* (AdaGrad). These serve as foundational tools for the developments in the subsequent chapters.

Chapter 3 is devoted to the concept of subgradients. Following the exposition in Nesterov's *Lectures on Convex Optimization* [2], we present a rigorous mathematical treatment of subgradients and their fundamental properties. We also recall the subgradient method and discuss its convergence behaviour in convex online settings.

Chapters 4 and 5 present two extensions of the AdaGrad algorithm from Chapter 2 to the case of subgradients based on Duchi, Hazan and Singer [3]. In Chapter 4, following the presentation in [3] we consider an algorithm that integrates subgradients into the Online Mirror Descent framework. Chapter 5 introduces a second variant based on an extension of Nesterov's primal-dual method. Both approaches are formulated and analyzed within the OCO setting.

Chapter 6 presents the resulting algorithms in detail and discusses their implementation aspects. In particular, we address the role of projection operators and explain how to compute and incorporate projections efficiently within the proposed methods.

Chapter 7 reports numerical experiments that illustrate and compare the performance of the Ada(Sub-)Gradient variants developed in Chapters 4 and 5. The results demonstrate the practical behavior of the algorithms and highlight the influence of different design choices on convergence and stability.

Together, these chapters provide a coherent progression from theoretical foundations to algorithmic development and empirical evaluation, with an emphasis on online optimization methods for convex and subgradient-based settings.

Notation

Let $a, b \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$.

- $a \odot b$: the Hadamard (element-wise) product of vectors.
- $\mathrm{Diag}(a) \in \mathbb{R}^{n \times n}$: the diagonal matrix with entries of a on the diagonal.
- $\mathrm{diag}(A) \in \mathbb{R}^n$: the vector of diagonal entries of A.
- $a_t \in \mathbb{R}^n$: a vector indexed by time $t \in \{1, \dots, T\}$, where $T \in \mathbb{N}$.
- $(a_t)_i \in \mathbb{R}$: the i-th entry of the vector a_t.

1 Basic definitions from Convex Optimization

This chapter reviews the concepts from *offline* convex optimization that we will need to formulate the Online Convex Optimization (OCO) model in Chapter 2. In addition, we want to introduce some concepts from calculus and linear algebra that will be helpful for analyzing online algorithms. At the end of this chapter, we want to present a few algorithms that we will refer back to later. Throughout Chapters 1 and 2 we restrict attention to differentiable functions, starting with Chapter 3 all subsequent chapters drop the differentiability assumption and rely on subgradient calculus. The concepts in this chapter are mainly taken from [1] and [2].

We define the domain of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{\pm\infty\}$ with

$$\operatorname{dom}(f) := \{x \in \mathbb{R}^n : |f(x)| < \infty\}$$

Throughout this thesis, we assume $\operatorname{dom}(f) \neq \emptyset$.

1.1 Convex Sets and Convex Functions

Definition 1.1: Convex set

A set $\mathcal{K} \subseteq \mathbb{R}^n$ is convex if for all $x, y \in \mathcal{K}$ and for all $\lambda \in [0, 1]$ it holds

$$\lambda x + (1 - \lambda)y \in \mathcal{K}.$$

Convex sets are central in optimization because they guarantee that local information e.g. (sub-) gradients can be reliably used to make global decisions.

Definition 1.2: Convex function

A function $f : \mathcal{K} \to \mathbb{R} \cup \{\infty\}$ is called convex if its domain is convex and for all $x, y \in \operatorname{dom}(f)$ and for all $\lambda \in [0, 1]$ holds

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \tag{1}$$

Besides the definition above, we list three equivalent characterisations of convex functions.

Theorem 1.1

A function $f : \mathcal{K} \to \mathbb{R} \cup \{\infty\}$ is convex if and only if for all $x, y \in \operatorname{dom}(f)$ and $\beta \geq 0$ such that $y + \beta(y - x) \in \operatorname{dom}(f)$, we have

$$f\big(y + \beta(y - x)\big) \geq f(y) + \beta\big(f(y) - f(x)\big). \tag{2}$$

Proof: Fix $x, y \in \operatorname{dom}(f)$ and $\beta \geq 0$ such that $z := y + \beta(y - x) \in \operatorname{dom}(f)$, and set

$$\lambda := \frac{\beta}{1 + \beta} \in [0, 1), \qquad 1 - \lambda = \frac{1}{1 + \beta}.$$

Then $y = (1 - \lambda)z + \lambda x$. Hence the convexity inequality for the pair (x, z) with weight λ reads

$$\begin{aligned} f(y) \leq (1 - \lambda)f(z) + \lambda f(x) &\iff (1 + \beta)f(y) \leq f(z) + \beta f(x) \\ &\iff f(z) \geq f(y) + \beta\big(f(y) - f(x)\big) \\ &\iff f\big(y + \beta(y - x)\big) \geq f(y) + \beta\big(f(y) - f(x)\big). \end{aligned}$$

Thus convexity implies (2). Conversely, given (2), take any $x, z \in \operatorname{dom}(f)$ and any $\lambda \in [0, 1)$. Define $y := (1 - \lambda)z + \lambda x$ and $\beta := \lambda/(1 - \lambda)$, so that $y + \beta(y - x) = z$. Applying (2) to (x, y, β) yields the convexity inequality $f\big((1 - \lambda)z + \lambda x\big) \leq (1 - \lambda)f(z) + \lambda f(x)$. For $\lambda = 1$ the inequality is trivial. □

For the other definition of convexity, we define the following sets

Definition 1.3: Epigraph

We denote the epigraph of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ by

$$\text{epi}(f) = \{(x, t) \in \text{dom}(f) \times \mathbb{R} \mid t \geq f(x)\}.$$

Furthermore, we denote the constrained epigraph with

$$\text{epi}_{\mathcal{K}}(f) = \{(x, t) \in \mathcal{K} \times \mathbb{R} \mid t \geq f(x)\},$$

where $\mathcal{K} \subseteq \text{dom}(f)$.

The epigraph provides us with an additional criterion for convexity.

Theorem 1.2: Convexity per Epigraph

A function $f : \mathcal{K} \to \mathbb{R} \cup \{\infty\}$ is convex if and only if its epigraph

$$\text{epi}(f) := \{(x, t) \in \text{dom}(f) \times \mathbb{R} | \ t \geq f(x)\} \tag{3}$$

is a convex set.

Proof: We derive the proof from Theorem 3.1.2 of [2].
($\Rightarrow$) Let f be convex and (x_1, t_1), $(x_2, t_2) \in \text{epi}(f)$ and let $\lambda \in [0, 1]$ be arbitrary, then it follows

$$\lambda t_1 + (1 - \lambda)t_2 \overset{(3)}{\geq} \lambda f(x_1) + (1 - \lambda)f(x_2) \overset{(1)}{\geq} f(\lambda x_1 + (1 - \lambda)x_2)$$

By definition, $(\lambda x_1 + (1 - \lambda)x_2, \lambda t_1 + (1 - \lambda)t_2) \in \text{epi}(f)$.
($\Leftarrow$) Let $\text{epi}(f)$ be a convex set. Since obviously $f(x) \geq f(x)$ for all $x \in \text{dom}(f)$ it holds

$$(x_1, f(x_1)) \in \text{epi}(f) \quad (x_2, f(x_2)) \in \text{epi}(f).$$

Since $\text{epi}(f)$ convex we get $(\lambda x_1 + (1 - \lambda)x_2, \lambda f(x_1) + (1 - \lambda)f(x_2)) \in \text{epi}(f)$. Per definition

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2),$$

so f is convex.

□

If f is differentiable, an equivalent condition is the first-order condition.

Theorem 1.3: First order condition of convex function

A differentiable function $f : \mathcal{K} \to \mathbb{R} \cup \{\infty\}$ is convex if and only if for all $x, y \in \text{dom}(f)$ holds

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x). \tag{4}$$

The first order condition, expressed in inequality (4), is both a necessary and sufficient condition for the convexity of differentiable functions.

A useful definition that will be helpful later is the following

Definition 1.4: α-strongly convex function

A differentiable function $f : \mathcal{K} \to \mathbb{R} \cup \{\infty\}$ is called α-strongly convex for $\alpha > 0$ if for all $x, y \in \text{dom}(f)$ holds

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\alpha}{2}\|y - x\|^2. \tag{5}$$

If $\alpha = 0$, we get a convex function in sense of (1).

1.2 Constrained Convex Optimization

The goal of convex optimization is to find a point $x^* \in \mathcal{K} \subseteq \mathbb{R}^n$ that minimizes a convex function $f : \mathcal{K} \to \mathbb{R} \cup \{\infty\}$, where $\mathcal{K}$ is a convex feasible set. Formally, the problem is stated as

$$x^* \in \arg\min_{x \in \mathcal{K}} f(x),$$

since the optimal point may not be unique.

Unconstrained Convex Optimization:

In the special case where there are no explicit constraints, i.e., $\mathcal{K} = \mathbb{R}^n$, the problem reduces to

$$x^* := \arg\min_{x \in \mathbb{R}^n} f(x).$$

If f is convex and differentiable, any local minimum is also a global minimum. A key optimality condition is that the gradient of f at x^* vanishes

$$\nabla f(x^*) = 0.$$

This condition is necessary and sufficient that x^* to be a global minimizer.

Constrained Convex Optimization:

When the feasible set $\mathcal{K} \subseteq \mathbb{R}^n$ is convex and closed, the problem is

$$\min_{x \in \mathcal{K}} f(x).$$

Since f is convex and differentiable, the optimality condition generalizes as follows

Theorem 1.4: First-Order Optimality Condition for constrained problems

A point $x^* \in \mathcal{K}$ is a global minimizer of f over $\mathcal{K}$ if and only if

$$\nabla f(x^*)^\top (x - x^*) \geq 0 \quad \text{for all } x \in \mathcal{K}. \tag{6}$$

We refer the proof to [6].

1.3 Projection onto Convex Sets

In many optimization algorithms, especially when iterates must remain feasible, a projection step is employed to ensure that updates stay within the feasible convex set $\mathcal{K}$.

Definition 1.5: Projection onto convex sets

Given a closed convex set $\mathcal{K} \subseteq \mathbb{R}^n$ and a point $y \in \mathbb{R}^n$, the projection of y onto $\mathcal{K}$ is defined as:

$$\Pi_{\mathcal{K}}(y) := \arg\min_{x \in \mathcal{K}} \|x - y\|.$$

This definition finds the closest point in $\mathcal{K}$ to y in the Euclidean norm. The projection has several useful properties.

Lemma 1.1: Properties of the projection

Let $\mathcal{K} \subseteq \mathbb{R}^n$ be a non-empty, closed, convex set, and let $y \in \mathbb{R}^n$. Then the projection $z = \Pi_{\mathcal{K}}(y)$ satisfies the following properties

1. **Uniqueness:** The projection z is unique.
2. **Optimality condition:** For all $x \in \mathcal{K}$, it holds that

$$(y - z)^\top (x - z) \leq 0. \tag{7}$$

3. **Non-expansiveness:** For any $x, y \in \mathbb{R}^n$, it holds that

$$\|\Pi_{\mathcal{K}}(x) - \Pi_{\mathcal{K}}(y)\| \leq \|x - y\|. \tag{8}$$

Proof: We provide the proof here for completeness.
(1) Uniqueness: Since $\mathcal{K}$ is closed and convex, and the Euclidean norm is strictly convex, the minimization problem

$$\Pi_{\mathcal{K}}(y) := \arg\min_{x \in \mathcal{K}} \|x - y\|$$

has a unique solution. The strict convexity of the norm ensures that the objective function has a unique minimizer over a convex set.
(2) Optimality condition: By definition, the projection $z = \Pi_{\mathcal{K}}(y)$ is the solution to

$$\min_{x \in \mathcal{K}} \|x - y\|^2.$$

This is a convex minimization problem. Using first-order optimality conditions (6), z is optimal if and only if for all $x \in \mathcal{K}$,

$$\nabla \left(\|w - y\|^2 \right) \Big|_{w=z}^{\top} (x - z) = 2(z - y)^{\top}(x - z) \geq 0.$$

Rewriting this yields

$$(y - z)^{\top}(x - z) \leq 0.$$

(3) Non-expansiveness: Let $x, y \in \mathbb{R}^n$ and define $u = \Pi_{\mathcal{K}}(x)$, $v = \Pi_{\mathcal{K}}(y)$. From the optimality condition (7) applied to both projections:

$$(x - u)^{\top}(v - u) \leq 0 \quad \text{and} \quad (y - v)^{\top}(u - v) \leq 0.$$

Adding these inequalities yields:

$$(x - u)^{\top}(v - u) + (y - v)^{\top}(u - v) \leq 0.$$

Note that $(a - b)^{\top}(b - a) = -\|a - b\|^2$, so rearranging and simplifying gives:

$$\|u - v\|^2 \leq (x - y)^{\top}(u - v) \leq \|x - y\| \cdot \|u - v\|,$$

where the last inequality follows from the Cauchy–Schwarz inequality. Dividing both sides by $\|u - v\|$ (if nonzero), we obtain

$$\|u - v\| \leq \|x - y\|.$$

□

These properties are essential in projected gradient methods, ensuring that feasibility is maintained without sacrificing convergence.

1.4 Some basic concepts from Analysis and Linear Algebra

We will use distances later, and for that we need the concept of the norm.

Definition 1.6: Norm, normed space

A mapping $\|\cdot\| \colon \mathcal{K} \to [0, \infty)$ is called a *norm* if $\mathcal{K}$ is a vector space over $\mathbb{R}$ and the following holds for all $x, y \in \mathcal{K}$ and $\lambda \in \mathbb{R}$:

1. $\|x\| = 0 \iff x = 0$,
2. $\|\lambda x\| = |\lambda| \, \|x\|$,
3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

The pair $(\mathcal{K}, \|\cdot\|)$ is called a *normed space*.

Examples:

1. For $p \in [1, \infty)$, $\|x\|_p = (\sum_{i=1}^{d} |x_i|^p)^{1/p}$ is the ℓ_p norm.

2. $\|x\|_\infty = \max_i |x_i|$ is the ℓ_∞ norm.

3. If $A \succ 0$ is a positive definite matrix, then $\|x\|_A = \sqrt{x^\top Ax}$ is a norm, the Mahalanobis norm.

In finite dimensions, all norms are equivalent, i.e., for two norms $\|\cdot\|_a, \|\cdot\|_b$, there exist constants $c_1, c_2 > 0$ such that $c_1\|x\|_a \leq \|x\|_b \leq c_2\|x\|_a$ for all $x \in \mathcal{K}$. Later on, when we talk about a function with respect to a norm, a given norm is used in this function.

What will be helpful for later arguments is the concept of dual norms.

Definition 1.7: dualspace, dualnorm

For the normed space $(\mathcal{K}, \|\cdot\|)$ with $\mathcal{K} \subseteq \mathbb{R}^n$, the *dual space* is defined by

$$\mathcal{K}^* = \{f : \mathcal{K} \to \mathbb{R} \mid f(x) = \langle y, x\rangle \text{ for a } y \in \mathbb{R}^n\} \tag{9}$$

and the *dual norm* $\|\cdot\|^*$ is defined by

$$\|y\|^* := \sup_{\|x\|\leq 1} \langle x, y\rangle.$$

Example: ℓ_2 is self-dual; ℓ_1 and ℓ_∞ are dual to each other; in general, ℓ_p and ℓ_q are dual when $1/p + 1/q = 1$. The dual norm of the Mahalanobis norm is $\|x\|_A^* = \|x\|_{A^{-1}}$.

The following lemma will become very useful.

Lemma 1.2: Hölder/Cauchy in general norm

For all $x, y \in \mathbb{R}^n$, the following applies:

$$|\langle x, y\rangle| \leq \|x\| \, \|y\|_*. \tag{10}$$

We will not provide a proof and refer to well-known literature.

For the analysis of online algorithms, we need to define the trace.

Definition 1.8: Trace

Let $A \in \mathbb{R}^{n\times n}$ be an $n \times n$ matrix. The *trace* of A, denoted by $\mathrm{Tr}(A)$, is defined as the sum of the entries on the main diagonal of A, that is,

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} a_{ii}.$$

Equivalently, if A has complex eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ (counted with multiplicity), then

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} \lambda_i.$$

The following theorem will be helpful later on.

Theorem 1.5

Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive semidefinite (PSD), and let $\lambda_{\max}(A)$ denote its largest eigenvalue. Then, for every $x \in \mathbb{R}^n$,

$$x^\top Ax \leq \lambda_{\max}(A) \, \|x\|_2^2. \tag{11}$$

Proof: We provide the proof here for completeness.
Since A is real symmetric, by the spectral theorem there exists an orthogonal matrix Q and a real

diagonal matrix $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ with eigenvalues $\lambda_i \geq 0$ (PSD) such that

$$A \;=\; Q\,\Lambda\,Q^\top.$$

For any $x \in \mathbb{R}^n$, set $y := Q^\top x$. Because Q is orthogonal, $\|y\|_2 = \|x\|_2$. Then

$$x^\top Ax \;=\; x^\top Q\Lambda Q^\top x \;=\; y^\top \Lambda y \;=\; \sum_{i=1}^n \lambda_i\, y_i^2 \;\leq\; \big(\max_{1\leq i\leq n} \lambda_i\big) \sum_{i=1}^n y_i^2 \;=\; \lambda_{\max}(A)\,\|y\|_2^2 \;=\; \lambda_{\max}(A)\,\|x\|_2^2.$$

□

We would like to introduce the following two estimates, which will be useful in our later analysis of online algorithms.

Theorem 1.6

Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive semidefinite (PSD), with largest eigenvalue $\lambda_{\max}(A)$, then it follows

$$\lambda_{\max}(A) \;\leq\; \mathrm{Tr}(A). \tag{12}$$

Proof: We provide the proof here for completeness.
Since A is PSD, its eigenvalues $\lambda_1, \ldots, \lambda_n$ are all nonnegative and

$$\mathrm{Tr}(A) = \sum_{i=1}^n \lambda_i \geq \max_i \lambda_i = \lambda_{\max}(A).$$

□

Theorem 1.7

Let $A \in \mathbb{R}^{n\times n}$ be a positive semidefinite (PSD) matrix and let $B = \mathrm{diag}(A)$ denote the diagonal matrix whose diagonal entries are those of A. Then for all $x \in \mathbb{R}^n$,

$$x^\top Bx \leq \mathrm{Tr}(B)\,\|x\|_\infty^2. \tag{13}$$

Proof: We provide the proof here for completeness.
Since $A \succeq 0$, we have for each standard basis vector $e_i \in \mathbb{R}^n$

$$e_i^\top Ae_i = a_{ii} \geq 0.$$

Hence, all diagonal entries of $B = \mathrm{diag}(A)$ are nonnegative, i.e. $b_i = a_{ii} \geq 0$ for all $i = 1, \ldots, n$.
Let $x \in \mathbb{R}^n$ be arbitrary and set $M = \|x\|_\infty^2 = \max_i x_i^2$. Then

$$x^\top Bx = \sum_{i=1}^n b_i x_i^2 \leq \sum_{i=1}^n b_i M = M \sum_{i=1}^n b_i = \mathrm{Tr}(B)\,\|x\|_\infty^2.$$

□

1.5 Projected Gradient Descent (PGD)

Many practically relevant optimization problems are convex, but constrained by a set $\mathcal{K} \subseteq \mathbb{R}^n$. While the classical (unconstrained) gradient method moves a point exclusively in direction of the negative gradient, in the presence of constraints, each new candidate must then be projected back into the admissible set. This simple additional projection onto $\mathcal{K}$ leads to the Projected Gradient Descent (PGD).

Algorithm 1: Projected Gradient Descent (PGD)

Data: convex set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, initial point $w_1 \in \mathcal{K}$, step sizes $\eta_t > 0$ at iteration t

for *t=1 to T* **do**
  $\mathrm{w}_{t+1} = \Pi_{\mathcal{K}}(w_t - \eta_t \nabla f(w_t))$
end
return w_{T+1}

1.6 Generalised Project Gradient Descent (GPGD)

As we noted in the last subsection, the update step of the (PGD) is performed using a projection in Euclidean space. In this subsection we want to generalize this step. Consider again at the update step from (PGD)

$$w_{t+1} = \Pi_{\mathcal{K}}(w_t - \eta_t \nabla f(w_t)) = \arg\min_{w \in \mathcal{K}} \|w_t - \eta_t \nabla f(w_t) - w\|$$

Since the solution does not change under squares, we can also write

$$w_{t+1} = \arg\min_{w \in \mathcal{K}} \|w_t - \eta_t \nabla f(w_t) - w\|^2.$$

We want to omit the terms that do not depend on w, as they have no influence on w_{k+1}. It follows

$$\|w_t - \eta_t \nabla f(w_t) - w\|^2 = \|w - w_t\|^2 + 2\eta_t \langle w, \nabla f(w_t)\rangle - 2\eta_t \langle w_t, \nabla f(w_t)\rangle + \eta_t^2 \|\nabla f(w_t)\|^2$$

Rearranging the terms, we are then left with the equivalent iteration

$$w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\{ \langle w, \nabla f(w_t)\rangle + \frac{1}{\eta_t} \frac{\|w - w_t\|^2}{2} \right\}. \tag{14}$$

The objective function (14) makes it clear that the projected gradient descent can be seen as an optimization problem that creates a compromise. On the one hand, one should move along the direction of the negative gradient (first term), on the other hand, the iterates may only deviate from the current point to a limited extent (second term), while all steps remain strictly within the permissible region $\mathcal{K}$.

Based on this, we replace the second term of (14) with other distance measures and obtain

Definition 1.9: Generalised Projected Gradient Descent (GPGD)

Let a function

$$d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}$$

be given with $d(x, y) > 0$ for all $x \neq y$. The generalised projected gradient descent (GPGD) is defined by the following iteration

$$w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\{ \langle w, \nabla f(w_t)\rangle + \frac{1}{\eta_t} d(w, w_t) \right\}. \tag{15}$$

The update step (15) now offers a variety of algorithms depending on the selection of the function D.

1.7 Bregman Divergence and Mirror Descent Algorithm

Before introducing the Mirror Descent Algorithm, we would like to point out that there are two different derivations, both of which we will present. One is the mirror map view, which is the original view presented by Arkadi Nemirovski and David Yudin [4], and the other is the proximal view, which can be derived from Bregman Divergence. Bregman divergence is a distance function that can be used in (GPGD). We would like to begin with the proximal view and start with the definition of Bregman divergence.

Definition 1.10: Bregman Divergence

Let $\mathcal{K} \subset \mathbb{R}^n$ be closed and convex. Let $\psi : \mathcal{K} \to \mathbb{R}$ be a strictly convex function and continuously differentiable on the interior of $\mathcal{K}$. Then the Bregman divergence from $B_\psi(x\|y)$ is defined as

$$B_\psi(x\|y) = \psi(x) - \psi(y) - \langle \nabla\psi(y), x - y\rangle. \tag{16}$$

In this context, we refer to the function ψ as the generator function. Furthermore, we can see that the Bregman divergence calculates the distance between $\psi(x)$ and its first Taylor approximation. The Bregman Divergence also has the following properties.

Lemma 1.3

Let B_ψ, $\mathcal{K}$ be defined as in (16), then the following applies

1. $\forall x \in X$ it holds
$$B_\psi(x\|x) = 0 \tag{17}$$
2. $\forall x, y \in X$ with $x \neq y$ it holds $B_\psi(x\|y) > 0$
3. If ψ is α-strongly convex, then
$$B_\psi(x\|y) \geq \frac{\alpha}{2}||x-y||_2^2 \tag{18}$$
4. For all $x, y, z \in \mathcal{K}$, the following identity holds:
$$\langle \nabla\psi(y) - \nabla\psi(z),\, x-y\rangle = B_\psi(x\|z) - B_\psi(x\|y) - B_\psi(y\|z). \tag{19}$$
5. Let $z \in \mathcal{K}$, $y \in \mathbb{R}^n$ and $u = \arg\min_{w\in\mathcal{K}} B_\psi(w\|y)$, then the following inequality holds
$$B_\psi(u\|y) \leq B_\psi(z\|y) - B_\psi(z\|u). \tag{20}$$

Proof:We provide the proofs of (1)-(4) here for completeness.

1. Let $x \in X$ then $B_\psi(x\|x) = \psi(x) - \psi(x) - \langle\nabla\psi(x), x-x\rangle = 0$
2. From strict convexity it follows that for $x \neq y$:
$$\psi(x) > \psi(y) + \langle\nabla\psi(y),\, x-y\rangle.$$
Hence,
$$B_\psi(x\|y) > 0 \quad \text{for } x \neq y.$$
3. By α-strong convexity of ψ, for all x, y,
$$\psi(x) \overset{(5)}{\geq} \psi(y) + \langle\nabla\psi(y),\, x-y\rangle + \frac{\alpha}{2}\|x-y\|_2^2.$$
Rearranging yields
$$\psi(x) - \psi(y) - \langle\nabla\psi(y),\, x-y\rangle \geq \frac{\alpha}{2}\|x-y\|_2^2.$$
By definition $B_\psi(x\|y) = \psi(x) - \psi(y) - \langle\nabla\psi(y), x-y\rangle$, hence
$$B_\psi(x\|y) \geq \frac{\alpha}{2}\|x-y\|_2^2.$$
4. By the definition of Bregman divergence, we can expand each term:
$$B_\psi(x\|z) = \psi(x) - \psi(z) - \langle\nabla\psi(z), x-z\rangle,$$
$$B_\psi(x\|y) = \psi(x) - \psi(y) - \langle\nabla\psi(y), x-y\rangle,$$
$$B_\psi(y\|z) = \psi(y) - \psi(z) - \langle\nabla\psi(z), y-z\rangle.$$
Now consider the difference
$$B_\psi(x\|z) - B_\psi(x\|y) - B_\psi(y\|z).$$

Substituting the expansions above, we obtain

$$= (\psi(x)-\psi(z)-\langle\nabla\psi(z), x-z\rangle)-(\psi(x)-\psi(y)-\langle\nabla\psi(y), x-y\rangle)-(\psi(y)-\psi(z)-\langle\nabla\psi(z), y-z\rangle).$$

The terms in $\psi(x)$, $\psi(y)$, and $\psi(z)$ cancel out, leaving

$$= \langle\nabla\psi(y), x-y\rangle - \langle\nabla\psi(z), x-z\rangle + \langle\nabla\psi(z), y-z\rangle.$$

Rearranging the last two terms yields

$$B_\psi(x\|z) - B_\psi(x\|y) - B_\psi(y\|z) = \langle\nabla\psi(y), x-y\rangle - \langle\nabla\psi(z), (x-y)\rangle = \langle\nabla\psi(y) - \nabla\psi(z),\, x-y\rangle.$$

This is exactly the claimed identity.

5. We follow the proof of [9]. We create the function

$$\begin{aligned} G(z) &= B_\psi(z\|y) - B_\psi(z\|u) \\ &= \psi(u) - \psi(y) + \langle\nabla\psi(y), y\rangle - \langle\nabla\psi(u), u\rangle + \langle\nabla\psi(u) - \nabla\psi(y), z\rangle. \end{aligned}$$

As can be clearly seen, the function G is convex in z since it is a linear function in z. For this reason, the following applies for $\lambda \in [0, 1]$

$$\begin{aligned} B_\psi(\lambda z + (1-\lambda)u\|y) - B_\psi(\lambda z + (1-\lambda)u\|u) &\leq \lambda\big(B_\psi(z\|y) - B_\psi(z\|u)\big) + (1-\lambda)\big(B_\psi(u\|y) - B_\psi(u\|u)\big) \\ &\overset{(17)}{=} \lambda\big(B_\psi(z\|y) - B_\psi(z\|u)\big) + (1-\lambda)B_\psi(u\|y) \end{aligned}$$

After slight transformations, we obtain for $\lambda > 0$

$$B_\psi(u\|y) \leq B_\psi(z\|y) - B_\psi(z\|u) + \frac{1}{\lambda}\Big(B_\psi(\lambda z + (1-\lambda)u\|u) - B_\psi(\lambda z + (1-\lambda)u\|y) + B_\psi(u\|y)\Big).$$

If we now let $\lambda \to 0$ and apply (17), we obtain the desired statement. □

As we can see, Bregman divergence is suitable for measuring distance. It is important to note that Bregman divergence is not symmetric in general. Now the question arises as to which choice is appropriate for the generator function ψ. This choice depends heavily on the geometry of the data, and there is no "best" choice. One example we would like to give is $\psi(x) = \frac{1}{2}\|x\|_2^2$. This leads to

$$B_\psi(x\|y) = \frac{1}{2}\|x\|_2^2 - \frac{1}{2}\|y\|_2^2 - \langle y, x-y\rangle = \frac{1}{2}\|x\|_2^2 + \frac{1}{2}\|y\|_2^2 - \langle y, x\rangle = \frac{1}{2}\|x-y\|_2^2,$$

which is a good example for the function $d(x, y)$ from (15). We obtain exactly (14), the projected gradient descent.

Now let's explain the mirror map perspective. The *mirror map* provides a way to perform optimization in a dual space that better reflects the geometry of the problem than the original, or *primal*, space. Nemirovski and Yudin introduced this idea to generalize gradient-based methods to settings where the Euclidean geometry is not appropriate. In particular, the gradients $g_t = \nabla f(w_t)$ are naturally elements of the dual space $\mathcal{K}^*$, since they represent linear functionals (9) acting on the primal space $\mathcal{K}$. Their method proceeds as follows

1. **Mapping to the dual space:** Choose a differentiable and strongly convex function $\psi : \mathcal{K} \to \mathbb{R}$, called the *mirror map* or *potential function*. The current iterate in the primal space w_k is mapped to the dual space via the gradient of ψ:

$$y_k = \nabla\psi(w_k).$$

2. **Gradient step in the dual space:** A standard gradient step is then performed in the dual space:

$$y_{k+1} = y_k - \eta\nabla f(w_k),$$

where $\eta > 0$ is the step size.

3. **Mapping back to the primal space:** The next iterate in the primal space is obtained by applying the inverse of the mirror map:

$$w_{k+1} = (\nabla\psi)^{-1}(y_{k+1}).$$

4. **Projection onto the feasible set:** If the optimization problem is constrained, the new point w_{k+1} is projected back onto the feasible region $\mathcal{K}$.

Geometric interpretation: When the space $\mathcal{K}$ is Euclidean with the L^2 norm, the primal and dual spaces coincide, and $\nabla\psi(w) = w$. In this case, the method reduces to ordinary gradient descent. However, for non-Euclidean geometries, the primal and dual spaces differ. The choice of the mirror map ψ determines how information is transferred between these spaces and how the geometry of the optimization domain is incorporated into the update.
For example

- If $\psi(w) = \frac{1}{2}\|w\|_2^2$, the updates are Euclidean (classical gradient descent).
- If $\psi(w) = \sum_i w_i \log w_i$, the updates follow an entropic geometry suitable for probability simplices.

Before specifying the algorithm, we would like to clarify once again why both perspectives yield the same algorithm. Consider (15) and apply the Bregman divergence.

$$w_{t+1} = \arg\min_{w\in\mathcal{K}} \left\{ \langle w, \nabla f(w_t)\rangle + \frac{1}{\eta_t} B_\psi(w\|w_t) \right\}$$

This exactly the proximal view. Consider the gradient of B_ψ

$$\nabla_w B_\psi(w\|w_t) = \nabla\psi(w) - \nabla\psi(w_t)$$

We obtain

$$\nabla_w \Big(\langle w, \nabla f(w_t)\rangle + \frac{1}{\eta_t} B_\psi(w\|w_t) \Big)\Big|_{w=w_{t+1}} = 0$$

which is

$$\nabla f(w_t) + \frac{1}{\eta_t}\Big(\nabla\psi(w_{t+1}) - \nabla\psi(w_t)\Big) = 0.$$

Rearranging terms gives a step of update in the dual space

$$\nabla\psi(w_{t+1}) = \nabla\psi(w_t) - \eta_t \nabla f(w_t).$$

Even though the Bregman divergence was not used in the work of Yudin and Nemirovski, we will use it to project into the feasible region. We provide the following algorithm

Algorithm 2: Mirror Descent (MD)

Data: convex set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, initial point $w_1 \in \mathcal{K}$, step sizes $\eta_t > 0$ at iteration t, generator function ψ

for *t=1 to T* **do**
  $y_{t+1} = (\nabla\psi)^{-1}(\nabla\psi(w_t) - \eta_t \nabla f(w_t))$
  $w_{t+1} = \arg\min_{w\in\mathcal{K}} B_\psi(w\|y_{t+1})$
end
return w_{T+1}

Remark:

- The mirror descent algorithm was developed primarily for online settings and with non-differentiable convex functions. For the sake of completeness, we have included the variant for offline settings and with the availability of differentiability. In the next chapter, we will introduce the online version.
- We have not clarified how to calculate he inverse of ψ. Later on, when we deal with general convex functions, we will see that this can be calculated using the convex conjugate of ψ.
- A convergence analysis for the online setting will be given in Chapter 2 in section 2.5.

2 Online Convex Optimization (OCO)

The framework of Online Convex Optimization (OCO) provides a robust and versatile approach to solving optimization problems in the context of online learning. In this setting, data arrives sequentially over time, and future observations are unavailable during the decision-making process. Importantly, OCO extends beyond this specific scenario and is capable of addressing a wide range of online optimization problems.

To develop a comprehensive understanding of OCO, we begin by introducing the problem through its formulation in game theory. This perspective serves as the foundation for rigorously defining the OCO model and understanding its key components. We then introduce the concept of regret, a central metric used to evaluate the performance of OCO algorithms. Finally, we discuss fundamental OCO algorithms and analyze their regret bounds.

2.1 Definition and Procedure

Online Convex Optimization can be conceptualized as a repeated game between a player and an environment, where the player makes decisions based on limited information. At each iteration, the environment reveals a cost function, and the player incurs a loss based on their decision. Formally, the process unfolds as follows:

1. At each time $t \in \{1, 2, ..., T\}$, where $T \in \mathbb{N}$ is the number of iterations, the player selects a decision $w_t \in \mathcal{K}$ where $\mathcal{K} \subseteq \mathbb{R}^n$ is a convex feasible set.
2. After committing to the decision w_t, the environment reveals a convex cost function $f_t : \mathcal{K} \to \mathbb{R}$.
3. The player incurs the cost $f_t(w_t)$ associated with their decision at time t.

This iterative interaction models a broad range of practical problems in online learning and optimization. Zinkevich provided a formal definition of OCO in [5], which we adopt here with modifications inspired by Hazan's book [1].

Definition 2.1: OCO (Zinkevich)

An **OCO model** consists of:

- A nonempty, convex and closed feasible set $\mathcal{K} \subseteq \mathbb{R}^n$,
- An infinite sequence of convex cost functions $\{f_1, f_2, ...\} \subset \mathcal{F}$, where $\mathcal{F} = \{f : \mathcal{K} \to \mathbb{R} |\ f \text{ convex}\}$. The respective function f_t is only known *after* the selection of w_t in step t.

At each time step t, an **OCO algorithm** selects a decision $w_t \in \mathcal{K}$. After selecting w_t, the algorithm observes the cost function f_t and incurs the cost $f_t(w_t)$.

To analyze the performance of OCO algorithms, we adopt the following assumptions, slightly modifying the original framework in [3]:

1. **Bounded Feasible Set**: The feasible set $\mathcal{K}$ is bounded, and $D > 0$ is an upper bound of the diameter, such that for all $x, y \in \mathcal{K}$,

$$d(x, y) \leq D, \tag{21}$$

   where $d(\cdot, \cdot)$ denotes a distance metric (e.g., Euclidean norm).

2. **Differentiable Cost Functions**: Each cost function f_t is differentiable for all t.
3. **Bounded Gradient Norms**: There exists a $G > 0$ such that for all t, for all $w \in \mathcal{K}$,

$$||\nabla f_t(w)|| \leq G. \tag{22}$$

These assumptions ensure the feasibility of algorithmic updates and allow for regret analysis using well-established mathematical tools.

Remarks:

- **Projection Onto the Convex Set**: Since $\mathcal{K}$ is convex and closed, projection onto $\mathcal{K}$ is well-defined and computationally efficient for many practical problems.
- **Comparison with Zinkevich's Original Framework**: We have explicitly incorporated convexity and non-emptiness of $\mathcal{K}$ into the model definition, which were treated as assumptions in Zinkevich's original formulation.
- **Differentiability of Cost Functions**: The differentiability assumption is made only in this chapter, facilitating the analysis of gradient-based algorithms. In the setting of the paper [3] we will only assume the existence of subgradients.

These remarks clarify the role of assumptions and emphasize the practical implications of the OCO model.

2.2 Regret: A Metric for Evaluating OCO Models

Having established the OCO framework, we now turn to the central question: how can we evaluate the performance of an OCO algorithm? The concept of regret serves as a key metric for this purpose, quantifying the difference between the algorithm's cumulative cost and that of the best fixed decision in hindsight.

Definition 2.2: Regret (Zinkevich)

Let an OCO algorithm $\mathcal{A}$ be given and an OCO model $(\mathcal{K}, \{f_1, f_2, ...\})$. Furthermore, let $\{w_1, w_2, ...\} \subset \mathcal{K}$ be the decisions chosen by the algorithm $\mathcal{A}$, then the costs up to time T are defined as:

$$C_T(\mathcal{A}) = \sum_{t=1}^{T} f_t(w_t).$$

The cost of a static feasible solution $w \in \mathcal{K}$ up to time T is

$$C_T(w) = \sum_{t=1}^{T} f_t(w).$$

The regret of algorithm $\mathcal{A}$ until time T is

$$\begin{aligned} Regret_T^{\mathcal{A}} &= \sup_{\{f_1, f_2, ..., f_T\} \subseteq \mathcal{F}} \Big\{ C_T(\mathcal{A}) - \min_{w \in \mathcal{K}} C_T(w) \Big\} \\ &= \sup_{\{f_1, f_2, ..., f_T\} \subseteq \mathcal{F}} \Big\{ \sum_{t=1}^{T} f_t(w_t) - \min_{w \in \mathcal{K}} \sum_{t=1}^{T} f_t(w) \Big\} \end{aligned}$$

Here, $\min_{w \in \mathcal{K}} \sum_{t=1}^{T} f_t(w)$ describes the cumulative costs of the best fixed decision in hindsight.

The regret quantifies how well an online algorithm performs relative to the best fixed decision that could have been chosen in hindsight. Minimizing regret is the central goal of OCO, and ideally, the asymptotic growth of the regret should be sublinear in T, i.e.,

$$Regret_T(\mathcal{A}) = o(T),$$

which implies that, asymptotically, the algorithm performs as well as the best fixed decision in hindsight as the number of iterations T grows large.

In classical optimization, one often analyzes the convergence behavior of a sequence (w_t) toward

an optimal point w^*. In the online learning setting, however, this goal is less meaningful, since the objective functions f_t can change from round to round and a fixed optimum point may not exist. Instead, the performance of the algorithm is evaluated using regret, i.e., the cumulative additional loss compared to the best fixed predictor in retrospect.

To fully understand the regret definition, we now provide a formal description of an OCO algorithm. In general, an OCO algorithm $\mathcal{A}$ can be viewed as a mapping that, at each time step t, selects a decision based on the sequence of previously revealed cost functions. Formally, for each $t > 0$, the algorithm can be represented as a function:

$$\mathcal{A} : \bigotimes_{k=1}^{t-1} \mathcal{F} \to \mathcal{K}$$
$$\{f_1, f_2, ..., f_{t-1}\} \mapsto w_t = \mathcal{A}(f_1, f_2, ..., f_{t-1})$$

where $\bigotimes_{k=1}^{t-1} \mathcal{F}$ denotes the product space of all previous cost functions $\{f_1, f_2, ..., f_{t-1}\}$ and the output $w_t = \mathcal{A}(f_1, f_2, ..., f_{t-1})$ is the decision made by the algorithm at time t.
In cases where the algorithm $\mathcal{A}$ is understood from the context, we will omit the explicit reference to $\mathcal{A}$ as the input parameter.

Having laid the foundation of Online Convex Optimization (OCO) and established the definition of the regret as a key performance indicator, we now turn to one of the most fundamental algorithms in this framework: the Online Gradient Descent (OGD).

2.3 Online Gradient Descent (OGD)

The algorithm was first introduced by Zinkevich (2003) and is a central component of online learning algorithms due to its simplicity and effectiveness in a variety of applications. It provides a robust way to sequentially optimize decisions by using gradient information that is made available at each time step. We give the algorithm which is an analogy to the gradient descent

Algorithm 3: Online gradient descent (OGD)

Data: convex set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, initial point $w_1 \in \mathcal{K}$, step sizes $\{\eta_t\}$
for *t=1 to T* **do**
  Observe cost $f_t(w_t)$
  Update: $y_{t+1} = w_t - \eta_t \nabla f_t(w_t)$
  Project: $w_{t+1} = \Pi_{\mathcal{K}}(y_{t+1})$
end
return w_{T+1}

We would now like to analyze the regret of (OGD). We will see that the regret of OGD is sublinear.

Theorem 2.1: Regret of (OGD)

Let $\{w_t\}$ be a sequence generated by Algorithm 3 and let f_t be convex and differentiable for all $t \in \{1, 2, ..., T\}$ with $g_t = \nabla f_t(w_t)$. Online gradient descent (OGD) with step sizes $\{\eta_t = \frac{D}{G\sqrt{t}}, t \in \{1, 2, ..., T\}\}$, where D defined as (21) and G defined as (22), guarantees the following for all $T \geq 1$

$$\text{Regret}_T^{\text{OGD}} = \sum_{t=1}^{T} f_t(w_t) - \min_{w \in \mathcal{K}} \sum_{t=1}^{T} f_t(w) \leq \frac{3}{2} GD\sqrt{T}$$

Proof: The proof follows directly from Theorem 3.1 in [1].
Let $u \in \arg\min_{w \in \mathcal{K}} \sum_{t=1}^{T} f_t(w)$ and let $g_t \coloneqq \nabla f_t(w_t)$. By convexity,

$$f_t(w_t) - f_t(u) \overset{(4)}{\leq} \langle g_t, w_t - u \rangle. \tag{23}$$

We upper bound the right-hand side using the update and the non-expansiveness of the projection. Since projection onto a closed convex set is non-expansive,

$$\|w_{t+1} - u\|_2^2 = \big\|\Pi_{\mathcal{K}}(w_t - \eta_t g_t) - u\big\|_2^2 \overset{(8)}{\leq} \big\|(w_t - \eta_t g_t) - u\big\|_2^2.$$

Expanding the square gives

$$\|w_{t+1} - u\|_2^2 \leq \|w_t - u\|_2^2 + \eta_t^2 \|g_t\|_2^2 - 2\eta_t \langle g_t,\, w_t - u\rangle.$$

Rearranging yields

$$2\langle g_t,\, w_t - u\rangle \;\leq\; \frac{\|w_t - u\|_2^2 - \|w_{t+1} - u\|_2^2}{\eta_t} \;+\; \eta_t \|g_t\|_2^2. \tag{24}$$

Combining (23) and (24) and summing over $t = 1, \ldots, T$ gives

$$2\sum_{t=1}^{T}\big(f_t(w_t) - f_t(u)\big) \;\leq\; \sum_{t=1}^{T} \frac{\|w_t - u\|_2^2 - \|w_{t+1} - u\|_2^2}{\eta_t} \;+\; \sum_{t=1}^{T} \eta_t \|g_t\|_2^2.$$

For the first sum with varying step sizes, write it as a telescoping series

$$\begin{aligned}\sum_{t=1}^{T} \frac{\|w_t - u\|_2^2 - \|w_{t+1} - u\|_2^2}{\eta_t} &= \frac{\|w_1 - u\|_2^2}{\eta_1} + \sum_{t=2}^{T}\Big(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}}\Big)\|w_t - u\|_2^2 - \frac{\|w_{T+1} - u\|_2^2}{\eta_T}. \\ &\overset{(21)}{\leq} \frac{D^2}{\eta_1} + \sum_{t=2}^{T}\Big(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}}\Big)D^2 = \frac{D^2}{\eta_T}.\end{aligned}$$

Therefore

$$2\,\mathrm{Regret}_T \overset{(22)}{\leq} \frac{D^2}{\eta_T} + G^2 \sum_{t=1}^{T} \eta_t \overset{\eta_t = \frac{D}{G\sqrt{t}}}{\leq} 3GD\sqrt{T}$$

□

2.4 Follow the Regularized Leader (FTRL)

As we have seen in online convex optimization, the learner selects a point $w_t \in \mathcal{K}$ in each round t, sees a convex loss function f_t, and pays $f_t(w_t)$. An obvious approach is to take the minimizer of the losses observed so far,

$$w_{t+1} = \arg\min_{w \in \mathcal{K}} \sum_{s=1}^{t} f_s(w).$$

This is known as the "Follow The Leader" algorithm (FTL). FTL is notorious for jumps and poor behavior in non-smooth or adversarial sequences. In [1] provides an example that even small changes to the last function can shift the minimizer significantly and thus generate high regret. The idea behind RFTL is to dampen this instability through curvature, a strictly convex regularization term ψ is added to the cumulative loss function, which "holds the updates together". Formally, the update is

$$w_{t+1} \in \arg\min_{w \in \mathcal{K}} \Big\{\eta \sum_{s=1}^{t} f_s(w) + \psi(w)\Big\},$$

where $\eta > 0$ is a step size and $\psi : \mathcal{K} \to \mathbb{R}$ should be differentiable and strongly convex with respect to a chosen norm $\|\cdot\|$. Intuitively, ψ is an smoothing "memory function." Even if the last losses are very flat or jagged, ψ stabilizes the updates by penalizing large deviations from previous iterates and ensuring strong convexity of the optimization objective.
Because of (4), we can estimate the convex cost function with

$$f_t(w_t) - f_t(w^*) \overset{(4)}{\leq} \langle g_t, w_t - w^*\rangle, \tag{25}$$

so we use the following update step

$$w_{t+1} \in \arg\min_{w\in\mathcal{K}} \Big\{\eta \sum_{t=1}^{T} \langle g_t, w - w^* \rangle + f_t(w^*) + \psi(w)\Big\} = \arg\min_{w\in\mathcal{K}} \Big\{\eta \sum_{t=1}^{T} \langle g_t, w\rangle + \psi(w)\Big\}.$$

This gives us the following pseudocode for FTRL

Algorithm 4: Follow the Regularized Leader (FTRL)

Data: convex set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, regularization function ψ, initial point $w_1 \in \arg\min_{w\in\mathcal{K}}\{\psi(w)\}$, step sizes $\{\eta_t\}$

for *t=1 to T* **do**
  Observe cost $f_t(w_t)$
  Update: $w_{t+1} = \arg\min_{w\in\mathcal{K}} \Big\{\eta \sum_{s=1}^{t} \langle g_s, w\rangle + \psi(w)\Big\}$
end
return w_{T+1}

The approach has proven particularly effective in online advertising, for example for click predictions and decisions in real-time auctions. FTRL is also used successfully in personalized recommendation systems for e-commerce, news, and streaming services, as well as in search and autocomplete ranking.

We would like to mention that there are two names for this algorithm in the literature: Regularized Follow The Leader (RFTL) and Follow The Regularized Leader (FTRL).

Now we come to analyze the regret.

Theorem 2.2

Let $\{w_t\}$ be a sequence generated by Algorithm 4 and let f_t be convex and differentiable for all $t \in \{1, 2, ..., T\}$ with $g_t = \nabla f_t(w_t)$. Furthermore let $\psi : \mathcal{K} \to \mathbb{R}$ be differentiable and strongly convex. The FTRL Algorithm attains for every $u \in \mathcal{K}$ the following bound on the regret

$$\text{Regret}_T^{\text{FTRL}} \leq 2\eta \sum_{t=1}^{T} \|g_t\|_*^2 + \frac{\psi(u) - \psi(w_1)}{\eta}$$

Proof: We derive the proof ourselves. Define

$$\Phi_t := \min_{w\in\mathcal{K}} \{\eta \sum_{s=1}^{t} \langle g_s, w\rangle + \psi(w)\},$$

where, for reasons of optimality, the following applies

$$\Phi_t := \eta \sum_{s=1}^{t} \langle g_s, w_{t+1}\rangle + \psi(w_{t+1}) \quad \text{and} \quad \Phi_{t-1} := \eta \sum_{s=1}^{t-1} \langle g_s, w_t\rangle + \psi(w_t).$$

By optimality of w_t, the following holds

$$\eta \sum_{s=1}^{t-1} \langle g_s, w_{t+1}\rangle + \psi(w_{t+1}) \geq \eta \sum_{s=1}^{t-1} \langle g_s, w_t\rangle + \psi(w_t). \tag{26}$$

Now consider

$$\begin{aligned}\Phi_t - \Phi_{t-1} &= \eta \langle \sum_{s=1}^{t-1} g_s + g_t, w_{t+1}\rangle + \psi(w_{t+1}) - \eta \sum_{s=1}^{t-1} \langle g_s, w_t\rangle - \psi(w_t) \\ &\overset{(26)}{\geq} \eta \langle g_t, w_{t+1}\rangle.\end{aligned}$$

By rearranging and using the telescopic sum, we obtain for all $u \in \mathcal{K}$

$$\sum_{t=1}^{T} \langle g_t, w_{t+1} \rangle \leq \frac{\Phi_T - \Phi_0}{\eta} \leq \sum_{s=1}^{T} \langle g_s, u \rangle + \frac{\psi(u) - \psi(w_1)}{\eta}. \tag{27}$$

Now we estimate the regret

$$\begin{aligned}
\text{Regret}_T &\leq \sum_{t=1}^{T} \langle g_t, w_t - u \rangle = \sum_{t=1}^{T} \langle g_t, w_t - w_{t+1} \rangle + \sum_{t=1}^{T} \langle g_t, w_{t+1} - u \rangle \\
&\overset{(10)}{\leq} \sum_{t=1}^{T} \|g_t\|_* \|w_t - w_{t+1}\| + \sum_{t=1}^{T} \Big(\langle g_t, w_{t+1} \rangle - \sum_{s=1}^{T} \langle g_s, u \rangle \Big) \\
&\overset{(5),(27)}{\leq} 2\eta \sum_{t=1}^{T} \|g_t\|_*^2 + \frac{\psi(u) - \psi(w_1)}{\eta}
\end{aligned}$$

□

2.5 Online Mirror Descent (OMD)

With the preparations from Chapter 1, we can directly specify the online version of the Mirror Descent Algorithm.

Algorithm 5: Online Mirror Descent (OMD) with Agile version

Data: closed convex set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, initial point $w_1 \in \arg\min_{w \in \mathcal{K}} \{B_\psi(w \| y_1) \mid 0 = \nabla \psi(y_1)\}$, step size $\eta > 0$, generator function ψ

1 **for** *t=1 to T* **do**
2 Observe cost $f_t(w_t)$.
3 $y_{t+1} = \nabla \psi^{-1}(\nabla \psi(w_t) - \eta \nabla f_t(w_t))$
4 $w_{t+1} = \arg\min_{w \in \mathcal{K}} B_\psi(w \| y_{t+1})$
5 **end**
6 **return** w_{T+1}

We would now like to analyze the regret of OMD.

Theorem 2.3

Let $\{w_t\}$ be a sequence generated by Algorithm 5 and let f_t be convex and differentiable for all $t \in \{1, 2, ..., T\}$ with $g_t = \nabla f_t(w_t)$. Furthermore let $\psi : \mathcal{K} \to \mathbb{R}$ be differentiable and strongly convex. The online mirror descent algorithm (OMD) attains for every $u \in \mathcal{K}$ the following bound on the regret

$$\text{Regret}_T^{\text{OMD}} \leq \frac{\eta}{2} \sum_{t=1}^{T} \|g_t\|_*^2 + \frac{\psi(u) - \psi(w_1)}{\eta} \tag{28}$$

Proof: The proof follows directly from Theorem 5.6 in [1].
Since the functions f_t are convex, for any $u \in \mathcal{K}$,

$$\begin{aligned}
f_t(w_t) - f_t(u) &\overset{(25)}{\leq} \nabla f_t(w_t)^\top (w_t - u) \\
&= \tfrac{1}{\eta} (\nabla \psi(y_{t+1}) - \nabla \psi(w_t))^\top (u - w_t) \\
&\overset{(19)}{=} \tfrac{1}{\eta} \big[B_\psi(u \| w_t) - B_\psi(u \| y_{t+1}) + B_\psi(w_t \| y_{t+1}) \big] \\
&\overset{(20)}{\leq} \tfrac{1}{\eta} \big[B_\psi(u \| w_t) - B_\psi(u \| w_{t+1}) + B_\psi(w_t \| y_{t+1}) \big]
\end{aligned}$$

where the last inequality follows from (20), as w_{t+1} is the projection w.r.t the Bregman divergence of w_{t+1} and $u \in \mathcal{K}$ is in the convex set. The first equality in line 2 is obtained by rearranging step 3 in the OMD algorithm according to $\nabla f_t(w_t)$. Summing over all iterations,

$$\text{Regret} \leq \tfrac{1}{\eta} \big[B_\psi(u \| w_1) - B_\psi(u \| w_{T+1}) \big] + \sum_{t=1}^{T} \tfrac{1}{\eta} B_\psi(w_t \| y_{t+1}). \tag{29}$$

We proceed to bound $B_\psi(w_t\|y_{t+1})$. By definition of Bregman divergence, and the generalized Cauchy inequality,

$$\begin{aligned} B_\psi(w_t\|y_{t+1}) + B_\psi(y_{t+1}\|w_t) &= (\nabla\psi(w_t) - \nabla\psi(y_{t+1}))^\top (w_t - y_{t+1}) \\ &= \eta \nabla f_t(w_t)^\top (w_t - y_{t+1}) \\ &\overset{(10)}{\leq} \eta \|\nabla f_t(w_t)\|^* \|w_t - y_{t+1}\| \\ &\leq \tfrac{1}{2}\eta^2 \|\nabla f_t(w_t)\|^{*2} + \tfrac{1}{2}\|w_t - y_{t+1}\|^2. \end{aligned}$$

where the last inequality is Young's inequality $ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2$. Thus, because of the strong convexity of ψ we have $B_\psi(w_t\|y_{t+1}) \geq \frac{1}{2}\|w_t - y_{t+1}\|^2$ and obtain

$$B_\psi(w_t\|y_{t+1}) \leq \tfrac{1}{2}\eta^2 \|\nabla f_t(w_t)\|^{*2} + \tfrac{1}{2}\|w_t - y_{t+1}\|^2 - B_\psi(y_{t+1}\|w_t) \overset{(18)}{\leq} \tfrac{1}{2}\eta^2 \|\nabla f_t(w_t)\|^{*2} \tag{30}$$

Now we want to bound the other term $B_\psi(u\|w_1) - B_\psi(u\|w_T)$ in (29). Recall that w_1 is chosen as a minimizer of $B_\psi(x\|y_1)$. Since by assumption $\nabla\psi(y_1) = 0$, the Bregman divergence reduces to

$$B_\psi(x\|y_1) = \psi(x) - \psi(y_1).$$

Thus minimizing $B_\psi(x\|y_1)$ over x is equivalent to minimizing $\psi(x)$ itself. In particular, w_1 is a minimizer of ψ, and therefore $\nabla\psi(w_1) = 0$. Using this property, we obtain

$$B_\psi(u\|w_1) = \psi(u) - \psi(w_1) - \langle \nabla\psi(w_1),\, u - w_1\rangle = \psi(u) - \psi(w_1).$$

Since $B_\psi(u\|w_T) \geq 0$, it follows that

$$B_\psi(u\|w_1) - B_\psi(u\|w_T) \;\leq\; B_\psi(u\|w_1) \;=\; \psi(u) - \psi(w_1). \tag{31}$$

Substitute (30) and (31) into (29) to obtain

$$\mathrm{Regret}_T \leq \frac{\eta}{2}\sum_{t=1}^T \|g_t\|^{*2} + \frac{\psi(u) - \psi(w_1)}{\eta}$$

□

2.6 Adaptive Gradient Method (AdaGrad)

In Online Mirror Descent (OMD), the choice of regularizer ψ is problem dependent, ideally, it should minimize regret given the decision set and the loss geometry. The central idea behind *adaptive* methods is to *learn* this geometry online. Consider a quadratic regularizer with a positive definite matrix $H \succ 0$:

$$\psi(x) = \tfrac{1}{2}\|x\|_H^2 = \tfrac{1}{2}x^\top H x.$$

We derive the idea behind the algorithm ourselves. By (28), the regret of OMD with step size $\eta > 0$ satisfies

$$\mathrm{Regret}_T \;\leq\; \frac{\eta}{2}\sum_{t=1}^T g_t^\top H^{-1} g_t \;+\; \frac{1}{2\eta}\,\|u - w_1\|_H^2. \tag{32}$$

Using the trace identity $a^\top M a = \mathrm{Tr}(M\, aa^\top)$ and linearity of the trace,

$$\sum_{t=1}^T g_t^\top H^{-1} g_t \;=\; \mathrm{Tr}\Big(H^{-1}\sum_{t=1}^T g_t g_t^\top\Big) \;=\; \mathrm{Tr}\big(H^{-1} G_T\big), \qquad G_T := \sum_{t=1}^T g_t g_t^\top.$$

Hence

$$\mathrm{Regret}_T \;\leq\; \frac{1}{2\eta}\,\|u - w_1\|_H^2 \;+\; \frac{\eta}{2}\,\mathrm{Tr}(H^{-1}G_T). \tag{33}$$

As u is unknown, we upper bound the first term in (33) by its worst case over $\mathcal{K}$. For any x and any PSD H,

$$x^\top H x \;\overset{(11)}{\leq}\; \lambda_{\max}(H)\,\|x\|_2^2 \;\overset{(12)}{\leq}\; \mathrm{Tr}(H)\,\|x\|_2^2.$$

Recalling (21) and defining $D_2^2 := \sup_{u,v\in\mathcal{K}} \|u-v\|_2^2$, we obtain

$$\sup_{u\in\mathcal{K}} \|u-w_1\|_H^2 \;\le\; D_2^2\ \mathrm{Tr}(H).$$

Plugging this into (33) yields the *geometry-selection* objective

$$\mathrm{Regret}_T \;\le\; \frac{D_2^2}{2\eta}\ \mathrm{Tr}(H) \;+\; \frac{\eta}{2}\ \mathrm{Tr}(H^{-1}G_T). \tag{34}$$

For fixed $\eta > 0$, minimizing (34) over $H \succeq 0$ reduces (up to the positive factor $\frac{\eta}{2}$) to

$$\min_{H\succeq 0}\ \Phi(H) := \mathrm{Tr}(H^{-1}G_T) + \alpha\ \mathrm{Tr}(H), \qquad \alpha := \frac{D_2^2}{\eta^2} > 0.$$

Let $G_T = U\Lambda U^\top$ with $\Lambda = \mathrm{diag}(\lambda_i) \succeq 0$ and set $K = U^\top H U$. By unitary invariance of the trace,

$$\Phi(H) = \mathrm{Tr}(K^{-1}\Lambda) + \alpha\ \mathrm{Tr}(K) =: \Psi(K).$$

A direct calculation gives $\nabla_K \Psi(K) = -K^{-1}\Lambda K^{-1} + \alpha I$. Any minimizer $K^\star \succ 0$ must satisfy $-K^{\star-1}\Lambda K^{\star-1} + \alpha I = 0$, hence $\Lambda = \alpha K^{\star 2}$. Because Λ is diagonal, $K^\star$ is diagonal and commutes with Λ, yielding the unique minimizer

$$K^\star = \alpha^{-1/2}\Lambda^{1/2} \qquad \Longrightarrow \qquad H^\star = UK^\star U^\top = \alpha^{-1/2} G_T^{1/2}.$$

Thus, the optimal Mahalanobis geometry aligns with the empirical second-moment matrix G_T.

Of course, G_T is not available online. The adaptive-gradient idea is to *learn* the geometry by replacing G_T with the cumulative sum up to time t,

$$G_t := \sum_{s=1}^{t} g_s g_s^\top, \qquad \boxed{H_t := G_t^{1/2}} \quad \text{(full-matrix AdaGrad)},$$

and performing the OMD update in the time-varying H_t-geometry:

$$y_{t+1} = w_t - \eta\, H_t^{-1} g_t, \qquad w_{t+1} = \Pi_{\mathcal{K}}^{H_t}(y_{t+1}),$$

where $\Pi_{\mathcal{K}}^{H_t}$ denotes the projection onto $\mathcal{K}$ in the norm induced by H_t.
In practical applications, H_t is a full matrix and is computationally prohibitive for large dimensions. To solve this, the problem is simplified by using only the diagonal elements of G. It yields

$$G_t := \sum_{s=1}^{t} g_s g_s^\top, \qquad \boxed{H_t := \mathrm{diag}(G_t)^{1/2}} \quad \text{(diagonal version AdaGrad)}.$$

With these considerations in mind, we can specify the AdaGrad algorithm.

Algorithm 6: Adaptive Gradient Method (AdaGrad)

Data: convex set $\mathcal{K}$, iteration number $T \in \mathbb{N}$, initial point $w_1 \in \mathcal{K}$, step size $\eta > 0$

1 **Initialize:** $0 = G_0 \in \mathbb{R}^{n \times n}$, $0 = s_0 \in \mathbb{R}^n$.

2 **for** $t = 1$ **to** T **do**

3 Play w_t and observe f_t; set $g_t = \nabla f_t(w_t)$.

4 Update:

- *Diagonal AdaGrad:*
  $s_t = s_{t-1} + g_t \odot g_t$
  $h_t = (\sqrt{s_{t1}}, ..., \sqrt{s_{tn}})^\top$
- *Full-matrix AdaGrad:*
  $G_t = G_{t-1} + g_t g_t^\top$
  $H_t = G_t^{1/2}$.

5 Compute:

- *Diagonal AdaGrad:*
  $y_{t+1} = w_t - \eta \, (\frac{1}{h_{t1}}, ..., \frac{1}{h_{tn}})^\top \odot g_t$
- *Full-matrix AdaGrad:*
  $y_{t+1} = w_t - \eta \, H_t^{-1} g_t$

6 Set:

- *Diagonal AdaGrad:*
  $w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\| \left(\frac{1}{\sqrt{h_{t1}}}, ..., \frac{1}{\sqrt{h_{tn}}}\right)^\top \odot (w - y_{t+1}) \right\|^2$
- *Full-matrix AdaGrad:*
  $w_{t+1} = \arg\min_{w \in \mathcal{K}} \|w - y_{t+1}\|_{H_t}^2$

7 **end**

8 **return** w_{T+1}

Remark:

- If the matrix in step 5 is singular, the Moore-Penrose pseudoinverse of H_t is used.
- AdaGrad can be seen as a preconditioned first-order method: instead of a scalar step size, it rescales the gradient by a diagonal or full matrix, i.e.,
  $$w_{t+1} = w_t - \eta \, P_t g_t.$$
  This is analogous to quasi-Newton methods, except that quasi-Newton chooses $P_t \approx (\nabla^2 f(w_t))^{-1}$ from curvature information, whereas AdaGrad builds P_t from gradient second moments $G_t = \sum_{\tau \leq t} g_\tau g_\tau^\top$.

In order to fully specify the regret of the AdaGrad algorithm, some preparations must be made. The following lemma provides an initial estimate.

Lemma 2.1

Let $\{w_t\}$ be a sequence generated by algorithm 6 and let f_t be convex and differentiable for all $t \in \{1, 2, ..., T\}$ with $g_t = \nabla f_t(w_t)$. If $0 = H_0 \in \mathbb{R}^{n \times n}$, then the following applies to both versions of the AdaGrad algorithm for all $u \in \mathcal{K}$

$$\text{Regret}_T^{\text{AdaGrad}} \leq \frac{1}{2\eta} \sum_{t=1}^{T} \|w_t - u\|_{H_t - H_{t-1}}^2 + \frac{\eta}{2} \sum_{t=1}^{T} g_t^\top H_t^{-1} g_t. \tag{35}$$

Proof: The proof follows directly from Lemma 5.13 in [1].
By the definition of y_{t+1} it holds

$$(y_{t+1} - u)^\top H_t (y_{t+1} - u) = (w_t - u)^\top H_t (w_t - u) - 2\eta \, g_t^\top (w_t - u) + \eta^2 g_t^\top H_t^{-1} g_t.$$

We rearrange the equation and obtain

$$g_t^\top(w_t - u) = \frac{1}{2\eta}\Big((w_t-u)^\top H_t(w_t-u) - (y_{t+1}-u)^\top H_t(y_{t+1}-u)\Big) + \frac{\eta}{2} g_t^\top H_t^{-1} g_t$$

Since w_{t+1} is the projection of y_{t+1} in the norm induced by H_t, we have

$$g_t^\top(w_t - u) \overset{(8)}{\leq} \frac{1}{2\eta}\Big((w_t-u)^\top H_t(w_t-u) - (w_{t+1}-u)^\top H_t(w_{t+1}-u)\Big) + \frac{\eta}{2} g_t^\top H_t^{-1} g_t,$$

where here we use the projection in the H_t norm. Summing over all iterations $t \in \{1, 2, 3, .., T\}$ we get

$$\begin{aligned}\sum_{t=1}^{T} g_t^\top(w_t - u) &\leq -\frac{1}{2\eta}\|w_{T+1} - u\|_{H_T}^2 + \frac{1}{2\eta}\sum_{t=1}^{T}\|w_t - u\|_{H_t}^2 - \|w_t - u\|_{H_{t-1}}^2 + \frac{\eta}{2}\sum_{t=1}^{T} g_t^\top H_t^{-1} g_t \\ &\leq \frac{1}{2\eta}\sum_{t=1}^{T}\|w_t - u\|_{H_t - H_{t-1}}^2 + \frac{\eta}{2}\sum_{t=1}^{T} g_t^\top H_t^{-1} g_t.\end{aligned}$$

□

We want to further estimate the term $\sum_{t=1}^{T} g_t^\top H_t^{-1} g_t$ and introduce the following lemma.

Lemma 2.2

Let $(g_t)_{t=1}^T \subset \mathbb{R}^n$, $G_0 = 0$, and

$$G_t \;=\; \sum_{s=1}^{t} g_s g_s^\top \quad \text{and} \quad H_t \;=\; G_t^{1/2} \;\; (t \geq 1).$$

Then

$$\sum_{t=1}^{T} g_t^\top H_t^{-1} g_t \;\leq\; 2\,\mathrm{Tr}(H_T). \tag{36}$$

Here H_t^{-1} is understood as the inverse of H_t (or the Moore–Penrose inverse of H_t).

Proof:We provide the proof here for completeness.
For $\varepsilon > 0$ define

$$\Phi_\varepsilon(X) \;:=\; \mathrm{Tr}\big((X + \varepsilon I)^{1/2}\big), \qquad X \succeq 0.$$

The map $X \mapsto \mathrm{Tr}(X^{1/2})$ is concave on the PSD cone (Theorem 4.2.3 in [13]), hence Φ_ε is concave and differentiable with

$$\nabla\Phi_\varepsilon(X) \;=\; \tfrac{1}{2}\,(X + \varepsilon I)^{-1/2},$$

where gradients are taken with respect to the trace inner product $\langle A, B\rangle = \mathrm{Tr}(A^\top B)$. Since $-\Phi_\varepsilon$ is convex, the first-order condition (with reversed sign) gives

$$\Phi_\varepsilon(G_t) - \Phi_\varepsilon(G_{t-1}) \;\geq\; \big\langle \nabla\Phi_\varepsilon(G_t),\, G_t - G_{t-1}\big\rangle.$$

where gradients are taken with respect to the trace inner product $\langle A, B\rangle := \mathrm{Tr}(A^\top B)$. By the first order condition for differentiable convex functions (change sign) we conclude,

$$\Phi_\varepsilon(G_t) - \Phi_\varepsilon(G_{t-1}) \;\overset{(4)}{\geq}\; \big\langle \nabla\Phi_\varepsilon(G_t),\, G_t - G_{t-1}\big\rangle.$$

Using $G_t - G_{t-1} = g_t g_t^\top$,

$$\Phi_\varepsilon(G_t) - \Phi_\varepsilon(G_{t-1}) \;\geq\; \tfrac{1}{2}\,\big\langle (G_t + \varepsilon I)^{-1/2},\, g_t g_t^\top\big\rangle \;=\; \tfrac{1}{2}\, g_t^\top (G_t + \varepsilon I)^{-1/2} g_t.$$

Summing over $t = 1, \ldots, T$ and noting $G_0 = 0$ gives

$$\sum_{t=1}^{T} g_t^\top (G_t + \varepsilon I)^{-1/2} g_t \;\leq\; 2\Big(\mathrm{Tr}\big((G_T + \varepsilon I)^{1/2}\big) - \mathrm{Tr}\big((G_0 + \varepsilon I)^{1/2}\big)\Big) = 2\Big(\mathrm{Tr}\big((G_T + \varepsilon I)^{1/2}\big) - n\sqrt{\varepsilon}\Big).$$

Finally, let $\varepsilon \downarrow 0$. For each t, the function $\varepsilon \mapsto g_t^\top (G_t + \varepsilon I)^{-1/2} g_t$ decreases to $g_t^\top G_t^{-1/2} g_t$, the limit is finite because $g_t \in \text{range}(G_t) = \text{span}\{g_1, \ldots, g_t\}$, hence it has no component in $\ker(G_t)$ where $(G_t + \varepsilon I)^{-1/2}$ would blow up. On the right, continuity of $X \mapsto \text{Tr}(X^{1/2})$ yields

$$2\Big(\text{Tr}\big((G_T + \varepsilon I)^{1/2}\big) - n\sqrt{\varepsilon}\Big) \;\longrightarrow\; 2\,\text{Tr}(G_T^{1/2}).$$

Therefore

$$\sum_{t=1}^{T} g_t^\top G_t^{-1/2} g_t \;\le\; 2\,\text{Tr}(G_T^{1/2}),$$

which is (36) since $H_t = G_t^{1/2}$. □

Now we can specify the regret bound for both variants of the AdaGrad algorithm.

Theorem 2.4

Let $D_2 := \sup_{u,v \in \mathcal{K}} \|u - v\|_2$ and $\eta = \frac{D_2}{\sqrt{2}}$ then the following regret bound hold

$$\text{Regret}_T^{\text{Full Adagrad}} \le \sqrt{2} D_2 \text{Tr}(H_T)$$

.

Proof: The proof follows directly from Lemma 5.15 in [1].
By definition $G_t \succeq G_{t-1} \Rightarrow H_t \succeq H_{t-1}$ (square root is operator–monotone Theorem 1.5.9 in [13]), hence $H_t - H_{t-1} \succeq 0$. For every $H_t - H_{t-1} \succeq 0$ and any $v \in \mathbb{R}^n$, $v^\top (H_t - H_{t-1}) v \le \lambda_{\max}(H_t - H_{t-1}) \|v\|_2^2 \le \text{Tr}(H_t - H_{t-1}) \|v\|_2^2$. Thus for all $u \in \mathcal{K}$,

$$\begin{aligned} \sum_{t=1}^{T} (w_t - u)^\top (H_t - H_{t-1})(w_t - u) &\le D_2^2 \sum_{t=1}^{T} \text{Tr}(H_t - H_{t-1}) \\ &= D_2^2 \sum_{t=1}^{T} \big(\text{Tr}(H_t) - \text{Tr}(H_{t-1})\big) \\ &= D_2^2\, \text{Tr}(H_T). \end{aligned} \tag{37}$$

Now put (36) and (37) into (35) and get

$$\text{Regret}_T(\text{Full matrix Adagrad}) \le \frac{D_2^2}{2\eta} \text{Tr}(H_T) + \eta \text{Tr}(H_T) \overset{\eta = \frac{D_2}{\sqrt{2}}}{=} \sqrt{2} D_2 \text{Tr}(H_T)$$

□

Theorem 2.5

Let $D_\infty := \sup_{u,v \in \mathcal{K}} \|u - v\|_\infty$ and $\eta = \frac{D_\infty}{\sqrt{2}}$ then the following regret bound hold

$$\text{Regret}_T^{\text{Diag Adagrad}} \le \sqrt{2} D_\infty \text{Tr}(H_T)$$

.

Proof: The proof follows directly from Lemma 5.15 in [1].
For PSD matrices, the diag operator is monotonic, so by definition $G_t \succeq G_{t-1} \Rightarrow H_t \succeq H_{t-1}$. Since for a diagonal matrix H it holds that $x^\top H x \le \|x\|_\infty^2 \text{Tr}(H)$, we have for all $u \in \mathcal{K}$

$$\begin{aligned} \sum_{t=1}^{T} (w_t - u)^\top (H_t - H_{t-1})(w_t - u) &\le D_\infty^2 \sum_{t=1}^{T} \text{Tr}(H_t - H_{t-1}) \\ &= D_\infty^2 \sum_{t=1}^{T} \big(\text{Tr}(H_t) - \text{Tr}(H_{t-1})\big) \\ &= D_\infty^2\, \text{Tr}(H_T). \end{aligned} \tag{38}$$

Now put (36) and (38) into (35) and get

$$\text{Regret}_T(\text{Diagonal Adagrad}) \leq \frac{D_\infty^2}{2\eta}\text{Tr}(H_T) + \eta\text{Tr}(H_T) \overset{\eta=\frac{D_\infty}{\sqrt{2}}}{=} \sqrt{2}D_\infty\text{Tr}(H_T)$$

□

3 Subgradients

In the previous chapter, we studied Online Convex Optimization (OCO) under the assumption that all cost functions are convex and differentiable. This setting allowed us to apply gradient-based algorithms such as Online Gradient Descent and analyze their performance using regret minimization. However, many functions of practical importance, including the hinge loss, the absolute value function, and ℓ_1-regularization objectives, and certain robustness terms, are *not differentiable* everywhere. In such cases, the standard gradient is no longer defined, and classical first-order methods cannot be applied directly.
To overcome this limitation, we introduce the concept of *subgradients*, a generalization of the gradient that enables the use of first-order methods even in the absence of differentiability. Subgradients play a fundamental role in modern convex analysis and online optimization, particularly in the design of algorithms that operate under minimal assumptions.
This chapter provides a rigorous theoretical foundation for subgradients and prepares their use in adaptive online learning methods discussed later in this thesis. To do so, we proceed step by step.

- **Closed Convex Functions**: We begin with the notion of closed convex functions, which ensure well-posedness of optimization problems through their epigraphical structure. Closedness also guarantees the existence of minimizers under mild assumptions.
- **Continuity and Directional Derivatives**: We then explore the regularity of convex functions. While convex functions may not be differentiable, they exhibit well-behaved properties such as continuity in the interior of their domain and the existence of directional derivatives in every direction.
- **Separation Theorems**: Classical separation theorems from convex geometry are presented, which guarantee the existence of supporting hyperplanes to convex sets. These geometric tools are essential for motivating the definition of subgradients.
- **Subgradients and Subdifferentials**: With this foundation, we formally define subgradients—both analytically and geometrically—as generalizations of gradients that define valid descent directions even for nonsmooth convex functions.
- **Optimality Conditions**: Based on subgradients, we formulate general first-order optimality conditions that remain valid even when the function is not differentiable.
- **The Subgradient Method**: Finally, we introduce the subgradient method, a simple yet powerful iterative algorithm for minimizing convex, possibly nondifferentiable, functions. We explain under which assumptions this method converges and how it fits naturally into the online optimization framework.

Altogether, this chapter extends the theory of convex optimization to nonsmooth settings and provides the mathematical machinery needed for robust and adaptive learning algorithms. Subgradients thus serve as the central tool for optimization in complex, real-world scenarios where smoothness cannot be guaranteed.

We would like to note that almost all of the following proofs was based on [2], but we have provided more detail for the sake of clarity.

3.1 Closed convex functions

In convex optimization—particularly when dealing with nondifferentiable functions—it is essential to impose minimal yet sufficient regularity conditions that ensure the existence of solutions and the well-posedness of optimization problems. One such foundational property is *closedness*. To motivate this concept, we begin by examining *level sets*—the sets on which a function takes values below a given threshold. Level sets are a fundamental tool in convex analysis and algorithm design.

Definition 3.1: Level sets

We define the level sets of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ with

$$\mathscr{L}_f(\beta) = \{x \in \mathrm{dom}(f) | f(x) \leq \beta\},$$

where $\beta \in \mathbb{R}$.

The following classical result guarantees that level sets of convex functions preserve convexity.

Theorem 3.1: Level sets of convex functions

If a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ is convex then all level sets

$$\mathscr{L}_f(\beta) = \{x \in \mathrm{dom}(f) | f(x) \leq \beta\}, \ \beta \in \mathbb{R} \tag{39}$$

are either convex or empty.

Proof: The proof comes from Theorem 3.1.3 in [2]. Let $x_1 \in \mathscr{L}_f(\beta)$ and $x_2 \in \mathscr{L}_f(\beta)$, then choose $\lambda \in [0, 1]$ we get

$$f(\lambda x_1 + (1-\lambda)x_2) \overset{(1)}{\leq} \lambda f(x_1) + (1-\lambda) f(x_2) \overset{(39)}{\leq} \lambda\beta + (1-\lambda)\beta = \beta.$$

This means $\lambda x_1 + (1-\lambda)x_2 \in \mathscr{L}_f(\beta)$ and therefore $\mathscr{L}_f(\beta)$ is convex.

□

To ensure convergence of optimization algorithms, we need to understand how a function behaves under limiting operations. This leads to the concept of *closed convex functions*, defined via their epigraphs.

Definition 3.2: Closed convex functions

A convex function $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ is called closed and convex on a convex set $\mathcal{K} \subset \mathrm{dom}(f)$ if its constrained epigraph

$$\mathrm{epi}_{\mathcal{K}}(f) = \{(x, t) \in \mathcal{K} \times \mathbb{R} | t \geq f(x)\}$$

is a closed convex set. If $\mathcal{K} = \mathrm{dom}(f)$, we call f a closed convex function.

The next result summarizes several key topological properties that make closed convex functions analytically robust.

Theorem 3.2: Topological properties of closed convex functions

Let a convex function $f : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ be closed.

1. For any sequence $\{x_k\}_{k\in\mathbb{N}} \subset \mathrm{dom}(f)$ convergent to a point $\bar{x} \in \mathrm{dom}(f)$ we have
$$\liminf_{k\to\infty} f(x_k) \geq f(\bar{x}). \tag{40}$$
2. For any sequence $\{x_k\}_{k\in\mathbb{N}} \subset \mathrm{dom}(f)$ convergent to a point $\bar{x} \notin \mathrm{dom}(f)$ we have
$$\lim_{k\to\infty} f(x_k) = \infty.$$
3. All level sets of the function f are either empty or closed and convex.
4. Let f be closed and convex on a closed set $\mathcal{K}$ and its constrained level sets be bounded. Then problem
$$\min_{x\in\mathcal{K}} f(x)$$
is solvable.
5. Let f be closed and convex on $\mathcal{K}$. If the optimal set $\arg\min_{x\in\mathcal{K}} f(x)$ is nonempty and bounded, then all level sets of the function f on $\mathcal{K}$ are either empty or bounded.

Proof: The proof comes from Theorem 3.1.4 in [2]. It should be noted that we present the proofs of the statements in much more detail.

1. Consider the epigraph
$$\mathrm{epi}(f) \;=\; \{(x,t) \in \mathbb{R}^n \times \mathbb{R} \mid f(x) \leq t\}.$$
Since f is closed, $\mathrm{epi}(f)$ is a closed convex set. For the sequence
$$P_k = (x_k,\, f(x_k)) \;\in\; \mathrm{epi}(f),$$
we have $x_k \to \bar{x}$. By closedness of $\mathrm{epi}(f)$, every convergent subsequence of $\{P_k\}$ must have its limit in $\mathrm{epi}(f)$.

   Case 1: $\{f(x_k)\}$ has a convergent subsequence.
   Then there exists a subsequence k_j with
$$P_{k_j} = (x_{k_j}, f(x_{k_j})) \;\longrightarrow\; (\bar{x}, \bar{t}),$$
   and $(\bar{x}, \bar{t}) \in \mathrm{epi}(f)$. Hence $\bar{t} \geq f(\bar{x})$ and
$$\liminf_{k\to\infty} f(x_k) = \lim_{j\to\infty} f(x_{k_j}) = \bar{t} \geq f(\bar{x}).$$

   Case 2: $\{f(x_k)\}$ has no convergent subsequence.
   Then either
$$\liminf_{k\to\infty} f(x_k) = -\infty \quad \text{or} \quad \liminf_{k\to\infty} f(x_k) = +\infty.$$

   - *Case 2a:* $\liminf f(x_k) = -\infty$.
     Then there exists a subsequence $\{x_{k_j}\}$ with $f(x_{k_j}) \to -\infty$. In particular, for any $\varepsilon > 0$ there exists $J \in \mathbb{N}$ such that for all $j \geq J$,
$$f(x_{k_j}) < f(\bar{x}) - \varepsilon,$$
     i.e. $f(\bar{x}) - \varepsilon \geq f(x_{k_j})$, which means $(x_{k_j},\, f(\bar{x}) - \varepsilon) \in \mathrm{epi}(f)$. Since $x_{k_j} \to \bar{x}$ (as a subsequence of $x_k \to \bar{x}$), we obtain
$$(x_{k_j},\, f(\bar{x}) - \varepsilon) \;\longrightarrow\; (\bar{x},\, f(\bar{x}) - \varepsilon),$$
     but $(\bar{x},\, f(\bar{x}) - \varepsilon) \notin \mathrm{epi}(f)$, contradicting the closedness of $\mathrm{epi}(f)$.

- *Case 2b:* $\liminf f(x_k) = +\infty$.
  Then trivially $\liminf f(x_k) = +\infty \geq f(\bar{x})$.

Since Case 2a is impossible, only Case 2b remains, and in both main cases we conclude

$$\liminf_{k\to\infty} f(x_k) \;\geq\; f(\bar{x}).$$

2. We prove the contrapositive. Assume, for contradiction, that there exists a subsequence $\{x_{k_l}\}$ and a finite constant $M < \infty$ such that

$$f\big(x_{k_l}\big) \;\leq\; M \quad \text{for all } \ell.$$

Set $\tau_l = M + 1$. Then each point

$$\big(x_{k_l},\, \tau_l\big) \;=\; \big(x_{k_l},\, M+1\big)$$

lies in the epigraph of f, since $\tau_l = M + 1 \geq f(x_{k_l})$. But $x_{k_l} \to \bar{x}$ and τ_l is constant, so

$$\big(x_{k_l},\, \tau_l\big) \;\longrightarrow\; \big(\bar{x},\, M+1\big) \in \mathbb{R}^{n+1}.$$

Because f is closed, $\mathrm{epi}(f)$ is a closed set, and therefore its limit point $\big(\bar{x}, M+1\big)$ must also lie in $\mathrm{epi}(f)$. Hence

$$M + 1 \;\geq\; f(\bar{x}),$$

which contradicts $\bar{x} \notin \mathrm{dom}(f)$. This contradiction shows that no bounded subsequence of $\{f(x_k)\}$ can exist. Therefore

$$\lim_{k\to\infty} f(x_k) \;=\; +\infty,$$

as claimed.

3. Recall that the *epigraph* of f is

$$\mathrm{epi}(f) \;=\; \{\,(x,t) \in \mathbb{R}^n \times \mathbb{R} \mid t \geq f(x)\},$$

which by assumption is a closed convex set in $\mathbb{R}^{n+1}$.

Consider the hyperplane

$$H_\beta \;=\; \{\,(x,t) \in \mathbb{R}^n \times \mathbb{R} \mid t = \beta\}.$$

Since H_β is an affine (hence closed and convex) subset of $\mathbb{R}^{n+1}$, the intersection

$$\mathrm{epi}(f) \;\cap\; H_\beta$$

is also closed and convex. But

$$\mathrm{epi}(f) \;\cap\; H_\beta \;=\; \{\,(x,\beta) \mid \beta \geq f(x)\} \;=\; \{\,(x,\beta) \mid x \in \mathscr{L}_f(\beta)\} \;\cong\; \{(\mathscr{L}_f(\beta), \beta)\},$$

Therefore $\mathscr{L}_f(\beta)$ is closed and convex (or empty, if no x satisfies $f(x) \leq \beta$).

4. We divide the proof into four steps.
Step 1: Construction of a minimizing sequence.
Let

$$f^* := \inf_{x\in\mathcal{K}} f(x).$$

Since $f^* \in \mathbb{R} \cup \{-\infty\}$, choose a sequence $\{x_k\} \subseteq \mathcal{K}$ such that

$$f(x_k) \to f^*.$$

Step 2: Boundedness and convergence of subsequence.
Since $f(x_k) \leq f(x_1)$ for all k, the sequence $\{x_k\}$ lies in the level set

$$\mathscr{L}_f(f(x_1)) = \{x \in \mathcal{K} : f(x) \leq f(x_1)\},$$

which is bounded by assumption. Thus, $\{x_k\}$ is a bounded sequence. By the Bolzano–Weierstrass theorem, there exists a convergent subsequence. Without loss of generality, assume:

$$x_k \to x^* \in \mathcal{K}.$$

Step 3: Show that $f^* > -\infty$.
Suppose, for contradiction, that $f^* = -\infty$. Define a sequence of convex combinations:

$$y_k := (1 - \alpha_k)x_0 + \alpha_k x_k,$$

where $x_0 \in \mathcal{K}$ is fixed and $\alpha_k \downarrow 0$ is a sequence of slowly decreasing positive numbers.
Since $\mathcal{K}$ is convex, $y_k \in \mathcal{K}$. By convexity of f, we have:

$$f(y_k) \leq (1 - \alpha_k)f(x_0) + \alpha_k f(x_k) = f(x_0) + \alpha_k(f(x_k) - f(x_0)).$$

As $f(x_k) \to -\infty$, we can choose $\alpha_k := \frac{1}{\sqrt{|f(x_k)|+1}} \to 0$ slowly enough so that

$$\alpha_k(f(x_k) - f(x_0)) \to -\infty.$$

Therefore, $f(y_k) \to -\infty$.
This implies

$$(y_k, f(y_k)) \to (x_0, -\infty),$$

which contradicts the closedness of the epigraph

$$\mathrm{epi}_{\mathcal{K}}(f) := \{(x, t) \in \mathcal{K} \times \mathbb{R} : f(x) \leq t\}.$$

Thus, our assumption must be false. Therefore,

$$f^* > -\infty.$$

Step 4: Use closedness of the epigraph to conclude optimality.
Since $f(x_k) \to f^*$ and $x_k \to x^*$, it follows that:

$$(x_k, f(x_k)) \to (x^*, f^*).$$

By closedness of the epigraph $\mathrm{epi}_{\mathcal{K}}(f)$, we have:

$$(x^*, f^*) \in \mathrm{epi}_{\mathcal{K}}(f) \Rightarrow f(x^*) \leq f^*.$$

But by definition, $f^* = \inf_{x \in \mathcal{K}} f(x) \leq f(x^*)$. Hence:

$$f(x^*) = f^*.$$

Thus, the minimum is attained at $x^* \in \mathcal{K}$, and the problem is solvable.

5. We prove the claim by contradiction.
Step 1: Assume level set is unbounded.
Suppose that there exists $\beta > f^*$ such that the level set $\mathscr{L}_f(\beta)$ is unbounded. Then there exists a sequence $\{x_k\} \subset \mathscr{L}_f(\beta)$ such that

$$\|x_k - x^*\| \to \infty,$$

for some fixed $x^* \in X^*$. Define $\rho_k := \|x_k - x^*\| \to \infty$.

Step 2: Construct bounded sequence.
Since X^* is bounded, choose $R > \max_{y \in X^*} \|y - x^*\|$. Define the sequence:

$$y_k := x^* + \frac{R}{\rho_k}(x_k - x^*).$$

Then

- $y_k \in \mathcal{K}$, because $\mathcal{K}$ is convex and y_k is a convex combination of x^* and x_k,
- $\|y_k - x^*\| = R$, so $\{y_k\}$ lies on the boundary of a fixed ball.

Step 3: Use convexity to estimate $f(y_k)$.
By convexity of f, we have:

$$f(y_k) \overset{(1)}{\leq} (1 - \tfrac{R}{\rho_k})f(x^*) + \tfrac{R}{\rho_k} f(x_k).$$

Since $f(x^*) = f^*$ and $f(x_k) \leq \beta$, we obtain:

$$f(y_k) \leq f^* + \tfrac{R}{\rho_k}(\beta - f^*) \to f^* \quad \text{as } k \to \infty.$$

Step 4: Derive contradiction via compactness.
Since $\{y_k\} \subset \mathcal{K}$ is bounded (with fixed norm R), the sequence has a convergent subsequence. Assume $y_k \to \bar{y} \in \mathcal{K}$.

Then

$$f(y_k) \to f^* \quad \Rightarrow \quad f(\bar{y}) = f^* \quad \Rightarrow \quad \bar{y} \in X^*.$$

However,

$$\|\bar{y} - x^*\| = \lim_{k\to\infty} \|y_k - x^*\| = R > \sup_{y \in X^*} \|y - x^*\|,$$

which contradicts the fact that X^* is bounded and all points in it lie within distance less than R from x^*.

□

Closedness thus emerges as a minimal regularity condition that ensures well-posedness, stability under limits, and feasibility of iterative optimization schemes. It provides the analytical structure required to generalize gradient-based methods to the nondifferentiable setting.
To better understand why the notion of closedness is essential, consider the following motivating example.

Example: The example comes from Example 3.1.1 (6) in [2]. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined on the closed unit disk as follows

$$f(x, y) = \begin{cases} 0, & \text{if } x^2 + y^2 < 1, \\ \phi(x, y), & \text{if } x^2 + y^2 = 1, \end{cases}$$

where $\phi(x, y)$ is an *arbitrary nonnegative function real-valued* defined only on the boundary of the unit circle, i.e., for all (x, y) such that $x^2 + y^2 = 1$.

The domain of this function is the *closed unit disk*, which is both closed and convex. It is easy to verify that the function f is convex, the epigraph of f is a convex set. However, f can exhibit completely arbitrary (and potentially pathological) behavior on the boundary due to the arbitrariness of ϕ.

In particular, unless $\phi(x, y) \equiv 0$, the function is *not closed*, because the epigraph may fail to be closed at boundary points. This example motivates the introduction of the concept of a *closed convex function* in convex analysis.

We next state two structural results that preserve convexity and closedness under common operations.

Theorem 3.3: Convexity Preservation under Infimum

Let $\mathcal{K}_1 \times \mathcal{K}_2 := \mathcal{K} \subset \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ with $n_1, n_2 \in \mathbb{N}$ be a convex set, and let the function $\phi : \mathcal{K} \to \mathbb{R} \cup \{\infty\}$ be convex with $\mathcal{K} \subset \text{dom}(\phi)$. Then the function $f : \mathcal{K}_1 \to \mathbb{R}$ defined by

$$f(x) = \inf_{y \in \mathcal{K}_2} \{\phi(x, y)\} \tag{41}$$

is convex on $\hat{\mathcal{K}} = \{x \in \mathcal{K}_1 : \exists\, y \in \mathcal{K}_2 \text{ such that } (x, y) \in \mathcal{K}_1 \times \mathcal{K}_2\}$

Proof: The proof comes from Theorem 3.1.7 in [2]. Let $x_1, x_2 \in \hat{\mathcal{K}}$. By definition of $\hat{\mathcal{K}}$, there exist y_1, y_2 such that $(x_1, y_1), (x_2, y_2) \in \mathcal{K}$. Since $f(x) = \inf\{\phi(x, y) : (x, y) \in \mathcal{K}\}$, the infimum may not be attained.
So, for each $i \in \{1, 2\}$, there exists a sequence $\{y_{i,k}\}_{k\in\mathbb{N}}$ such that $(x_i, y_{i,k}) \in \mathcal{K}$ and

$$\lim_{k\to\infty} \phi(x_1, y_{1,k}) = f(x_1), \quad \lim_{k\to\infty} \phi(x_2, y_{2,k}) = f(x_2).$$

Now let $\alpha \in [0, 1]$. Because $\mathcal{K}$ is convex, and $(x_1, y_{1,k}), (x_2, y_{2,k}) \in \mathcal{K}$ for all k, the convex combination

$$(\alpha x_1 + (1-\alpha)x_2,\ \alpha y_{1,k} + (1-\alpha)y_{2,k}) \in \mathcal{K}.$$

Since ϕ is convex in (x, y) on $\mathcal{K}$, we have:

$$\begin{aligned} f(\alpha x_1 + (1-\alpha)x_2) &\overset{(41)}{\leq} \phi\left(\alpha x_1 + (1-\alpha)x_2,\ \alpha y_{1,k} + (1-\alpha)y_{2,k}\right) \\ &\overset{(1)}{\leq} \alpha\phi(x_1, y_{1,k}) + (1-\alpha)\phi(x_2, y_{2,k}). \end{aligned}$$

Taking limits on the right-hand side as $k \to \infty$, we get:

$$f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha)f(x_2).$$

This proves that f is convex on $\hat{\mathcal{K}}$.

□

Theorem 3.4: Supremum preservation of convexity and closedness

Let $M \subseteq \mathbb{R}^{n_1}$ with $n_1 \in \mathbb{N}$ be an arbitrary set, $\phi : \mathbb{R}^{n_2} \times M \to \mathbb{R} \cup \{\infty\}$, $n_2 \in \mathbb{N}$ and

$$f(x) = \sup_y \{\phi(x, y) | y \in M\}. \tag{42}$$

Suppose that for any $y \in M$ the functions $\phi(\cdot, y)$ are closed and convex on some closed set $\mathcal{K} \subseteq \mathbb{R}^{n_2}$. Then $f(\cdot)$ is a closed convex function on the set

$$\hat{\mathcal{K}} = \{x \in \mathcal{K} | \sup_{y\in M} \phi(x, y) < \infty\}.$$

Proof: The proof comes from Theorem 3.1.8 in [2]. It should be noted that we present the proofs of the statements in much more detail. Let $f(x) = \sup_{y\in M} \phi(x, y)$, and assume that for each $y \in M$, the function $\phi(\cdot, y)$ is closed and convex on some set $\mathcal{K}$. Define the domain:

$$\hat{\mathcal{K}} = \left\{x \in \mathcal{K} \mid \sup_{y\in M} \phi(x, y) < +\infty\right\}.$$

Let $Q \subseteq \hat{\mathcal{K}}$ be arbitrary. Consider the epigraph of f over the set Q:

$$\text{epi}_Q(f) = \{(x, t) \in Q \times \mathbb{R} \mid t \geq f(x)\}.$$

Since $f(x) = \sup_{y\in M} \phi(x, y)$, we have:

$$(x, t) \in \text{epi}_Q(f) \quad \Leftrightarrow \quad x \in Q \text{ and } t \geq \phi(x, y) \text{ for all } y \in M.$$

Thus, the epigraph of f can be written as the intersection of the epigraphs of $\phi(\cdot, y)$:

$$\text{epi}_Q(f) = \bigcap_{y\in M} \text{epi}_Q(\phi(\cdot, y)).$$

Each set $\text{epi}_Q(\phi(\cdot, y))$ is closed and convex since $\phi(\cdot, y)$ is closed and convex for every $y \in M$. Because the intersection of any collection of closed convex sets is also closed and convex, it follows that

$$\text{epi}_Q(f) \text{ is closed and convex.}$$

Therefore, f is a closed convex function on Q. Since $Q \subseteq \hat{\mathcal{K}}$ was arbitrary, this implies that f is closed and convex on the entire set $\hat{\mathcal{K}}$.

□

We conclude this subsection with a motivating example that highlights an important subtlety: although closed convex functions behave well under limits, they may still be discontinuous.

Example: The example comes from Example 3.1.2 (5) in [2]. Let Q be a set in $\mathbb{R}^n$. Consider the function

$$\psi(g, \gamma) = \sup_{y \in Q} \phi(y, g, \gamma),$$

where

$$\phi(y, g, \gamma) = \langle g, y \rangle - \frac{\gamma}{2}\|y\|^2.$$

The function $\psi(g, \gamma)$ is closed and convex in (g, γ) by (42). Let us study its properties.
Case 1: If Q is bounded, then $\text{dom}(\psi) = \mathbb{R}^n \times \mathbb{R} = \mathbb{R}^{n+1}$.
Case 2: Consider the unbounded case $Q = \mathbb{R}^n$.

- If $\gamma < 0$, then for any $g \neq 0$, we can set $y_\alpha = \alpha g$. Then,

$$\phi(y_\alpha, g, \gamma) = \alpha\|g\|^2 - \frac{\gamma}{2}\alpha^2\|g\|^2 \to \infty \quad \text{as } \alpha \to \infty.$$

  So, $\psi(g, \gamma) = +\infty$ and such $(g, \gamma) \notin \text{dom}(\psi)$. Therefore, $\text{dom}(\psi) \subseteq \{(g, \gamma) : \gamma \geq 0\}$.

- If $\gamma = 0$, then $\phi(y, g, 0) = \langle g, y \rangle$. This is unbounded unless $g = 0$. So the only point with finite value is $(g, \gamma) = (0, 0)$, where we define $\psi(0, 0) = 0$.

- If $\gamma > 0$, then $\phi(y, g, \gamma)$ is strongly concave in y. The supremum is attained at

$$y^*(g, \gamma) = \frac{1}{\gamma} g,$$

  giving

$$\psi(g, \gamma) = \phi\left(\frac{1}{\gamma} g, g, \gamma\right) = \frac{\|g\|^2}{2\gamma}.$$

Thus, we can write

$$\psi(g, \gamma) = \begin{cases} 0, & \text{if } g = 0,\ \gamma = 0, \\ \frac{\|g\|^2}{2\gamma}, & \text{if } \gamma > 0. \end{cases}$$

The domain is

$$\text{dom}(\psi) = (\mathbb{R}^n \times \{\gamma > 0\}) \cup \{(0, 0)\}.$$

This set is convex but neither open nor closed. Nevertheless, ψ is a closed convex function. Note, however, that ψ is discontinuous at the origin. For example, along the path $g_\epsilon = \sqrt{\epsilon} g$, $\gamma = \epsilon$, we find

$$\psi(\sqrt{\gamma} g, \gamma) = \frac{1}{2}\|g\|^2, \quad \gamma \neq 0.$$

This shows discontinuity at $(0, 0)$. Now consider the closed convex set:

$$Q = \{(g, \gamma) \in \mathbb{R}^n \times \mathbb{R} : \gamma \geq \|g\|^2\}.$$

On this set, ψ is a closed convex function with bounded values. However, it remains discontinuous at the origin.

This example highlights the distinction between closedness and continuity—demonstrating that even closed convex functions may be discontinuous, particularly at boundary points. This observation motivates a closer study of continuity and directional derivatives in the next subsection.

3.2 Continuity and Differentiability of convex functions

In the previous subsection, we saw through examples that convex functions can exhibit irregular behavior, especially near the boundaries of their domains. However, within the interior of their domain, convex functions tend to behave much more regularly. In this subsection, we study these local regularity properties, focusing on continuity and differentiability.

Since convex functions are not necessarily differentiable everywhere, we introduce weaker but still meaningful notions of regularity, such as Lipschitz continuity, directional derivatives, and positive homogeneity. These concepts are foundational in convex analysis and are especially important in optimization contexts involving nonsmooth functions.

We begin by introducing a condition that bounds how fast a function can change between two points.

Definition 3.3: Lipschitz continous

A function f is Lipschitz continuous with a Lipschitz constant $L > 0$ if the following applies for all $x, y \in \text{dom}(f)$

$$|f(x) - f(y)| \leq L||x - y||.$$

This condition guarantees that f does not oscillate too wildly. However, for functions defined on unbounded or non-convex domains, global Lipschitz continuity can be too restrictive. A weaker, more flexible version is

Definition 3.4: Locally lipschitz continous

A function $f : \mathbb{R}^n \to \mathbb{R}$ is locally Lipschitz-continuous if for each point $x \in \mathbb{R}^n$ an open ball $U(x)$ exists so that f is Lipschitz-continuous on $U(x)$. This means that there is a constant L_x for each x so that the following applies for all $y, z \in U(x)$

$$|f(y) - f(z)| \leq L_x||y - z||.$$

This property is especially useful for convex functions, which may fail to be globally Lipschitz but are almost always locally Lipschitz in the interior of their domain.

We now present a foundational result that ensures convex functions behave well at interior points of their domain.

Theorem 3.5

Let f be convex and $x_0 \in \text{int}(\text{dom}(f))$. Then f is locally bounded and locally Lipschitz continuous at x_0.

Proof: We refer to Theorem (Satz) 7.3.5 in [6].

This theorem is crucial as it ensures that convex functions, despite possible irregularities at boundary points, behave predictably in the interior of their domain.

To study generalized notions of differentiability, we introduce several geometric and functional concepts.

Definition 3.5: Cone

A set $\mathcal{K} \subset \mathbb{R}^n$ is called a cone if and only if $\forall \lambda \geq 0$ holds

$$x \in \mathcal{K} \Rightarrow \lambda x \in \mathcal{K}.$$

A cone $\mathcal{K} \subset \mathbb{R}^n$ is convex if additionally satisfies

$$x, y \in \mathcal{K} \Rightarrow x + y \in \mathcal{K}.$$

Definition 3.6: (Positively) homogeneous function of degree p

A function f is called (positively) homogeneous of degree $p \geq 0$ if $\mathrm{dom}(f)$ is a cone and

$$f(\tau x) = \tau^p f(x) \quad \forall\, x \in \mathrm{dom}(f),\ \forall \tau \geq 0$$

These properties naturally appear when studying directional derivatives—a notion of slope valid even when classical gradients do not exist.

Definition 3.7

Let $x \in \mathrm{dom}(f)$. We call f directionally differentiable at the point x in direction $p \neq 0$ if the following limit exists

$$f'(x; p) = \lim_{\alpha \downarrow 0} \frac{1}{\alpha} \big(f(x + \alpha p) - f(x)\big) \tag{43}$$

The value $f'(x; p)$ is called the *directional derivative* of f at x.

The directional derivative quantifies the infinitesimal rate of change of a function in the direction p. Unlike classical derivatives, it does not require differentiability in all directions simultaneously.

The following theorem confirms that this generalized slope exists for all convex functions at interior points.

Theorem 3.6

A convex function f is directionally differentiable in the sense of (43) in any direction at any interior point of its domain.

Proof: The proof comes from Theorem 3.1.12 in [2]. It should be noted that we present the proofs of the statements in much much detail. Let $x \in \mathrm{int}(\mathrm{dom}(f))$, and fix a direction $p \in \mathbb{R}^n$. Since x is an interior point, there exists $\varepsilon > 0$ such that $x + \alpha p \in \mathrm{dom}\, f$ for all $\alpha \in [0, \varepsilon]$.
Define the function

$$\phi(\alpha) := \frac{f(x + \alpha p) - f(x)}{\alpha}, \quad \alpha \in (0, \varepsilon].$$

Our goal is to show that the limit $\lim_{\alpha \downarrow 0} \phi(\alpha)$ exists.

Step 1: Monotonicity of ϕ
Let $0 < \alpha \leq \varepsilon$ and let $\beta \in (0, 1]$. Then $\alpha\beta \in (0, \varepsilon]$, and we can write

$$x + \alpha\beta p = (1 - \beta)x + \beta(x + \alpha p),$$

which is a convex combination of x and $x + \alpha p$.
Since f is convex, we have

$$f(x + \alpha\beta p) \overset{(1)}{\leq} (1 - \beta) f(x) + \beta f(x + \alpha p).$$

Subtract $f(x)$ and divide both sides by $\alpha\beta$

$$\frac{f(x+\alpha\beta p)-f(x)}{\alpha\beta} \leq \frac{\beta f(x+\alpha p)-\beta f(x)}{\alpha\beta} = \frac{f(x+\alpha p)-f(x)}{\alpha}.$$

This shows

$$\phi(\alpha\beta) \leq \phi(\alpha),$$

so ϕ is decreasing on $(0, \varepsilon]$.
Step 2: Lower Bound of ϕ
Let us now show that $\phi(\alpha)$ is bounded below. Since $x \in \text{int}(\text{dom}(f))$, there exists $\gamma > 0$ such that $x - \gamma p \in \text{dom}(f)$.
It holds

$$x + \alpha p = x + \frac{\alpha}{\gamma}(x - (x - \gamma p))$$

Therefore

$$\begin{aligned}\phi(\alpha) = \frac{f(x+\alpha p)-f(x)}{\alpha} &= \frac{f(x+\frac{\alpha}{\gamma}(x-(x-\gamma p)))-f(x)}{\alpha} \\ &\overset{(2)}{\geq} \frac{\frac{\alpha}{\gamma}(f(x)-f(x-\gamma p))}{\alpha} = \frac{(f(x)-f(x-\gamma p))}{\gamma}\end{aligned}$$

This lower bound is independent of α, so $\phi(\alpha)$ is bounded below.

Since $\phi(\alpha)$ is decreasing and bounded below on $(0, \varepsilon]$, the limit

$$\lim_{\alpha \downarrow 0} \phi(\alpha)$$

exists (possibly infinite). Hence, the directional derivative $f'(x; p)$ exists. This concludes the proof that every convex function is directionally differentiable at interior points of its domain.

□

This is a powerful result, every convex function is directionally differentiable in the interior of its domain, even if it is not differentiable in the classical sense.

Directional derivatives of convex functions are remarkably well-structured.

Lemma 3.1

Let the function f be convex and $x \in \text{int}(\text{dom}(f))$. Then $f'(x; \cdot)$ is a convex function which is positively homogeneous of degree one. It holds $\forall y \in \text{dom}(f)$

$$f(y) \geq f(x) + f'(x; y - x). \tag{44}$$

Proof: The proof comes from Lemma 3.1.5 in [2]. It should be noted that we present the proofs of the statements in much more detail.
Positive Homogeneity
Let $\tau > 0$. By definition of the directional derivative:

$$f'(x; \tau p) = \lim_{\alpha \downarrow 0} \frac{f(x+\alpha\tau p)-f(x)}{\alpha}.$$

Substitute $\beta = \tau\alpha$, so that $\alpha = \beta/\tau$:

$$f'(x; \tau p) = \lim_{\beta \downarrow 0} \frac{f(x+\beta p)-f(x)}{\beta/\tau} = \tau \lim_{\beta \downarrow 0} \frac{f(x+\beta p)-f(x)}{\beta} = \tau f'(x; p).$$

Thus, $f'(x; \tau p) = \tau f'(x; p)$. This proves that $f'(x; \cdot)$ is positively homogeneous of degree one.

Convexity of the Directional Derivative
Let $p_1, p_2 \in \mathbb{R}^n$ and $\beta \in [0, 1]$. Consider:

$$f'(x; \beta p_1 + (1-\beta)p_2) = \lim_{\alpha \downarrow 0} \frac{f(x+\alpha(\beta p_1 + (1-\beta)p_2))-f(x)}{\alpha}.$$

Let $z_1 = x + \alpha p_1$ and $z_2 = x + \alpha p_2$, so that

$$x + \alpha(\beta p_1 + (1-\beta)p_2) = \beta z_1 + (1-\beta) z_2.$$

By convexity of f,

$$f(\beta z_1 + (1-\beta) z_2) \leq \beta f(z_1) + (1-\beta) f(z_2).$$

So,

$$\begin{aligned}
\frac{f(x + \alpha(\beta p_1 + (1-\beta)p_2)) - f(x)}{\alpha} &\leq \frac{\beta f(x+\alpha p_1) + (1-\beta) f(x+\alpha p_2) - f(x)}{\alpha} \\
&= \frac{\beta f(x+\alpha p_1) + (1-\beta) f(x+\alpha p_2) - \beta f(x) - (1-\beta) f(x)}{\alpha} \\
&= \beta \frac{f(x+\alpha p_1) - f(x)}{\alpha} + (1-\beta) \frac{f(x+\alpha p_2) - f(x)}{\alpha}.
\end{aligned}$$

Taking the limit $\alpha \downarrow 0$, we obtain

$$f'(x; \beta p_1 + (1-\beta)p_2) \leq \beta f'(x; p_1) + (1-\beta) f'(x; p_2),$$

so $f'(x; \cdot)$ is convex.

First-Order Condition for Convexity
Let $y \in \mathrm{dom}(f)$, and define $y_\alpha := x + \alpha(y - x)$ for $\alpha \in (0, 1]$. Then $y_\alpha \to x$ as $\alpha \downarrow 0$, and by convexity

$$f(y_\alpha) \leq (1-\alpha) f(x) + \alpha f(y).$$

Rewriting

$$\frac{f(y_\alpha) - f(x)}{\alpha} \leq f(y) - f(x).$$

Note that $y_\alpha = x + \alpha(y - x)$ implies

$$\frac{f(x + \alpha(y-x)) - f(x)}{\alpha} \leq f(y) - f(x).$$

Taking the limit as $\alpha \downarrow 0$

$$f'(x; y - x) \leq f(y) - f(x) \quad \Rightarrow \quad f(y) \geq f(x) + f'(x; y - x).$$

□

This lemma reveals the fundamental structure of directional derivatives for convex functions. In particular, the inequality (44) serves as a first-order condition for convexity, encapsulating the key idea behind subgradients. This concept will form the foundation of one of the next subsections, where we formally introduce subdifferentials as a generalization of gradients in nonsmooth convex analysis.

3.3 Separation Theorems

Separation theorems are a foundational tool in convex analysis, with key applications in optimization. In particular, they enable the definition of subgradients for convex, possibly non-differentiable functions, and serve as a basis for optimality conditions and subgradient methods. The central idea is that convex sets, and by extension, the epigraphs of convex functions, can be separated or supported by hyperplanes. This geometric perspective leads to important analytical tools such as subgradients and descent directions.

Definition 3.8

Let $\mathcal{K}$ be a convex set. We say that the hyperplane

$$\mathscr{H}(g,\gamma) = \{x \in \mathbb{R}^n \mid \langle g, x\rangle = \gamma\}, \quad g \neq 0,$$

is *supporting* to $\mathcal{K}$ if any $x \in \mathcal{K}$ satisfies inequality $\langle g, x\rangle \leq \gamma$. The hyperplane $\mathscr{H}(g,\gamma) \not\supseteq \mathcal{K}$ *separates* a point x_0 from $\mathcal{K}$ if

$$\langle g, x\rangle \leq \gamma \leq \langle g, x_0\rangle \tag{45}$$

for all $x \in \mathcal{K}$. If one of the inequalities in (45) is strict, then we call the separation *strong*.

This geometric notion plays a key role in the characterization of subgradients. For a convex function, the epigraph is a convex set, and supporting hyperplanes to this epigraph at a point yield valid subgradients. This perspective also justifies the use of subgradients in descent methods and optimality conditions.
The following theorem is proved in [6].

Theorem 3.7

Let $\mathcal{K}$ be a closed convex set and $x \notin \mathcal{K}$. Then x is strongly separable from $\mathcal{K}$.

Proof: We refer to Theorem (Satz) 7.2.2 in [6].

We come to an important theorem which later helps to ensure the existence of subgradients.

Theorem 3.8: Supporting Hyperplane Theorem

Let $\mathcal{K}$ be a closed convex set. If the point x_0 belongs to the boundary of $\mathcal{K}$, then there exists a supporting to $\mathcal{K}$ hyperplane $\mathscr{H}(g,\gamma)$ which contains x_0.
Such a vector g is called *supporting to* $\mathcal{K}$ at the point x_0.

Proof: The proof comes from Theorem 3.1.14 in [2]. It should be noted that we present the proofs of the statements in much more detail. Since $x_0 \in \partial\mathcal{K}$, for every $\varepsilon > 0$ there exists a point $y_\varepsilon \notin \mathcal{K}$ with $\|y_\varepsilon - x_0\| < \varepsilon$. Thus, we can construct a sequence $\{y_k\} \subset \mathbb{R}^n \setminus \mathcal{K}$ such that $y_k \to x_0$ as $k \to \infty$.
For each $y_k \notin \mathcal{K}$, the projection $\Pi_{\mathcal{K}}(y_k) \in \mathcal{K}$ is uniquely defined (Lemma 1.1 (1)), since $\mathcal{K}$ is closed and convex. Define the normalized vector

$$g_k := \frac{y_k - \Pi_{\mathcal{K}}(y_k)}{\|y_k - \Pi_{\mathcal{K}}(y_k)\|},$$

which points from $\Pi_{\mathcal{K}}(y_k)$ to y_k.
By the optimality condition for projections onto convex sets (Lemma 1.1 (2), set $y = y_k$ and $z = \Pi_{\mathcal{K}}(y_k)$), we have

$$\langle g_k, x - \Pi_{\mathcal{K}}(y_k)\rangle \leq 0 \quad \text{for all } x \in \mathcal{K},$$

which implies

$$\langle g_k, x\rangle \leq \langle g_k, \Pi_{\mathcal{K}}(y_k)\rangle \quad \text{for all } x \in \mathcal{K}. \tag{46}$$

Now, since $y_k \to x_0$ and projection is non-expansive (Lemma 1.1 (3)), we also have $\Pi_{\mathcal{K}}(y_k) \to x_0$. The vectors $g_k \in \mathbb{S}^{n-1}$ (the unit sphere), which is compact, so there exists a convergent subsequence (Bolzano–Weierstrass Theorem) $g_{k_j} \to g^*$ with $\|g^*\| = 1$.
Let $\gamma := \langle g^*, x_0\rangle$. Taking the limit in inequality (46) with g_{k_j} as $j \to \infty$, we get for all $x \in \mathcal{K}$,

$$\langle g_{k_j}, x\rangle \leq \langle g_{k_j}, \Pi_{\mathcal{K}}(y_{k_j})\rangle \to \langle g^*, x_0\rangle = \gamma,$$

and $\langle g_{k_j}, x\rangle \to \langle g^*, x\rangle$, so

$$\langle g^*, x\rangle \leq \gamma \quad \text{for all } x \in \mathcal{K}.$$

Moreover, since $\Pi_{\mathcal{K}}(y_{k_j}) \to x_0$, we also have

$$\langle g^*, x_0 \rangle = \gamma.$$

Hence, the hyperplane $\mathscr{H}(g^*, \gamma)$ supports $\mathcal{K}$ at x_0.

□

3.4 Subgradients

Subgradients provide a powerful generalization of gradients to nonsmooth convex functions. While classical differentiability fails for many convex functions—such as piecewise linear or ℓ_1-regularized functions—subgradients offer an effective way to retain directional and optimality information.

Subgradients play a key role in convex analysis and optimization. They allow for the formulation of first-order optimality conditions, duality theory, and gradient-like iterative methods in the absence of smoothness. The core idea is to replace the tangent hyperplane with a *supporting hyperplane*, which touches the epigraph of a convex function from below.

Definition 3.9: Subgradient

A vector $g \in \mathbb{R}^n$ is called a *subgradient* of the function $f : \mathbb{R}^n \to \mathbb{R}$ at the point $x_0 \in \mathrm{dom}(f)$ if for any $y \in \mathrm{dom}(f)$ we have

$$f(y) \geq f(x_0) + \langle g, y - x_0 \rangle. \tag{47}$$

The set of all subgradients of f at x_0 is called the *subdifferential* $\partial f(x_0)$ of the function f at the point x_0. If inequality (47) only applies to $y \in \mathcal{K} \subset \mathrm{dom}(f)$ with $\mathcal{K}$ convex, then we use the constrained subdifferential $g \in \partial_{\mathcal{K}} f(x_0)$.

Remarks:

- The inequality (47) characterizes a family of half-spaces parameterized by $y \in \mathcal{K}$. Therefore, $\partial_{\mathcal{K}} f(x_0)$ is always a closed and convex set.
- For $x_1, x_2 \in \mathrm{dom}(f)$ with subgradients $g_1 \in \partial f(x_1)$ and $g_2 \in \partial f(x_2)$, the subgradient inequalities imply

  $$0 \leq \langle g_1 - g_2, x_2 - x_1 \rangle,$$

  which shows that the subdifferential is a *monotone set-valued mapping*.

The subgradient inequality (47) means that the affine function $y \mapsto f(x_0) + \langle g, y - x_0 \rangle$ lies below the graph of f and touches it at x_0. This is a geometric manifestation of a *supporting hyperplane* to the epigraph of f, allowing us to use linear approximations even for nondifferentiable functions.

We now present a key lemma that connects the existence of subgradients to the topological regularity of a function.

Lemma 3.2

Let $\mathcal{K}$ be a convex set. Assume that, for any $x \in \mathcal{K} \subset \mathrm{dom}(f)$, the constrained subdifferential $\partial_{\mathcal{K}} f(x)$ is nonempty. Then f is a closed convex function on $\mathcal{K}$.

Proof: The proof comes from Lemma 3.1.6 in [2]. It should be noted that we present the proofs of the statements in much more detail. For each $x \in \mathcal{K}$, define

$$\hat{f}(x) := \sup_{y \in \mathcal{K}} \{ f(y) + \langle g(y), x - y \rangle \},$$

where $g(y) \in \partial_{\mathcal{K}} f(y)$.
By Theorem 3.4 (42), the function $\hat{f}$ is convex and closed, since it is defined as the pointwise

supremum of affine functions over a convex set. We show that $f(x) \geq \hat{f}(x)$.
From the definition of the constrained subdifferential, for every $y \in \mathcal{K}$ and $g(y) \in \partial_{\mathcal{K}} f(y)$, we have

$$f(x) \geq f(y) + \langle g(y), x - y \rangle.$$

Taking the supremum over all $y \in \mathcal{K}$ yields:

$$f(x) \geq \sup_{y \in \mathcal{K}} \{f(y) + \langle g(y), x - y \rangle\} = \hat{f}(x). \tag{48}$$

Now we show that $f(x) \leq \hat{f}(x)$.
To establish the reverse inequality, fix $x \in \mathcal{K}$ and choose $y = x$. Then

$$\hat{f}(x) \geq f(x) + \langle g(x), x - x \rangle = f(x). \tag{49}$$

From (48) and (49), we conclude that $f(x) = \hat{f}(x)$ for all $x \in \mathcal{K}$. Since $\hat{f}$ is convex and closed, it follows that f is also convex and closed on $\mathcal{K}$.

□

We now prove a foundational result guaranteeing that subgradients exist in the interior of the domain.

Theorem 3.9

Let the function f be convex. If $x_0 \in \text{int}(\text{dom}(f))$, then $\partial f(x_0)$ is a nonempty bounded set.

Proof: The proof comes from Lemma 3.1.15 in [2]. It should be noted that we present the proofs of the statements in much more detail. We divide the proof into two parts, showing the nonemptiness and boundedness of $\partial f(x_0)$.

Part 1: Nonemptiness of $\partial f(x_0)$
Let us consider the epigraph of f,

$$\text{epi}(f) = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} \mid f(x) \leq t\},$$

which is a convex subset of $\mathbb{R}^{n+1}$ due to the convexity of f.
Since $x_0 \in \text{int}(\text{dom}(f))$, the point $(x_0, f(x_0)) \in \partial(\text{epi}(f))$. By the Supporting Hyperplane Theorem, there exists a nonzero vector $(d, \alpha) \in \mathbb{R}^n \times \mathbb{R}$, not both zero, such that for all $(x, t) \in \text{epi}(f)$,

$$\langle d, x - x_0 \rangle + \alpha(t - f(x_0)) \leq 0.$$

We now analyze the scalar α. Suppose $\alpha = 0$. Then,

$$\langle d, x - x_0 \rangle \leq 0, \quad \forall x \in \text{dom}(f).$$

Since x_0 is in the interior of the domain, there exists a ball around x_0 entirely within $\text{dom}(f)$, implying that $\langle d, x - x_0 \rangle \leq 0$ for all directions. This forces $d = 0$, contradicting that $(d, \alpha) \neq (0, 0)$. Hence, $\alpha \neq 0$. Suppose $\alpha > 0$, we can choose $x = x_0$, then

$$\langle d, 0 \rangle + \alpha(t - f(x_0) = \alpha(t - f(x_0) \leq 0.$$

Since t can be choosen that $f(x_0) \leq t$, we conclude $\alpha < 0$. Now, using $f(x) \leq t$ for $(x, t) \in \text{epi}(f)$, we substitute $t = f(x)$ in the inequality

$$\langle d, x - x_0 \rangle + \alpha(f(x) - f(x_0)) \leq 0.$$

Since $\alpha < 0$, we can divide through by $-\alpha > 0$ and rearrange:

$$f(x) \geq f(x_0) + \left\langle -\frac{d}{\alpha}, x - x_0 \right\rangle.$$

Define $g := -\frac{d}{\alpha}$. Then, this inequality is exactly the condition that $g \in \partial f(x_0)$. Thus, $\partial f(x_0) \neq \emptyset$.

Part 2: Boundedness of $\partial f(x_0)$

Since f is convex and $x_0 \in \text{int}(\text{dom}(f))$, by Theorem 3.5, f is locally Lipschitz continuous near x_0. Therefore, there exists $\varepsilon > 0$ and $L > 0$ such that for all $x \in B(x_0, \varepsilon)$,

$$|f(x) - f(x_0)| \leq L\|x - x_0\|.$$

Let $g \in \partial f(x_0)$ and consider the point

$$x := x_0 + \varepsilon \frac{g}{\|g\|} \in B(x_0, \varepsilon) \subseteq \text{dom}(f).$$

By the subgradient inequality,

$$f(x) \overset{(47)}{\geq} f(x_0) + \langle g, x - x_0 \rangle = f(x_0) + \varepsilon\|g\|.$$

By the Lipschitz condition,

$$f(x) \leq f(x_0) + L\|x - x_0\| = f(x_0) + L\varepsilon.$$

Combining the two,

$$f(x_0) + \varepsilon\|g\| \leq f(x) \leq f(x_0) + L\varepsilon.$$

Subtracting $f(x_0)$ and dividing by $\varepsilon > 0$, we get:

$$\|g\| \leq L.$$

Hence, every subgradient $g \in \partial f(x_0)$ has norm bounded by L, and so $\partial f(x_0)$ is a bounded set. □

Theorem 3.9 provides a crucial guarantee for nonsmooth optimization, subgradients always exist and remain bounded at interior points. This enables the use of subgradients in formulating optimality conditions, constructing descent methods, and developing duality theory. In essence, it ensures that even without smoothness, we retain a reliable first-order approximation within the domain.

We would now like to introduce the Fenchel dual of a function f, which is very interesting when f is convex.

Theorem 3.10: Fenchel Dual

For the function f, define its Fenchel dual

$$f_*(s) = \sup_{y \in \text{dom}(f)} \Big(\langle s, y \rangle - f(y) \Big), \tag{50}$$

and the dual of the Fenchel dual

$$f_{**}(x) = \sup_{s \in \text{dom}(f_*)} \Big(\langle s, x \rangle - f_*(s) \Big).$$

Then $f(x) \geq f_{**}(x)$ for all $x \in \text{dom}(f)$. Moreover, if $\partial f(x) \neq \emptyset$ for some $x \in \text{dom}(f)$, then $\partial f(x) \subseteq \text{dom}(f_*)$ and $f(x) = f_{**}(x)$.

Proof: The proof comes from Theorem 3.1.16 in [2]. Indeed, for any $x \in \text{dom}(f)$ we have

$$\begin{aligned} f_{**}(x) &= \sup_{s \in \text{dom}(f_*)} [\langle s, x \rangle - f_*(s)] \\ &\overset{(50)}{=} \sup_{s \in \text{dom}(f_*)} \inf_{y \in \text{dom}(f)} [\langle s, x \rangle - \langle s, y \rangle + f(y)] \\ &\leq \inf_{y \in \text{dom}(f)} \sup_{s \in \text{dom}(f_*)} [\langle s, x - y \rangle + f(y)] \\ &\overset{y=x}{\leq} f(x). \end{aligned}$$

Let us choose now an arbitrary $g \in \partial f(x)$. Then for any $y \in \text{dom}(f)$ we have

$$\langle g, y \rangle - f(y) \overset{(47)}{\leq} \langle g, y \rangle - f(x) - \langle g, y - x \rangle = \langle g, x \rangle - f(x).$$

Thus, $g \in \text{dom}(f_*)$. Therefore,

$$\begin{aligned} f_{**}(x) &= \sup_{s \in \text{dom}(f_*)} \inf_{y \in \text{dom}(f)} [\langle s, x\rangle - \langle s, y\rangle + f(y)] \\ &\geq \inf_{y \in \text{dom}(f)} [\langle g, x\rangle - \langle g, y\rangle + f(y)] \\ &\overset{(47)}{=} f(x). \end{aligned}$$

□

Remark: One useful application of the duals and these theorems is the calculation of the inverse of the derivative or its subgradient. More precisely, in the differentiable case, it can be shown that

$$(\nabla f)^{-1} = \nabla f_*,$$

and in the non-differentiable case

$$(\partial f)^{-1} = \partial f_*,$$

which are called *Legendre Transformation*. If ∇f_* or ∂f_* can be calculated then we can use it to get the inverse of the derivative of the generator function ψ to be calculated in the OMD algorithm Algorithm 5 from Chapter 2 .

The following lemma will be helpful later when we want the subdifferential of a supremum function.

Lemma 3.3

Let E be an arbitrary set, and

$$f(x) = \sup\{\phi(x, y) \mid y \in E\}.$$

Suppose that for any $y \in E$ the function $\phi(\cdot, y)$ is closed and convex on some convex set Q. Then f is closed convex on the set

$$\widehat{Q} = \left\{ x \in Q \,\middle|\, \sup_{y \in E} \phi(x, y) < +\infty \right\}.$$

Moreover, for any $x \in \widehat{Q}$ we have

$$\partial_{\widehat{Q}} f(x) \supseteq \text{Conv}\{\partial_{Q,x} \phi(x, y) \mid y \in I(x)\}, \tag{51}$$

where

$$I(x) = \{y \in E \mid \phi(x, y) = f(x)\},$$

and $\partial_{Q,x}$ denotes the subdifferential w.r.t x.

Proof: The proof comes from Lemma 3.1.14 in [2]. In view of (42), we have to prove only the inclusion (51). Indeed, for any $x \in \widehat{Q}$, $y_0 \in I(x_0)$, and $g_0 \in \partial_{Q,x}\phi(x_0, y_0)$, we have

$$f(x) \geq \phi(x, y_0) \overset{(47)}{\geq} \phi(x_0, y_0) + \langle g_0, x - x_0\rangle = f(x_0) + \langle g_0, x - x_0\rangle.$$

□

The next theorem gives a relationship between the subdifferential and the directional derivatives of a convex function.

Theorem 3.11

Let the function f be convex and $x_0 \in \text{int}(\text{dom}(f))$. Then

$$\partial_2 f'(x_0; 0) = \partial f(x_0), \tag{52}$$

where the subdifferential ∂_2 corresponds to the second argument of the function $f'(x_0; \cdot)$. Moreover, for any $p \in \mathbb{R}^n$, we have

$$f'(x_0; p) = \max\{\langle g, p\rangle \mid g \in \partial f(x_0)\} \tag{53}$$

Proof: The proof comes from Theorem 3.1.17 in [2]. Before proving (52), we note that equality $g \in \partial_2 f'(x_0;0)$ means that for all $x_0 \in \text{int}(\text{dom}(f))$ if it holds

$$f'(x_0;p) \overset{(47)}{\geq} f'(x_0;0) + \langle g, p-0\rangle = \lim_{\alpha \downarrow 0} \frac{f(x_0)-f(x_0)}{\alpha} + \langle g,p\rangle = \langle g,p\rangle \quad \forall p \in \mathbb{R}^n \tag{54}$$

Now we show $\partial f(x_0) \subset \partial_2 f'(x_0;0)$. By the definition of subgradient, for each $g \in \partial f(x_0)$ one has

$$f(x_0+\alpha p) \overset{(47)}{\geq} f(x_0) + \langle g, \alpha p\rangle \quad \forall\, \alpha > 0.$$

Dividing by α and letting $\alpha \downarrow 0$ gives

$$f'(x_0;p) \;=\; \lim_{\alpha \downarrow 0} \frac{f(x_0+\alpha p)-f(x_0)}{\alpha} \;\geq\; \langle g,p\rangle,$$

and this is (54), so $g \in \partial_2 f'(x_0;0)$. This means

$$\partial f(x_0) \subset \partial_2 f'(x_0;0).$$

Now we show $\partial_2 f'(x_0;0) \subset \partial f(x_0)$. Let $g \in \partial_2 f'(x_0;0)$ then we have for any $y \in \text{dom}(f)$

$$f(y) \overset{(44)}{\geq} f(x_0) + f'(x_0; y-x_0) \overset{(54)}{\geq} f(x_0) + \langle g, y-x_0\rangle,$$

so $g \in \partial f(x_0)$ and finally $\partial_2 f'(x_0;0) = \partial f(x_0)$.

Now we show

$$f'(x_0;p) = \max\{\langle g,p\rangle \mid g \in \partial f(x_0)\}.$$

In (54) we already have

$$f'(x_0;p) \geq \max_{g \in \partial f(x_0)} \langle g,p\rangle.$$

By Lemma 3.1 (44) we know $f'(x;\cdot)$ is a convex function in the second argument, so it follows for all $p \in \mathbb{R}^n$

$$f'(x_0;0) \overset{(47)}{\geq} f'(x_0,p) + \langle g, 0-p\rangle,$$

where $g \in \partial_2 f'(x_0;0)$. Rearrange and simplify this inequality gives us

$$f'(x_0,p) \leq \langle g,p\rangle.$$

Since (52) holds, we have $g \in \partial f(x_0)$. This proves (53).

□

3.5 Optimality conditions

In this subsection we will look at optimality conditions for unconstrained and constrained optimization problems. We start with the unconstrained part, i.e. we set optimality conditions for

$$\min_{x \in \mathbb{R}^n} f(x) \tag{55}$$

where f is a general convex function. The following object is very useful

Definition 3.10: Epigraph facet

Let the set $X \subset \text{dom}(f)$ be closed and convex. The set

$$\widehat{\partial f}(X) = \bigcap_{x \in X} \partial f(x) \tag{56}$$

is called the *epigraph facet* of the set X.

Let us show how the epigraph facets arise in optimality conditions for unconstrained optimization.

Theorem 3.12

Let f be convex and

$$X^* = \arg\min_{x \in \mathrm{dom}(f)} f(x).$$

Then a closed convex set X_* is a subset of X^* if and only if

$$0 \in \widehat{\partial f}(X_*).$$

Proof: The proof comes from Theorem 3.1.20 in [2]. We prove both directions of the equivalence.

($\Rightarrow$) Assume that $X_* \subseteq X^*$.
This means that every point $x^* \in X_*$ is a minimizer of f, i.e.,

$$f(x) \geq f(x^*) \quad \text{for all } x \in \mathrm{dom}(f).$$

This inequality is equivalent to

$$f(x) \geq f(x^*) + \langle 0, x - x^* \rangle,$$

which, by definition of the subdifferential for convex functions, implies that $0 \in \partial f(x^*)$. Since this is true for every $x^* \in X_*$, we conclude

$$0 \in \bigcap_{x^* \in X_*} \partial f(x^*) = \widehat{\partial f}(X_*).$$

($\Leftarrow$) Assume that $0 \in \widehat{\partial f}(X_*)$.
Then, by definition, $0 \in \partial f(x^*)$ for every $x^* \in X_*$. From the subdifferential inequality, we have:

$$f(x) \geq f(x^*) + \langle 0, x - x^* \rangle = f(x^*) \quad \text{for all } x \in \mathrm{dom}(f).$$

Therefore, $x^* \in X^*$, i.e., each point in X_* minimizes f. Hence

$$X_* \subseteq X^*.$$

□

Let us now look at constrained optimization problems with general objective functions. Consider the problem

$$\min_{x \in \mathcal{K}} f(x), \tag{57}$$

where $\mathcal{K} \subset \mathbb{R}^n$ is a closed convex set and f is a closed convex function with $\mathcal{K} \subset \mathrm{dom}(f)$. We now want to define two sets that will help us for the optimality conditions of subgradients.

Definition 3.11: Normal cone

Let $\mathcal{K}$ be a closed convex set. For a point $\bar{x} \in \mathcal{K}$ define the *normal cone*

$$\mathcal{N}(\bar{x}) = \{g \in \mathbb{R}^n \mid \langle g, x - \bar{x} \rangle \geq 0, \ \forall x \in \mathcal{K}\}. \tag{58}$$

Definition 3.12: Tangent cone

Let $\mathcal{K}$ be a closed convex set. For a point $\bar{x} \in \mathcal{K}$ define the *tangent cone*

$$\mathcal{T}(\bar{x}) = \{p \in \mathbb{R}^n \mid \langle g, p \rangle \geq 0, \ \forall g \in \mathcal{N}(\bar{x})\}. \tag{59}$$

To characterize the relationships between these sets, we introduce the following definition.

Definition 3.13: Dual cone

The dual cone $\mathcal{K}^*$ to a cone $\mathcal{K}$ with respect to the scalar product $\langle y, x\rangle$ is defined as

$$\mathcal{K}^* := \{y \in \mathbb{R}^n \mid \langle y, x\rangle \geq 0 \; \forall x \in \mathcal{K}\}. \tag{60}$$

We would now like to list a few properties of the normal cone and tangent cone.

Lemma 3.4

Let $\mathcal{K}$ be a closed convex set. For a point $\bar{x} \in \mathcal{K}$ it holds

1. The *normal cone* $\mathscr{N}(\bar{x}) = \{g \in \mathbb{R}^n \mid \langle g, x - \bar{x}\rangle \geq 0, \; \forall x \in \mathcal{K}\}$ is a cone.
2. The *tangent cone* $\mathscr{T}(\bar{x}) = \{p \in \mathbb{R}^n \mid \langle g, p\rangle \geq 0, \; \forall g \in \mathscr{N}(\bar{x})\}$ is a cone.
3. The tangent cone is the standard dual cone to $\mathscr{N}(\bar{x})$.
4. $\mathscr{N}(\bar{x}) = \{0_n\}$ for all $\bar{x} \in \operatorname{int}(\mathcal{K})$.
5. $\mathscr{T}(\bar{x}) = \mathbb{R}^n$ for all $\bar{x} \in \operatorname{int}(\mathcal{K})$.

Proof: We prove the statements ourselves.

(1) Let $g_1, g_2 \in \mathscr{N}(\bar{x})$ and $\lambda_1, \lambda_2 \geq 0$. Then for all $x \in \mathcal{K}$,

$$\langle \lambda_1 g_1 + \lambda_2 g_2, x - \bar{x}\rangle = \lambda_1 \langle g_1, x - \bar{x}\rangle + \lambda_2 \langle g_2, x - \bar{x}\rangle \geq 0,$$

since each term is nonnegative. Hence, $\lambda_1 g_1 + \lambda_2 g_2 \in \mathscr{N}(\bar{x})$. Therefore, $\mathscr{N}(\bar{x})$ is a cone.

(2) Let $p_1, p_2 \in \mathscr{T}(\bar{x})$ and $\lambda_1, \lambda_2 \geq 0$. Then for any $g \in \mathscr{N}(\bar{x})$,

$$\langle g, \lambda_1 p_1 + \lambda_2 p_2\rangle = \lambda_1 \langle g, p_1\rangle + \lambda_2 \langle g, p_2\rangle \geq 0.$$

Therefore, $\lambda_1 p_1 + \lambda_2 p_2 \in \mathscr{T}(\bar{x})$, and so $\mathscr{T}(\bar{x})$ is a cone.

(3) We observe that

$$\mathscr{T}(\bar{x}) = \{p \in \mathbb{R}^n \mid \langle g, p\rangle \geq 0, \; \forall g \in \mathscr{N}(\bar{x})\} \overset{(60)}{=} \mathscr{N}(\bar{x})^*.$$

Hence, $\mathscr{T}(\bar{x})$ is the dual cone of $\mathscr{N}(\bar{x})$.

(4) Suppose $g \in \mathscr{N}(\bar{x})$. Since $\bar{x} \in \operatorname{int}(\mathcal{K})$, there exists $\varepsilon > 0$ such that the ball $B(\bar{x}, \varepsilon) \subseteq \mathcal{K}$. In particular, for small enough $\delta > 0$, the point $x := \bar{x} - \delta g$ lies in $\mathcal{K}$. Then

$$\langle g, x - \bar{x}\rangle = \langle g, -\delta g\rangle = -\delta \|g\|^2.$$

If $g \neq 0$, this is strictly negative, contradicting $g \in \mathscr{N}(\bar{x})$. Hence, $g = 0_n$, and so $\mathscr{N}(\bar{x}) = \{0_n\}$.

(5) From Part (4), we have $\mathscr{N}(\bar{x}) = \{0_n\}$, hence its dual cone is:

$$\mathscr{T}(\bar{x}) = \{p \in \mathbb{R}^n \mid \langle 0_n, p\rangle \geq 0\} = \mathbb{R}^n.$$

□

After we have dealt with the corresponding cones, we still need two lemmas to set up our optimality conditions for (57).

Lemma 3.5

Let $\mathcal{K}$ be convex and closed and let $\bar{x} \in \partial\mathcal{K}$. Then $\mathcal{K} - \bar{x} \subset \mathscr{T}(\bar{x})$. Moreover

$$\mathscr{T}(\bar{x}) = \operatorname{cl}(\operatorname{cone}(\mathcal{K} - \bar{x})). \tag{61}$$

Thus, $\mathscr{T}(\bar{x})$ is the closure of the conic hull of the set $\mathcal{K} - \bar{x}$.

Proof: The proof comes from Lemma 3.1.18 in [2]. It should be noted that we present the proofs of the statements in much more detail. We prove each part of the lemma separately.
(1) Inclusion: $\mathcal{K} - \bar{x} \subset \mathscr{T}(\bar{x})$**.**
Let $x \in \mathcal{K}$. Then $v := x - \bar{x} \in \mathcal{K} - \bar{x}$. We want to show that $v \in \mathscr{T}(\bar{x})$. Recall Lemma 3.4 (3) that for a convex set $\mathcal{K}$, the tangent cone at $\bar{x}$ can be characterized by the dual relationship with the normal cone $\mathcal{N}(\bar{x})$, as follows

$$\mathscr{T}(\bar{x}) = \{d \in \mathbb{R}^n : \langle g, d\rangle \geq 0,\ \forall g \in \mathcal{N}(\bar{x})\}.$$

Also, from the definition of the normal cone (58)

$$g \in \mathcal{N}(\bar{x}) \iff \langle g, x - \bar{x}\rangle \geq 0 \quad \text{for all } x \in \mathcal{K}.$$

Therefore, for any $x \in \mathcal{K}$, the vector $v = x - \bar{x}$ satisfies

$$\langle g, v\rangle \geq 0 \quad \text{for all } g \in \mathcal{N}(\bar{x}),$$

which implies $v \in \mathscr{T}(\bar{x})$. Hence,

$$\mathcal{K} - \bar{x} \subset \mathscr{T}(\bar{x}).$$

(2) Equality: $\mathscr{T}(\bar{x}) = \mathrm{cl}\big(\mathrm{cone}(\mathcal{K} - \bar{x})\big)$**.**
Let us define the set

$$\mathcal{H} := \mathrm{cl}\left(\mathrm{cone}(\mathcal{K} - \bar{x})\right).$$

From part (1) of the proof, we already have $\mathcal{K} - \bar{x} \subset \mathscr{T}(\bar{x})$, and since $\mathscr{T}(\bar{x})$ is a closed convex cone, it must contain the conic hull of $\mathcal{K} - \bar{x}$, and hence also its closure

$$\mathcal{H} \subseteq \mathscr{T}(\bar{x}).$$

To prove equality, we suppose for contradiction that the inclusion is strict, i.e., there exists a vector $p \in \mathscr{T}(\bar{x}) \setminus \mathcal{H}$.
Since both $\mathcal{H}$ and $\mathscr{T}(\bar{x})$ are closed convex cones and $p \notin \mathcal{H}$, by the Supporting Hyperplane Theorem, there exists a nonzero vector $g \in \mathbb{R}^n$ and scalar $\gamma \in \mathbb{R}$ such that

$$\langle g, p\rangle < \gamma \leq \langle g, h\rangle \quad \text{for all } h \in \mathcal{H}.$$

Since $\mathcal{H}$ is a cone, for every $x \in \mathcal{K}$ and $\alpha \geq 0$ we have

$$\alpha\,(x - \bar{x}) \ \in\ \mathcal{H} \ \implies\ \langle g, \alpha(x - \bar{x})\rangle \ \geq\ \gamma.$$

By linearity of the inner product, this can be written as

$$\alpha\,\langle g,\, x - \bar{x}\rangle \ \geq\ \gamma \ \implies\ \langle g,\, x - \bar{x}\rangle \ \geq\ \frac{\gamma}{\alpha}.$$

Now letting $\alpha \to \infty$ gives $\gamma/\alpha \to 0$, and hence

$$\langle g,\, x - \bar{x}\rangle \ \geq\ 0 \quad \forall\, x \in \mathcal{K},$$

which implies $g \in \mathscr{N}(\bar{x})$. Now return to the original separation:

$$\langle g, p\rangle < \gamma \leq 0 \quad \Rightarrow \quad \langle g, p\rangle < 0.$$

But by the definition of the tangent cone

$$p \in \mathscr{T}(\bar{x}) \Rightarrow \langle g, p\rangle \geq 0 \quad \text{for all } g \in \mathscr{N}(\bar{x}).$$

This gives a contradiction, since $\langle g, p\rangle < 0$, so $p \notin \mathscr{T}(\bar{x})$. Thus, our assumption must be false, and

$$\mathscr{T}(\bar{x}) \subseteq \mathcal{H}.$$

Combining both inclusions

$$\mathscr{T}(\bar{x}) = \mathrm{cl}\left(\mathrm{cone}(\mathcal{K} - \bar{x})\right).$$

□

The next lemma gives us an optimality condition for a linearized version of problem (57).

Lemma 3.6

Let x^* be an optimal solution to problem (57). Then

$$f'(x^*;p) \geq 0 \ \ \forall p \in \mathscr{T}(x^*). \tag{62}$$

Proof: The proof comes from Lemma 3.1.19 in [2]. It should be noted that we present the proofs of the statements in much more detail. Suppose for the sake of contradiction that there exists $p \in \mathscr{T}(x^*)$ with $f'(x^*;p) < 0$. By (61),

$$\mathscr{T}(x^*) \ = \ \mathrm{cl}(\mathrm{cone}(\mathcal{K} - \bar{x})),$$

so we may choose sequences $\{x_k\} \subset \mathcal{K}$ and $\{\alpha_k\} \subset [0,\infty)$ such that

$$\alpha_k \, (x_k - x^*) \ \longrightarrow \ p \quad (k \to \infty).$$

Since the directional derivative is positively homogeneous and continuous (44), it follows that

$$f'(x^*;p) \ = \ \lim_{k\to\infty} f'\big(x^*;\, \alpha_k(x_k - x^*)\big) \ = \ \lim_{k\to\infty} \alpha_k \, f'\big(x^*; x_k - x^*\big).$$

By definition of the directional derivative,

$$f'\big(x^*; x_k - x^*\big) \ = \ \lim_{\beta \downarrow 0} \frac{f\big(x^* + \beta(x_k - x^*)\big) - f(x^*)}{\beta},$$

and hence

$$f'(x^*;p) = \lim_{k\to\infty} \lim_{\beta\downarrow 0} \frac{\alpha_k}{\beta}\Big[f\big(x^* + \beta(x_k - x^*)\big) - f(x^*)\Big].$$

For each $0 < \beta < 1$, convexity of $\mathcal{K}$ ensures $x^* + \beta(x_k - x^*) \in \mathcal{K}$, and optimality of x^* gives

$$f\big(x^* + \beta(x_k - x^*)\big) \ \geq \ f(x^*),$$

so the bracketed term is nonnegative. Since $\beta > 0$ and $\alpha_k \geq 0$ we get

$$\frac{\alpha_k}{\beta}\Big[f\big(x^* + \beta(x_k - x^*)\big) - f(x^*)\Big] \geq 0,$$

which yields $f'(x^*;p) \geq 0$, a contradiction. Therefore no such p can satisfy $f'(x^*;p) < 0$, and the lemma follows.

□

Now we come to the main statement which are the optimality conditions for problem (57). This theorem is one of the most powerful tools of Convex Analysis.

Theorem 3.13

Let $\mathcal{K}$ be convex and closed and f a closed convex function. Let

$$X^* = \arg\min_{x\in\mathcal{K}} f(x).$$

A point $x^* \in \mathcal{K}$ belongs to X^* if and only if there exists a $g^* \in \partial f(x^*)$ such that

$$\langle g^*, x - x^* \rangle \geq 0 \quad \forall x \in \mathcal{K}. \tag{63}$$

In this case, $g^* \in \widehat{\partial f}(X^*) \cap \widehat{\mathscr{N}}(X^*)$ with $\widehat{\mathscr{N}}(X^*) \overset{(56)}{=} \bigcap_{x\in X^*} \mathscr{N}(x)$.

Proof: The proof comes from Theorem 3.1.24 in [2]. It should be noted that we present the proofs of the statements in much more detail. First we show that (63) is sufficient for optimality. Suppose there is a subgradient $g^* \in \partial f(x^*)$ satisfying

$$\langle g^*,\, x - x^* \rangle \geq 0 \quad \forall\, x \in \mathcal{K}.$$

Then by the subgradient inequality,

$$f(x) \;\geq\; f(x^*) \;+\; \langle g^*,\, x - x^*\rangle \;\geq\; f(x^*) \quad \forall\, x \in \mathcal{K},$$

so x^* attains the minimum of f over $\mathcal{K}$, i.e. $x^* \in X^*$. Conversely, let $x^* \in X^*$ be optimal and assume for contradiction that no $g \in \partial f(x^*)$ satisfies (63). By (58) this is equivalent to

$$\partial f(x^*) \;\cap\; \mathscr{N}(x^*) \;=\; \emptyset,$$

where $\mathscr{N}(x^*)$ is the normal cone of $\mathcal{K}$ at x^*. Both $\partial f(x^*)$ and $\mathscr{N}(x^*)$ are nonempty, closed, convex sets and by Theorem 3.9 $\partial f(x^*)$ is bounded, so we conclude with Supporting Hyperplane Theorem that the projection problem

$$\min_{g_1 \in \partial f(x^*),\, g_2 \in N(x^*)} \; \frac{1}{2}\|g_1 - g_2\|^2$$

admits a unique solution (g_1^*, g_2^*) with positive optimal value $\rho^* = \frac{1}{2}\|g_1^* - g_2^*\|^2 > 0$. By the first-order optimality condition (4) we get

$$\langle \nabla_{(g_1,g_2)} \left(\frac{1}{2}\|g_1 - g_2\|^2\right)\Big|_{g_1=g_1^*, g_2=g_2^*}, (g_1 - g_1^*, g_2 - g_2^*)\rangle \geq 0 \quad \forall\, g_1 \in \partial f(x^*), \quad \forall\, g_2 \in \mathscr{N}(x^*)$$

This results to

$$\langle (g_1^* - g_2^*, g_2^* - g_1^*), (g_1 - g_1^*, g_2 - g_2^*)\rangle \geq 0 \quad \forall\, g_1 \in \partial f(x^*), \quad \forall\, g_2 \in \mathscr{N}(x^*)$$

Fix $g_2 = g_2^*$ then we get

$$\left\langle g_1^* - g_2^*,\, g_1 - g_1^*\right\rangle \;\geq\; 0 \quad \forall\, g_1 \in \partial f(x^*). \tag{64}$$

In the same way we fix $g_1 = g_1^*$ to get

$$\left\langle g_2^* - g_1^*,\, g_2 - g_2^*\right\rangle \;\geq\; 0 \quad \forall\, g_2 \in \mathscr{N}(x^*). \tag{65}$$

From (65), using that $\mathscr{N}(x^*)$ is a cone, substitute $g_2 = 2\, g_2^*$

$$0 \leq \langle g_2^* - g_1^*, 2g_2^* - g_2^*\rangle = \langle g_2^* - g_1^*, g_2^*\rangle. \tag{66}$$

Since $0 \in \mathscr{N}(x^*)$ we substitute $g_2 = 0$ and get

$$0 \geq \langle g_2^* - g_1^*, g_2^*\rangle. \tag{67}$$

Therefore with (66) and (67) we conclude

$$\langle g_2^*, g_2^*\rangle \;=\; \langle g_1^*, g_2^*\rangle. \tag{68}$$

It follows

$$0 \overset{(65)}{\leq} \langle g_2^* - g_1^*,\, g_2 - g_2^*\rangle = g_2^* g_2 - g_2^* g_2^* - g_1^* g_2 + g_1^* g_2^* \overset{(68)}{=} \langle g_2, g_2^* - g_1^*\rangle \quad \forall\, g_2 \in \mathscr{N}(x^*),$$

so by (59) of the tangent cone, setting $p := g_2^* - g_1^*$ yields

$$p \;\in\; \mathscr{T}(x^*).$$

Recall that by (68) it holds

$$\langle p, g_2^*\rangle = 0. \tag{69}$$

On the other hand, we have for every $g_1 \in \partial f(x^*)$

$$\left\langle g_1^* - g_2^*,\, g_1 - g_1^*\right\rangle \overset{(64)}{\geq} 0 \iff \langle g_1,\, p\rangle \leq \langle g_1^*, p\rangle \overset{(69)}{=} \langle g_1^* - g_2^*, p\rangle \;=\; -2\rho^*.$$

Therefore

$$f'(x^*; p) \overset{(53)}{=} \sup_{g \in \partial f(x^*)} \langle g, p\rangle \;\leq\; -2\rho^* < 0,$$

which contradicts Lemma 3.6.
Let $x_1^* \in X^*$. Since $f(x_1^*) = f(x^*)$ and $g^* \in \partial f(x^*)$, the subgradient inequality at x^* yields

$$f(x_1^*) \;=\; f(x^*) \;\geq\; f(x^*) + \langle g^*, x_1^* - x^*\rangle \implies \langle g^*, x_1^* - x^*\rangle = 0.$$

But then applying the subgradient inequality *at* x_1^* shows $g^* \in \partial f(x_1^*)$. Since x_1^* was arbitrary in X^*, we get $g^* \in \bigcap_{x \in X^*} \partial f(x) = \widehat{\partial} f(X^*)$.
Similarly, from $g^* \in \mathscr{N}(x^*)$ and the symmetry of the normal-cone condition one deduces $g^* \in \mathscr{N}(x_1^*)$ for every $x_1^* \in X^*$. Hence $g^* \in \bigcap_{x \in X^*} \mathscr{N}(x) = \widehat{\mathscr{N}}(X^*)$.

3.6 Subgradient method

In this subsection we would like to introduce the subgradient method for constrained problems. Before we do so, we would like to define some tools to later investigate the convergence of this method. Consider again the constrained minimization problem (57)

$$\min_{x\in\mathcal{K}} f(x), \tag{70}$$

where $\mathcal{K} \subset \mathbb{R}^n$ is a closed convex set, and $f : \mathbb{R}^n \to \mathbb{R}$ is a convex (but not necessarily smooth) function. Since f may not be differentiable, we rely on subgradients $g(x) \in \partial f(x)$ instead of gradients. This substitution introduces several challenges

- Unlike gradients, subgradients do not guarantee that f decreases along the direction $-g(x)$.
- Even if $x \to x^*$, we cannot assume that $g(x) \to 0$.

Nevertheless, subgradients possess a key property that underpins most nonsmooth optimization methods. As shown in (63), the following inequality holds

$$\langle g(x), x - x^* \rangle \geq 0 \quad \forall x \in \mathcal{K}, \tag{71}$$

where $x^* \in \mathcal{K}$ denotes a minimizer of f. This inequality implies two critical facts

- The distance between x and x^* decreases along the direction $-g(x)$.
- Inequality (71) divides $\mathbb{R}^n$ into two half-spaces, and we can identify which one contains the solution.

Localization function: To evaluate the quality of an approximate solution $x \in \mathbb{R}^n$, we define the following quantity for fix $\bar{x} \in \mathbb{R}^n$

$$v_f(\bar{x}, x) = \frac{1}{\|g(x)\|}\langle g(x), x - \bar{x}\rangle, \quad \text{if } g(x) \neq 0. \tag{72}$$

If $g(x) = 0$, we set $v_f(\bar{x}, x) := 0$. By the Cauchy–Schwarz inequality, we have

$$v_f(\bar{x}, x) \leq \|x - \bar{x}\|.$$

The value $v_f(\bar{x}, x)$ has a natural geometric interpretation, it represents the distance from the point $\bar{x}$ to the hyperplane

$$H = \{y \in \mathbb{R}^n : \langle g(x), x - y\rangle = 0\}.$$

To see this, consider the point

$$\bar{y} = \bar{x} + v_f(\bar{x}, x) \cdot \frac{g(x)}{\|g(x)\|}. \tag{73}$$

Then we have

$$\langle g(x), x - \bar{y}\rangle \overset{(73)}{=} \langle g(x), x - \bar{x}\rangle - v_f(\bar{x}, x) \cdot \|g(x)\| \overset{(72)}{=} 0.$$

Thus, $v_f(\bar{x}, x)$ can be interpreted as the orthogonal distance from $\bar{x}$ to the hyperplane H.

Growth Function of f: To further analyze the behavior of f around a fixed point $\bar{x}$, we define the local growth function

$$\omega_f(\bar{x}; t) = \max_{\|x-\bar{x}\|\leq t} [f(x) - f(\bar{x})], \quad \text{for } t \geq 0, \tag{74}$$

and set $\omega_f(\bar{x}; t) := 0$ for $t < 0$.
The function ω_f satisfies the following clearly properties

- $\omega_f(\bar{x}; t) = 0$ for all $t \leq 0$,
- $\omega_f(\bar{x}; t)$ is a non-decreasing function of t,

- For any $x \in \mathbb{R}^n$, it holds that

$$f(x) - f(\bar{x}) \leq \omega_f(\bar{x}; \|x - \bar{x}\|). \tag{75}$$

In nonsmooth convex optimization, it is often difficult to directly evaluate the quality of a candidate solution due to the absence of gradients and the non-availability of optimality conditions such as stationarity. However, using subgradient-based geometry and certain auxiliary functions, it is possible to estimate how far a point is from being optimal.
The following lemma provides a key result that relates the value of the objective function at a point $\bar{x} \in \mathbb{R}^n$ to its subgradient structure.

Lemma 3.7

Let $\bar{x} \in \mathbb{R}^n$ and f a convex function, then for any $x \in \mathbb{R}^n$ we have

$$f(x) - f(\bar{x}) \leq \omega_f(\bar{x}; v_f(\bar{x}; x)). \tag{76}$$

If f is Lipschitz continuous on $B_2(\bar{x}, R)$ with constant $M > 0$, then

$$f(x) - f(\bar{x}) \leq M \cdot \max\{0, v_f(\bar{x}; x)\} \tag{77}$$

for all $x \in \mathbb{R}^n$ with $v_f(\bar{x}; x) \leq R$.

Proof: The proof comes from Lemma 3.2.1 in [2]. We split the proof of (76) into two cases. Let $g \in \partial f(x)$ and $\bar{x} \in \mathbb{R}^n$.

Case $\langle g,\, x - \bar{x}\rangle < 0$: Then from the subgradient inequality

$$f(\bar{x}) \overset{(47)}{\geq} f(x) + \langle g,\, \bar{x} - x\rangle \;\geq\; f(x),$$

so $f(x) - f(\bar{x}) \leq 0$. By definition $v_f(\bar{x}; x) = \frac{1}{\|g\|}\langle g, x - \bar{x}\rangle \geq 0$, and the growth function of f $\omega_f(\bar{x}; t)$ vanishes for $t \leq 0$. Hence

$$f(x) - f(\bar{x}) \;\leq\; 0 = \omega_f\big(\bar{x}\,;\, v_f(\bar{x}; x)\big).$$

Case $\langle g,\, x - \bar{x}\rangle \geq 0$. Define the point

$$\bar{y} \;=\; \bar{x} \;+\; v_f(\bar{x}; x)\, \frac{g}{\|g\|}.$$

By construction,

$$\|\bar{y} - \bar{x}\| \;=\; v_f(\bar{x}; x), \qquad \langle g,\, \bar{y} - x\rangle = \langle g,\, \bar{x} - x\rangle + \; v_f(\bar{x}; x)\, \frac{\langle g, g\rangle}{\|g\|} = 0.$$

Convexity of f then gives

$$f(\bar{y}) \overset{(47)}{\geq} f(x) + \langle g,\, \bar{y} - x\rangle \;=\; f(x). \tag{78}$$

Therefore

$$f(x) - f(\bar{x}) \overset{(78)}{\leq} f(\bar{y}) - f(\bar{x}) \overset{(75)}{\leq} \omega_f\big(\bar{x}\,;\|\bar{y} - \bar{x}\|\big) \;=\; \omega_f\big(\bar{x}\,;v_f(\bar{x}; x)\big),$$

which proves (76).

We prove now (77). If f is M-Lipschitz on $B_2(\bar{x}, R)$ and $0 \leq v_f(\bar{x}; x) \leq R$, then $\bar{y} \in B_2(\bar{x}, R)$ and

$$f(x) - f(\bar{x}) \overset{(78)}{\leq} f(\bar{y}) - f(\bar{x}) \;\leq\; M\,\|\bar{y} - \bar{x}\| \;=\; M\, v_f(\bar{x}; x).$$

□

The function $v_f(x^*; x)$ can also be used to define so-called *localization sets*, which are guaranteed to contain the optimal solution. These sets are constructed based on subgradient inequalities.

Definition 3.14

Let $\{x_i\}_{i=0}^{\infty} \subset \mathcal{K}$ be a sequence. Then for $k \geq 0$, define the localization set

$$S_k = \{x \in \mathcal{K} \mid \langle g(x_i), x_i - x \rangle \geq 0, \quad \text{for } i = 0, \ldots, k\},$$

where $g(x_i) \in \partial f(x_i)$.

By (71), we know that the optimal solution $x^* \in S_k$ for all $k \geq 0$, hence

$$x^* \in \bigcap_{k=0}^{\infty} S_k.$$

Let us define the following values

$$v_i = v_f(x^*; x_i) \geq 0, \quad v_k^* = \min_{0 \leq i \leq k} v_i. \tag{79}$$

The following lemma provides a computable upper bound on the optimality gap $f_k^* - f^*$ based only on subgradient geometry and the growth function. It is of great practical value for termination criteria and convergence analysis of nonsmooth optimization algorithms.

Lemma 3.8

Let $\{x_i\}_{i=0}^{\infty} \subset \mathcal{K}$ be a sequence and f a convex function. Set

$$f_k^* = \min_{0 \leq i \leq k} f(x_i). \tag{80}$$

Then the following inequality holds

$$f_k^* - f(x^*) \leq \omega_f(x^*; v_k^*),$$

where x^* is the optimal solution to (70).

Proof: The proof comes from Lemma 3.2.2 in [2]. It holds

$$\omega_f(x^*; v_k^*) \overset{(79)}{=} \omega_f(x^*; \min_{0 \leq i \leq k} v_i) = \min_{0 \leq i \leq k} \omega_f(x^*; v_i) \overset{(76)}{\geq} \min_{0 \leq i \leq k} (f(x_i) - f(x^*)) \overset{(80)}{=} f_k^* - f(x^*)$$

□

We now want to impose additional conditions on the minimization problem (70). This is firstly that the set $\mathcal{K}$ is simple and secondly that the minimization problem is equipped with a first-order oracle.

The term *simple* refers to the tractability of projections, the set $\mathcal{K}$ is called simple if the Euclidean projection of an arbitrary point $x \in \mathbb{R}^n$ onto $\mathcal{K}$ can be computed efficiently and explicitly. This includes standard sets such as boxes, Euclidean balls, half-spaces, and simplices.

Oracle Model: We assume access to a first-order oracle, which at any query point $x \in \mathbb{R}^n$ returns both the function value $f(x)$ and a subgradient $g(x) \in \partial f(x)$.

Direction Selection: In classical gradient methods for smooth optimization, the gradient norm $\|\nabla f(x)\|$ is a useful measure of optimality and is often used to adapt the step size. However, in nonsmooth settings, the subgradient $g(x)$ may have a large or fluctuating norm even near the optimal solution, and thus the norm $\|g(x)\|$ is not a reliable indicator of progress.
Therefore, in subgradient methods, it is common practice to use the *normalized subgradient* direction

$$d_k = -\frac{g(x_k)}{\|g(x_k)\|},$$

which ensures that all steps are taken in directions of unit length. This normalization avoids instability caused by large subgradient norms and simplifies the design of step size rules.

In the following we give the subgradient method for simple sets

Algorithm 7: Subgradient Method for Simple Sets

Data: convex closed set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, initial point $w_1 \in \mathcal{K}$, step sizes $\{h_t\}$ with

$$h_k > 0, \quad h_k \to 0, \quad \sum_{k=1}^{\infty} h_k = \infty$$

for *t=1 to T* **do**
  Compute a subgradient $g_t \in \partial f(w_t)$
  Update: $y_{t+1} = w_t - h_t \frac{g_t}{||g_t||}$
  Project: $w_{t+1} = \Pi_{\mathcal{K}}(y_{t+1})$
end
return w_T

We will now estimate the convergence rate.

Theorem 3.14

Let a function f be Lipschitz continuous on $B_2(x^*, R)$ with constant M, where $R \geq \|x_0 - x^*\|$. Then

$$f_k^* - f(x^*) \leq M \frac{R^2 + \sum_{i=0}^{k} h_i^2}{2 \sum_{i=0}^{k} h_i}. \tag{81}$$

Proof: The proof comes from Theorem 3.2.2 in [2]. Set

$$r_i = \|x_i - x^*\|.$$

Using the nonexpansiveness of the projection $\Pi_{\mathcal{K}}$,

$$\begin{aligned} r_{i+1}^2 &= \left\|\Pi_{\mathcal{K}}(x_i - h_i \tfrac{g(x_i)}{\|g(x_i)\|}) - x^*\right\|^2 \\ &\leq \left\|x_i - h_i \tfrac{g(x_i)}{\|g(x_i)\|} - x^*\right\|^2. \end{aligned}$$

Expanding the square,

$$\left\|a - b\right\|^2 = \|a\|^2 - 2\langle a, b\rangle + \|b\|^2,$$

with

$$v_i = \left\langle \frac{g(x_i)}{\|g(x_i)\|}, x_i - x^* \right\rangle,$$

gives

$$r_{i+1}^2 \;\leq\; r_i^2 - 2\, h_i\, v_i + h_i^2.$$

Sum for $i = 0, \ldots, k$

$$\sum_{i=0}^{k} r_{i+1}^2 \leq \sum_{i=0}^{k} \left(r_i^2 - 2h_i v_i + h_i^2\right) \implies r_{k+1}^2 + 2\sum_{i=0}^{k} h_i v_i \;\leq\; r_0^2 + \sum_{i=0}^{k} h_i^2.$$

Since $r_{k+1}^2 \geq 0$ and $r_0^2 \leq R^2$, it follows that

$$\sum_{i=0}^{k} h_i\, v_i \;\leq\; \frac{R^2 + \sum_{i=0}^{k} h_i^2}{2}.$$

By Lipschitz-continuity,

$$f(x_i) - f(x^*) \;\leq\; M\, \|x_i - x^*\| \implies \frac{f(x_i) - f(x^*)}{M} \;\leq\; v_i.$$

Hence

$$\sum_{i=0}^{k} h_i \big(f(x_i) - f(x^*)\big) \;\leq\; M \sum_{i=0}^{k} h_i v_i \;\leq\; M\, \frac{R^2 + \sum_{i=0}^{k} h_i^2}{2}.$$

Because $f_k^* = \min_{0 \le i \le k} f(x_i)$, we have

$$\sum_{i=0}^{k} h_i \big(f(x_i) - f(x^*)\big) \;\ge\; \Big(\sum_{i=0}^{k} h_i\Big)\big(f_k^* - f^*\big).$$

Then it follows

$$\big(\sum_{i=0}^{k} h_i\big)\,(f_k^* - f^*) \;\le\; M\,\frac{R^2 + \sum_{i=0}^{k} h_i^2}{2} \;\Longrightarrow\; f_k^* - f^* \;\le\; M\,\frac{R^2 + \sum_{i=0}^{k} h_i^2}{2\,\sum_{i=0}^{k} h_i}.$$

This completes the proof. □

4 AdaGrad with Composite Mirror Descent

Before beginning this chapter, we would like to note that the algorithms from Chapter 2 (OGD, FTRL, OMD, AdaGrad) can all be extended to subgradients. Furthermore, the regret analyses still apply, whereby instead of $g_t = \nabla f_t(w_t)$, any vector $g_t \in \partial f_t(w_t)$ is used.

This chapter, together with the following one, focuses on the AdaGrad algorithm in the context of subgradient-based optimization, as presented in [3]. We discuss two modifications of the standard AdaGrad method. In this chapter, we derive the variant based on Composite Mirror Descent (COMID). We begin by introducing the COMID framework following [8], and subsequently formulate the AdaGrad algorithm using the Composite Mirror Descent update rule.

We begin by formulating the optimization problem that Composite Mirror Descent (COMID) was originally designed to address and then perform a corresponding regret analysis.

4.1 Problem Setting and Motivation of COMID

We build upon the Online Convex Optimization (OCO) framework introduced in chapter 2. Recall that, in OCO, at each round $t = 1, \dots, T$, an algorithm selects a decision $w_t \in \mathcal{K} \subseteq \mathbb{R}^n$, observes a convex loss function $f_t : \mathcal{K} \to \mathbb{R}$, and incurs loss $f_t(w_t)$. The goal is to minimize the cumulative regret with respect to the best fixed decision in hindsight.

While standard Mirror Descent provides an elegant and general approach for minimizing regret in this setting, it does not explicitly handle *composite objectives* that naturally arise in regularized learning problems. Many applications involve loss functions of the form

$$F_t(w) = f_t(w) + \varphi(w), \tag{82}$$

where f_t is a convex (possibly non-smooth) loss term that changes over time, and $\varphi \geq 0$ is a fixed convex regularizer that encodes structural properties such as sparsity or boundedness (e.g., via ℓ_1- or ℓ_2-regularization).

The *Composite Mirror Descent (COMID)* framework extends Mirror Descent to efficiently handle such objectives by incorporating the regularization term φ directly into the update step, rather than treating it as part of the loss. This modification allows the algorithm to maintain the advantages of mirror-based updates while achieving better adaptation to the geometry induced by φ.

Formally, COMID updates the iterates according to

$$w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\{ \eta \langle g_t, w \rangle + \eta \varphi(w) + B_\psi(w \| w_t) \right\}, \tag{83}$$

where $g_t \in \partial f_t(w_t)$ is a subgradient of the instantaneous loss, $\eta > 0$ is the step size, and $B_\psi(w \| w_t)$ denotes the Bregman divergence generated by a strictly convex function ψ.

This formulation generalizes Mirror Descent and provides the foundation for adaptive algorithms such as *AdaGrad with Composite Mirror Descent*, which will be derived in the following sections.

4.2 The Composite Mirror Descent Algorithm and Regret Analysis

In the following, we will describe the COMID algorithm.

Algorithm 8: Composite Mirror Descent Method (COMID)

Data: convex closed set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, initial point $w_1 \in \mathcal{K}$, step size $\eta > 0$

for *t=1 to T* **do**
  Observe convex loss f_t and compute subgradient $g_t \in \partial f_t(w_t)$
  Update: $w_{t+1} = \arg\min_{w \in \mathcal{K}} \{\eta \langle g_t, w \rangle + \eta \varphi(w) + B_\psi(w \| w_t)\}$
end
return w_T

Special cases: Several well-known algorithms arise as special cases of COMID.

- **Mirror Descent:** obtained when $\varphi \equiv 0$.
- **Online Gradient Descent:** obtained when $\psi(w) = \frac{1}{2}\|w\|_2^2$ and $\varphi \equiv 0$.
- **ℓ_1-regularized online learning:** obtained when $\psi(w) = \frac{1}{2}\|w\|_2^2$ and $\varphi(w) = \lambda\|w\|_1$.

In the following, we derive a regret bound for COMID and show that, under standard convexity and boundedness assumptions, the algorithm achieves sublinear regret, thereby generalizing the classical results of Mirror Descent. This lemma is the key to our later analysis.

Lemma 4.1

Let the sequence $\{w_t\}_{t=0}^T$ be defined by the update (83). Assume that $B_\psi(\cdot\|\cdot)$ is α-strongly convex with respect to a norm $\|\cdot\|$. For any $w^* \in \mathcal{K}$ it holds

$$\eta\big(f_t(w_t) - f_t(w^*)\big) + \eta\big(\varphi(w_t) - \varphi(w^*)\big) \leq B_\psi(w^*\|w_t) - B_\psi(w^*\|w_{t+1}) + \frac{\eta^2}{2\alpha}\|g_t(w_t)\|^{*2}, \tag{84}$$

where $g_t(w_t) \in \partial f_t(w_t)$.

Proof:The proof follows directly from Lemma 1 in [8].
Let $g_t \in \partial f_t(w_t)$. The optimality of w_{t+1} for (83) implies for all $w \in \mathcal{K}$ and $\tilde{\varphi}(w_{t+1}) \in \partial\varphi(w_{t+1})$ that

$$\langle w - w_{t+1},\, \eta g_t + \nabla\psi(w_{t+1}) - \nabla\psi(w_t) + \eta\tilde{\varphi}(w_{t+1})\rangle \overset{(63)}{\geq} 0. \tag{85}$$

From the subgradient inequality for convex functions, we have $f_t(w^*) \overset{(47)}{\geq} f_t(w_t) + \langle g_t, w^* - w_t\rangle$, or equivalently,

$$f_t(w_t) - f_t(w^*) \leq \langle g_t, w_t - w^*\rangle, \tag{86}$$

and likewise for $\varphi(w_{t+1})$. We thus have

$$\begin{aligned}
&\eta\big[f_t(w_t) + \varphi(w_{t+1}) - f_t(w^*) - \varphi(w^*)\big] \overset{(86)}{\leq} \eta\langle w_t - w^*, g_t\rangle + \eta\langle w_{t+1} - w^*, \tilde{\varphi}(w_{t+1})\rangle \\
&= \eta\langle w_{t+1} - w^*, g_t\rangle + \eta\langle w_{t+1} - w^*, \tilde{\varphi}(w_{t+1})\rangle + \eta\langle w_t - w_{t+1}, g_t\rangle \\
&= \langle w^* - w_{t+1}, \nabla\psi(w_t) - \nabla\psi(w_{t+1}) - \eta g_t - \eta\tilde{\varphi}(w_{t+1})\rangle \\
&\quad + \langle w^* - w_{t+1}, \nabla\psi(w_{t+1}) - \nabla\psi(w_t) + \eta\langle w_t - w_{t+1}, g_t\rangle. \\
&\overset{(85)}{\leq} \langle w^* - w_{t+1}, \nabla\psi(w_{t+1}) - \nabla\psi(w_t)\rangle + \eta\langle w_t - w_{t+1}, g_t\rangle \\
&= B_\psi(w^*\|w_t) - B_{\psi_t}(w_{t+1}\|w_t) - B_\psi(w^*\|w_{t+1}) + \eta\langle w_t - w_{t+1}, g_t\rangle \\
&= B_\psi(w^*\|w_t) - B_\psi(w_{t+1}\|w_t) - B_\psi(w^*\|w_{t+1}) + \eta\Big\langle\sqrt{\frac{\alpha}{\eta}}(w_t - w_{t+1}), \sqrt{\frac{\alpha}{\eta}}g_t\Big\rangle \\
&\leq B_\psi(w^*\|w_t) - B_\psi(w_{t+1}\|w_t) - B_\psi(w^*\|w_{t+1}) + \frac{\alpha}{2}\|w_t - w_{t+1}\|_\psi^2 + \frac{\eta^2}{2\alpha}\|g_t\|_{\psi^*}^2 \\
&\overset{(18)}{\leq} B_\psi(w^*\|w_t) - B_\psi(w^*\|w_{t+1}) + \tfrac{\eta^2}{2\alpha}\|g_t\|_{\psi^*}^2.
\end{aligned}$$

□

We establish a general regret bound for COMID.

Theorem 4.1

Let the sequence $\{w_t\}_{t=0}^{\infty}$ be defined by the update (83). Then for any $w^* \in \mathcal{K}$ it holds

$$R_T^{\text{COMID}} \leq \frac{1}{\eta} B_\psi(w^* \| w_1) + \varphi(w_1) + \frac{\eta}{2\alpha} \sum_{t=1}^{T} \|g_t(w_t)\|^{*2}, \tag{87}$$

where $g_t(w_t) \in \partial f_t(w_t)$ and $B_\psi(\cdot\|\cdot)$ is α-strongly convex. Furthermore if the functions f_t are Lipschitz continous with dual Lipschitz constant G^*, then

$$R_T^{\text{COMID}} \leq \frac{1}{\eta} B_\psi(w^* \| w_1) + \varphi(w_1) + \frac{T\eta}{2\alpha} G^{*2}. \tag{88}$$

Proof: The proof follows directly from Theorem 2 and Corollary 3 in [8].
It holds,

$$\begin{aligned}\eta \sum_{t=1}^{T} \Big(f_t(w_t) - f_t(w^*) + \varphi(w_t) - \varphi(w^*) \Big) &\overset{(84)}{\leq} \sum_{t=1}^{T} \big(B_\psi(w^* \| w_t) - B_\psi(w^* \| w_{t+1}) \big) + \frac{\eta^2}{2\alpha} \sum_{t=1}^{T} \|g_t\|_*^2 \\ &= B_\psi(w^* \| w_1) - B_\psi(w^* \| w_{T+1}) + \frac{\eta^2}{2\alpha} \sum_{t=1}^{T} \|g_t\|_*^2.\end{aligned}$$

Noting that Bregman divergences are always non-negative, and recalling our assumption that $\varphi(w) \geq 0$, we add $\eta\varphi(w_1)$ to the right side of the above inequality and drop the $-B_\psi(w^* \| w_{T+1})$ term. This gives

$$\eta \sum_{t=1}^{T} \Big(f_t(w_t) - f_t(w^*) + \varphi(w_t) - \varphi(w^*) \Big) \leq B_\psi(w^* \| w_1) + \eta\varphi(w_1) + \frac{\eta^2}{2\alpha} \sum_{t=1}^{T} \|g_t\|_*^2.$$

Dividing both sides by η yields (87). If f_t are Lipschitz continous then $g_t \leq G^*$ and we obtain (88).

□

It is easy to see that if we choose the step size η proportional to $\sqrt{T}$, we obtain sublinear regret.

4.3 Diagonal AdaGrad with COMID and Regret Analysis

We now combine the update step from AdaGrad with diagonal matrices (Algorithm 6 step 5) with the update step from COMID (Algorithm 8) and obtain the update step

$$w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\{ \eta \langle g_t, w \rangle + \eta\varphi(w) + B_{\psi_t}(w \| w_t) \right\}, \tag{89}$$

where $\psi_t(w) = \frac{1}{2}\langle w, h_t \odot w \rangle$. This gives us the following pseudocode

Algorithm 9: Diagonal AdaGrad with COMID

Data: convex closed set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$,step size $\eta > 0$,$\delta \geq 0$, regularizer φ
Initialize: $w_1 = 0 \in \mathbb{R}^n$ with $\varphi(w_1) = 0$, $\tilde{s}_0 = 0 \in \mathbb{R}^n$
for *t=1 to T* **do**
  Observe convex loss f_t and compute subgradient $g_t \in \partial f_t(w_t)$
  Update: $\tilde{s}_t = \tilde{s}_{t-1} + g_t \odot g_t$
  $s_t = (\sqrt{(\tilde{s}_t)_1}, \ldots \sqrt{(\tilde{s}_t)_n})^\top$
  Set: $h_t = \delta 1_n + s_t$, $\psi_t(w) = \frac{1}{2}\langle w, h_t \odot w \rangle$
  Update: $w_{t+1} = \arg\min_{w \in \mathcal{K}} \{\eta \langle g_t, w \rangle + \eta\varphi(w) + B_{\psi_t}(w \| w_t)\}$
end
return w_T

Remark:

- Note it applies

$$s_t = \begin{pmatrix} \sqrt{\sum_{i=1}^t (g_i)_1^2} \\ \cdots \\ \sqrt{\sum_{i=1}^t (g_i)_n^2} \end{pmatrix} \quad \text{and} \quad \sum_{i=1}^n (s_t)_i = \|s_t\|_1. \tag{90}$$

  We mention this because the paper [3] used a notation that was difficult to read, which we have chosen not to use. For better traceability, the term $\sum_{i=1}^n \|g_{1:t,i}\|_2$ from [3] corresponds exactly to $\|s_t\|_1$.

- Furthermore it holds $(s_{t+1})_i \geq (s_t)_i$ so also $\|s_{t+1} - s_t\|_1 = \|s_{t+1}\|_1 - \|s_t\|_1$.

The following theorem gives an initial estimate of regret.

Theorem 4.2

Let the sequence $\{w_t\}_{t=1}^\infty$ be defined by the update in Algorithm 9 and let $B_{\psi_t}(\cdot\|\cdot)$ be strongly convex with respect to a norm $\|\cdot\|_{\psi_t}$. Furthermore let $\|\cdot\|_{\psi_t^*}$ be the associated dual norm. For any $w^* \in \mathcal{K}$ holds

$$\begin{aligned} R_T^{\text{COMID Diag AdaGrad}} &= \sum_{t=1}^T f_t(w_t) + \varphi(w_t) - f_t(w^*) - \varphi(w^*) \\ &\leq \frac{1}{\eta} B_{\psi_1}(w^*\|w_1) + \frac{1}{\eta} \sum_{t=1}^{T-1} \Big(B_{\psi_{t+1}}(w^*\|w_{t+1}) - B_{\psi_t}(w^*\|w_{t+1}) \Big) + \frac{\eta}{2} \sum_{t=1}^T \|g_t\|_{\psi_t^*}^2, \end{aligned} \tag{91}$$

where $g_t \in \partial f_t(w_t)$.

Proof: The proof follows directly from Proposition 2 in [3].
We will show first with $g_t \in \partial f_t(w_t)$

$$\eta\big(f_t(w_t) - f_t(w^*)\big) + \eta\big(\varphi(w_{t+1}) - \varphi(w^*)\big) \leq B_{\psi_t}(w^*\|w_t) - B_{\psi_t}(w^*\|w_{t+1}) + \frac{\eta^2}{2}\|g_t\|_{\psi_t^*}^2 \tag{92}$$

The optimality of w_{t+1} for (89) implies for all $w \in \mathcal{K}$ and $\tilde{\varphi}(w_{t+1}) \in \partial\varphi(w_{t+1})$ that

$$\langle w - w_{t+1},\, \eta g_t + \nabla\psi_t(w_{t+1}) - \nabla\psi_t(w_t) + \eta\tilde{\varphi}(w_{t+1})\rangle \overset{(63)}{\geq} 0. \tag{93}$$

From the subgradient inequality for convex functions, we have $f_t(w^*) \overset{(47)}{\geq} f_t(w_t) + \langle g_t, w^* - w_t\rangle$, or equivalently,

$$f_t(w_t) - f_t(w^*) \leq \langle g_t, w_t - w^*\rangle, \tag{94}$$

and likewise for $\varphi(w_{t+1})$. We thus have

$$\begin{aligned} &\eta\big[f_t(w_t) + \varphi(w_{t+1}) - f_t(w^*) - \varphi(w^*)\big] \overset{(94)}{\leq} \eta\langle w_t - w^*, g_t\rangle + \eta\langle w_{t+1} - w^*, \tilde{\varphi}(w_{t+1})\rangle \\ &= \eta\langle w_{t+1} - w^*, g_t\rangle + \eta\langle w_{t+1} - w^*, \tilde{\varphi}(w_{t+1})\rangle + \eta\langle w_t - w_{t+1}, g_t\rangle \\ &= \langle w^* - w_{t+1}, \nabla\psi_t(w_t) - \nabla\psi_t(w_{t+1}) - \eta g_t - \eta\tilde{\varphi}(w_{t+1})\rangle \\ &\quad + \langle w^* - w_{t+1}, \nabla\psi_t(w_{t+1}) - \nabla\psi_t(w_t) + \eta\langle w_t - w_{t+1}, g_t\rangle. \\ &\overset{(93)}{\leq} \langle w^* - w_{t+1}, \nabla\psi_t(w_{t+1}) - \nabla\psi_t(w_t)\rangle + \eta\langle w_t - w_{t+1}, g_t\rangle \\ &= B_{\psi_t}(w^*\|w_t) - B_{\psi_t}(w_{t+1}\|w_t) - B_{\psi_t}(w^*\|w_{t+1}) + \eta\langle w_t - w_{t+1}, g_t\rangle \\ &= B_{\psi_t}(w^*\|w_t) - B_{\psi_t}(w_{t+1}\|w_t) - B_{\psi_t}(w^*\|w_{t+1}) + \eta\big\langle \eta^{-\frac{1}{2}}(w_t - w_{t+1}), \sqrt{\eta} g_t\big\rangle \\ &\leq B_{\psi_t}(w^*\|w_t) - B_{\psi_t}(w_{t+1}\|w_t) - B_{\psi_t}(w^*\|w_{t+1}) + \tfrac{1}{2}\|w_t - w_{t+1}\|_{\psi_t}^2 + \tfrac{\eta^2}{2}\|g_t\|_{\psi_t^*}^2 \\ &\overset{(18)}{\leq} B_{\psi_t}(w^*\|w_t) - B_{\psi_t}(w^*\|w_{t+1}) + \tfrac{\eta^2}{2}\|g_t\|_{\psi_t^*}^2. \end{aligned}$$

This proves (92). It follows

$$
\begin{aligned}
R_T &= \sum_{t=1}^{T} f_t(w_t) + \varphi(w_t) - f_t(w^*) - \varphi(w^*) \\
&\leq \sum_{t=1}^{T} \big(f_t(w_t) - f_t(w^*)\big) + \big(\varphi(w_{t+1}) - \varphi(w^*)\big) \overset{(92)}{\leq} \frac{1}{\eta}\sum_{t=1}^{T} B_{\psi_t}(w^* \| w_t) - B_{\psi_t}(w^* \| w_{t+1}) + \frac{\eta}{2}\|g_t\|_{\psi_t^*}^2 \\
&= \frac{1}{\eta} B_{\psi_1}(w^* \| w_1) - \frac{1}{\eta} B_{\psi_T}(w^* \| w_{T+1}) + \frac{1}{\eta}\sum_{t=1}^{T-1} B_{\psi_{t+1}}(w^* \| w_{t+1}) - B_{\psi_t}(w^* \| w_{t+1}) + \frac{\eta}{2}\sum_{t=1}^{T} \|g_t\|_{\psi_t^*}^2 \\
&\leq \frac{1}{\eta} B_{\psi_1}(w^* \| w_1) + \frac{1}{\eta}\sum_{t=1}^{T-1} B_{\psi_{t+1}}(w^* \| w_{t+1}) - B_{\psi_t}(w^* \| w_{t+1}) + \frac{\eta}{2}\sum_{t=1}^{T} \|g_t\|_{\psi_t^*}^2
\end{aligned}
$$

□

We would now like to introduce a lemma and a theorem to bring (91) into an accessible form.

Lemma 4.2

Let $g_t \in \partial f_t(w_t)$ and s_t defined as in Algorithm 9. Then

$$\sum_{t=1}^{T} \langle g_t, \mathrm{Diag}(s_t)^{-1} g_t \rangle \leq 2\|s_T\|_1 \tag{95}$$

Proof: The proof follows directly from Lemma 4 in [3].
We prove the lemma by considering an arbitrary real-valued sequence $\{a_i\}_{i=1}^T$ with the vector representation $a = (a_1, ..., a_T)^\top$. We are next going to show that

$$\sum_{t=1}^{T} \frac{a_t^2}{\left\| \left(a_1, ..., a_t\right)^\top \right\|_2} \leq 2\|a\|_2, \tag{96}$$

where we adopt the convention $\frac{0}{0} = 0$. We use induction on T to prove inequality (96), for this we define $a_{1:t} = \left(a_1, ..., a_t\right)^\top$. For $T = 1$, the inequality trivially holds. Assume the bound (96) holds true for $T - 1$, in which case

$$\sum_{t=1}^{T} \frac{a_t^2}{\|a_{1:t}\|_2} = \sum_{t=1}^{T-1} \frac{a_t^2}{\|a_{1:t}\|_2} + \frac{a_T^2}{\|a_{1:T}\|_2} \leq 2\|a_{1:T-1}\|_2 + \frac{a_T^2}{\|a_{1:T}\|_2},$$

where the inequality follows from the inductive hypothesis. We define $b_T = \sum_{t=1}^{T} a_t^2$ and use concavity to obtain that

$$\sqrt{b_T - a_T^2} \leq \sqrt{b_T} - a_T^2 \frac{1}{2\sqrt{b_T}} \quad \text{so long as } b_T - a_T^2 \geq 0.$$

Thus,

$$2\|a_{1:T-1}\|_2 + \frac{a_T^2}{\|a_{1:T}\|_2} = 2\sqrt{b_T - a_T^2} + \frac{a_T^2}{\sqrt{b_T}} \leq 2\sqrt{b_T} = 2\|a_{1:T}\|_2 = 2\|a\|_2.$$

Having proved the bound (96), we note that by construction $(s_t)_i = \sqrt{\sum_{j=1}^{t} (g_j)_i^2} = \left\| \left((g_1)_i, ..., (g_t)_i\right)^\top \right\|_2$, so

$$\sum_{t=1}^{T} \langle g_t, \mathrm{Diag}(s_t)^{-1} g_t \rangle = \sum_{t=1}^{T}\sum_{i=1}^{n} \frac{(g_t)_i^2}{\left\| \left((g_1)_i, ..., (g_t)_i\right)^\top \right\|_2} \leq 2\sum_{i=1}^{n} (s_T)_i \overset{(s_T)_i \geq 0}{=} 2\|s_T\|_1.$$

□

Note that (95) implies

$$\sum_{t=1}^{T} \|g_t\|_{\psi_t^*}^2 = \sum_{t=1}^{T} \langle g_t, \left(\delta I_n + \mathrm{Diag}(s_t)\right)^{-1} g_t \rangle$$
$$\overset{\delta I_n + \mathrm{Diag}(s_t) \succeq \mathrm{Diag}(s_t)}{\leq} \sum_{t=1}^{T} \langle g_t, \mathrm{Diag}(s_t)^{-1} g_t \rangle \overset{(95)}{\leq} 2 \sum_{i=1}^{n} (s_T)_i \tag{97}$$

Theorem 4.3

Let the sequences $\{w_t\}_{t=1}^T$ and $\{s_t\}_{t=1}^T$ be defined by the update in Algorithm 9. Then for any $w^* \in \mathcal{K}$

$$R_T^{\text{COMID Diag AdaGrad}} \leq \frac{1}{2\eta} \max_{t \leq T} \|w^* - w_t\|_\infty^2 \|s_T\|_1 + \eta \|s_T\|_1. \tag{98}$$

Furthermore set $D_\infty = \sup_{w \in \mathcal{K}} \|w - w^*\|_\infty$, $\gamma_T = \|s_T\|_1$ and $\eta = \frac{D_\infty}{\sqrt{2}}$, then we have

$$R_T \leq \sqrt{2} D_\infty \gamma_T. \tag{99}$$

Proof: The proof follows directly from Theorem 5 in [3].
Consider $\psi_t = \frac{1}{2}\langle w, h_t \odot w \rangle$, we get

$$B_{\psi_{t+1}}(w^* \| w_{t+1}) - B_{\psi_t}(w^* \| w_{t+1}) = \frac{1}{2} \langle w^* - w_{t+1}, \mathrm{Diag}(s_{t+1} - s_t)(w^* - w_{t+1}) \rangle$$
$$\leq \frac{1}{2} \max_{i \in \{1,2,\ldots,n\}} ((w)_i^* - (w_{t+1})_i)^2 \|s_{t+1} - s_t\|_1.$$

Since $\|s_{t+1} - s_t\|_1 = \langle s_{t+1} - s_t, 1_n \rangle$ and $\langle s_T, 1_n \rangle = \|s_T\|_1$, we have

$$\sum_{t=1}^{T-1} \left(B_{\psi_{t+1}}(w^* \| w_{t+1}) - B_{\psi_t}(w^* \| w_{t+1}) \right) \leq \frac{1}{2} \sum_{t=1}^{T-1} \|w^* - w_{t+1}\|_\infty^2 \langle s_{t+1} - s_t, 1_n \rangle$$
$$\leq \frac{1}{2} \max_{t \leq T} \|w^* - w_t\|_\infty^2 \|s_T\|_1 - \frac{1}{2} \|w^* - w_1\|_\infty^2 \langle s_1, 1_n \rangle. \tag{100}$$

Plug (100) and (97) in (91) and note that $B_{\psi_1}(w^* \| w_1) \leq \frac{1}{2} \|w^* - w_1\|_\infty^2 \|s_1\|_1$, we obtain (98).

Set $D_\infty = \sup_{w \in \mathcal{K}} \|w - w^*\|_\infty$, $\gamma_T = \|s_T\|_1$ and $\eta = \frac{D_\infty}{\sqrt{2}}$ and plug in (98)

$$R_T \leq \frac{\sqrt{2}}{2D_\infty} D_\infty^2 \gamma_T + \frac{D_\infty}{\sqrt{2}} \gamma_T = \sqrt{2} D_\infty \gamma_T$$

□

4.4 Full AdaGrad with COMID and Regret Analysis

We now combine the update step from AdaGrad with full matrices with the update step from COMID and obtain the update step

$$w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\{ \eta \langle g_t, w \rangle + \eta \varphi(w) + B_{\psi_t}(w \| w_t) \right\}, \tag{101}$$

where $\psi_t(w) = \frac{1}{2}\langle w, H_t w\rangle$. This gives us the following pseudocode

Algorithm 10: Full AdaGrad with COMID

Data: convex closed set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$,step size $\eta > 0$, $\delta \geq 0$,regularizer φ
Initialize: $w_1 = 0 \in \mathbb{R}^n$, $G_0 = 0 \in \mathbb{R}^{n\times n}$
for *t=1 to T* **do**
  Observe convex loss f_t and compute subgradient $g_t \in \partial f_t(w_t)$
  Update: $G_t = G_{t-1} + g_t g_t^\top$, $S_t = G_t^{\frac{1}{2}}$
  Set: $H_t = \delta I_n + S_t$, $\psi_t(w) = \frac{1}{2}\langle w, H_t w\rangle$
  Update: $w_{t+1} = \arg\min_{w\in\mathcal{K}} \{\eta\langle g_t, w\rangle + \eta\varphi(w) + B_{\psi_t}(w\|w_t)\}$
end
return w_T

We would like to point out that (91) also applies to Algorithm 10. We use it to prove the following theorem.

Theorem 4.4

Let G_t be the outer product matrix from Algorithm 10 and the sequence $\{w_t\}_{t=1}^T$ produced by Algorithm 10. If $w^* \in \mathcal{K}$ and $\delta \geq 0$, then

$$R_T^{\text{COMID Full AdaGrad}} \leq \frac{\delta}{\eta}\|w^*\|_2^2 + \frac{1}{2\eta}\max_{t\leq T}\|w^* - w_t\|_2^2 \mathrm{Tr}(G_T^{1/2}) + \eta\mathrm{Tr}(G_T^{1/2}). \tag{102}$$

Furthermore set $\delta = 0$, $\sup_{w\in\mathcal{K}}\|w - w^*\|_2 \leq D$ and $\eta = \frac{D}{\sqrt{2}}$, we have

$$R_T \leq \sqrt{2}D\mathrm{Tr}(G_T^{1/2}). \tag{103}$$

Proof: The proof follows directly from Theorem 7 in [3].
It holds (91)

$$R_T \overset{(91)}{\leq} \frac{1}{\eta}B_{\psi_1}(w^*\|w_1) + \frac{1}{\eta}\sum_{t=1}^{T-1}\Big(B_{\psi_{t+1}}(w^*\|w_{t+1}) - B_{\psi_t}(w^*\|w_{t+1})\Big) + \frac{\eta}{2}\sum_{t=1}^{T}\|g_t\|_{\psi_t^*}^2$$

By definition of the Bregman divergence for quadratic ψ:

$$B_{\psi_t}(w^*\|w_{t+1}) = \tfrac{1}{2}\langle w^* - w_{t+1},\, H_t(w^* - w_{t+1})\rangle.$$

Hence,

$$B_{\psi_{t+1}}(w^*\|w_{t+1}) - B_{\psi_t}(w^*\|w_{t+1}) = \tfrac{1}{2}\langle w^* - w_{t+1},\, (H_{t+1} - H_t)(w^* - w_{t+1})\rangle.$$

Because $H_t = \delta I + S_t$ and $S_t = G_t^{1/2}$, this simplifies to

$$B_{\psi_{t+1}}(w^*\|w_{t+1}) - B_{\psi_t}(w^*\|w_{t+1}) = \tfrac{1}{2}\langle w^* - w_{t+1},\, (S_{t+1} - S_t)(w^* - w_{t+1})\rangle.$$

Now, $S_{t+1} - S_t \succeq 0$, so its largest eigenvalue is at most its trace. Therefore,

$$B_{\psi_{t+1}} - B_{\psi_t} \overset{(11)}{\leq} \frac{1}{2}\lambda_{\max}(S_{t+1} - S_t)\|w^* - w_{t+1}\| \overset{(12)}{\leq} \tfrac{1}{2}\|w^* - w_{t+1}\|_2^2\,\mathrm{Tr}(S_{t+1} - S_t).$$

Summing from $t = 1$ to $T - 1$ gives

$$\sum_{t=1}^{T-1}(B_{\psi_{t+1}} - B_{\psi_t}) \;\leq\; \tfrac{1}{2}\max_{t\leq T}\|w^* - w_t\|_2^2\sum_{t=1}^{T-1}\mathrm{Tr}(S_{t+1} - S_t).$$

The traces telescope:

$$\sum_{t=1}^{T-1}\mathrm{Tr}(S_{t+1} - S_t) = \mathrm{Tr}(S_T) - \mathrm{Tr}(S_1) = \mathrm{Tr}(G_T^{1/2}) - \mathrm{Tr}(G_1^{1/2}),$$

and since $\mathrm{Tr}(G_1^{1/2}) \geq 0$ this term may be dropped. Hence,

$$\sum_{t=1}^{T-1} B_{\psi_{t+1}} - B_{\psi_t} \;\leq\; \tfrac{1}{2} \max_{t \leq T} \|w^* - w_t\|_2^2 \,\mathrm{Tr}(G_T^{1/2}). \tag{104}$$

Furthermore, the following holds by the same argument as in the proof of (36), with minor notational modifications

$$\begin{aligned} \sum_{t=1}^{T} \|g_t\|_{\psi_t^*}^2 = \sum_{t=1}^{T} g_t^\top H_t^{-1} g_t = \sum_{t=1}^{T} g_t^\top (\delta I_n + S_t)^{-1} g_t \\ \overset{\delta I_n + S_t \succeq S_t}{\leq} \sum_{t=1}^{T} g_t^\top S_t^{-1} g_t \overset{(36)}{\leq} 2\mathrm{Tr}(G_T^{1/2}). \end{aligned} \tag{105}$$

Recall that the first term (91) is

$$B_{\psi_1}(w^* \| w_1) = \tfrac{1}{2} \langle w^* - w_1,\, H_1(w^* - w_1) \rangle \overset{(11),(12)}{\leq} \tfrac{1}{2}(\delta + \mathrm{Tr}(S_1)) \|w^* - w_1\|_2^2.$$

Since $w_1 = 0$ this is upper-bounded by

$$B_{\psi_1}(w^* \| w_1) \overset{0 \leq \mathrm{Tr}(S_1)}{\leq} \delta \|w^*\|_2^2. \tag{106}$$

Finally plug (104), (105) and (106) in (91) and obtain (102). Now set $\delta = 0$, $\sup_{w \in \mathcal{K}} \|w - w^*\|_2 \leq D$ and $\eta = \frac{D}{\sqrt{2}}$ and we get

$$R_T \leq \frac{1}{\sqrt{2}D} D^2 \mathrm{Tr}(G_T^{1/2}) + \frac{D}{\sqrt{2}} \mathrm{Tr}(G_T^{1/2}) = \sqrt{2} D \mathrm{Tr}(G_T^{1/2}).$$

□

5 AdaGrad with Primal-Dual Subgradient

In the previous chapter, we introduced and analyzed the AdaGrad algorithm within the Composite Mirror Descent (COMID) framework. There, the adaptive geometry was incorporated through time-varying mirror maps, leading to regret bounds that generalize those of the classical Mirror Descent method.
In this chapter, we turn to a second formulation of the AdaGrad algorithm with subgradients, which is based on the *primal–dual subgradient method* introduced by Nesterov [10] and its extension, the *Regularized Dual Averaging (RDA)* framework proposed by Xiao [11]. Unlike COMID, which performs implicit updates in the primal space through mirror mappings, the primal–dual perspective maintains an explicit accumulation of subgradients in the dual space, with regularization introduced via strongly convex functions.
We begin by presenting the main ideas of the primal–dual subgradient method, followed by a presentation of the Regularized Dual Averaging algorithm. Finally, we derive the adaptive AdaGrad variant within this framework and establish corresponding regret bounds under standard convexity assumptions.

5.1 Problem Setting and Motivation of Primal-Dual method

We consider the convex, but not necessarily smooth, optimization problem of the form

$$\min_{w \in \mathcal{K}} f(w), \tag{107}$$

where f is convex but not differentiable and $\mathcal{K} \subseteq \mathbb{R}^n$ is closed and convex. In addition, we assume that the subgradient $g \in \partial f(w)$ can be calculated.
One way to solve (107) would be Algorithm 7, but it has a serious weakness. The classic subgradient step is

$$w_{t+1} = w_t - \lambda_t \frac{g_t}{\|g_t\|}$$

As can be seen in (81), in order to obtain convergence, the weights $\lambda_t \in \mathbb{R}$ must satisfy the following

$$\frac{R^2 + \sum_{i=0}^{k} \lambda_i^2}{2 \sum_{i=0}^{k} \lambda_i} \overset{k \to \infty}{\to} 0,$$

where $R \geq \|w_0 - w^*\|$. This only works if

$$\sum_{i=0}^{\infty} \lambda_i = \infty \quad \text{and} \quad \sum_{i=0}^{\infty} \lambda_i^2 < \infty.$$

That forces $\lambda_t \overset{t \to \infty}{\to} 0$. As a result, new subgradients with decreasing weights are incorporated into the model. Nesterov found this illogical, new gradients should be weighted more heavily, not less. This gave rise to the idea of using two sequences, one for the "primal" space and one for the "dual" space. Here, the sequence $\{w_t\}$ acts in the primal space, and the sequence in the dual space is to be generated by the mean of $\{g_t\}$. It should be noted that, according to (47), the subgradients form a linear functional and, according to (9), they live in the dual space $\mathcal{K}^*$, and therefore $g_t \in \partial f(w_t) \subseteq \mathcal{K}^*$ also applies.

5.2 The Gap Function and Quality of the Subgradient Sequence

In order to evaluate the quality of the iterations generated by a subgradient-based method, we need a quantitative measure that captures how far the current sequence is from satisfying the optimality condition of the convex problem (107). For convex functions, the first-order optimality condition (94) states that a point w^* is optimal if and only if

$$\langle g, w - w^* \rangle \geq 0 \quad \forall w \in \mathcal{K}, \;\; g \in \partial f(w^*). \tag{108}$$

Evaluating this inequality on the iterations w_i generated by a subgradient algorithm provides a natural way to test how close we are to optimality. In practice, however, the exact optimum w^* is

unknown, so we cannot directly check (108). Instead, we look for a proxy that measures how well the aggregated information collected so far (the subgradients $g_0, ..., g_k$) satisfy such conditions on average.

To define this term mathematically precisely, we introduce a prox function

$$\psi : \mathbb{R}^n \to \mathbb{R},$$

which is continuous and strongly convex with parameter σ and continuous function on $\mathcal{K}$. Without loss of generality, let

$$w_0 = \arg\min_{w \in \mathcal{K}} \{\psi(w)\} \quad \text{and} \quad \psi(w_0) = 0. \tag{109}$$

This function serves as a distance-like measure that defines the geometry of space. A typical example is

$$\psi(w) = \frac{1}{2}\|w - w_0\|_2^2.$$

Using ψ, we define a bounded level set

$$\mathcal{F}_D = \{w \in \mathcal{K} \mid \psi(w) \leq D\}, \qquad D > 0. \tag{110}$$

The constant D controls the size of this set and effectively limits our search range.
This restriction is necessary because in the subsequent analysis we will use linear functionals of the form $\langle g, w\rangle$ which are unbounded over $\mathbb{R}^n$. The level set $\mathcal{F}_D$ guarantees that maximization problems with such linear terms remain finite and thus analytically treatable.

Now assume we have performed k subgradient steps, producing pairs (w_i, g_i) with $g_i \in \partial f(w_i)$. Each subgradient inequality yields $\forall w \in \mathcal{K}$

$$f(w) \overset{(47)}{\geq} f(w_i) + \langle g_i, w - w_i\rangle.$$

By weighting these inequalities with positive coefficients $\lambda_i > 0$ and summing, we obtain

$$\sum_{i=0}^{k} \lambda_i f(w) \geq \sum_{i=0}^{k} \lambda_i \Big(f(w_i) + \langle g_i, w - w_i\rangle \Big).$$

Rearranging

$$\sum_{i=0}^{k} \lambda_i f(w_i) \leq \sum_{i=0}^{k} \lambda_i f(w) + \sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w\rangle$$

leads to the aggregate deviation

$$\sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w\rangle$$

which quantifies how much the iterates w_i violate the global optimality condition relative to a given w.
To measure the *worst-case violation* over the bounded region $\mathcal{F}_D$, we define the *gap function*

Definition 5.1: Gap function

Let $\mathcal{F}_D$ defined like in (110) and let (w_i, g_i) be a sequence generated by a subgradient method then we define the gap function

$$\delta_k(D) = \max_{w \in \mathcal{F}_D} \Big\{ \sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w\rangle \Big\} = \underbrace{\sum_{i=0}^{k} \lambda_i \langle g_i, w_i\rangle}_{\text{primal term}} - \underbrace{\min_{w \in \mathcal{F}_D} \Big\{ \sum_{i=0}^{k} \lambda_i \langle g_i, w\rangle \Big\}}_{\text{dual term}}. \tag{111}$$

Remark:

- If all iterates w_i satisfy the optimality condition exactly, i.e $w_i = w^*$ then $\langle g_i, w_i - w\rangle = 0$ for all $w \in \mathcal{K}$ and hence $\delta_k(D) = 0$. Thus, a small gap value indicates that the iterates are close to an optimal solution.
- The maximization over $w \in \mathcal{F}_D$ ensures that $\delta_k(D)$ captures the worst-case violation within the feasible region of interest.
- In primal–dual terms $\delta_k(D)$ corresponds to the duality gap between a primal candidate (the averaged iterate) and its associated dual variable (the aggregated subgradients).

5.3 Primal-dual subgradient method and convergence analysis

In the previous section, we introduced the gap function $\delta_k(D)$, which measures how well the accumulated subgradients of a subgradient method satisfy the convex optimality conditions on average. We now use this concept to derive Nesterov's primal-dual subgradient method, which solves the main limitation of the classical subgradient algorithm, namely the need for decreasing step sizes, which lead to decreasing weights of new subgradients. At iteration t, we obtain a subgradient $g_t \in \partial f(w_t)$. Instead of directly updating w_t via a step size, we form the dual accumulator

$$s_{t+1} = s_t + \lambda_t g_t = \sum_{i=0}^{t} \lambda_i g_i, \qquad s_0 = 0. \tag{112}$$

The variable $s_t \in \mathcal{K}^*$ lives in the dual space, where the subgradients naturally reside. The sequence $\{s_t\}$ can be interpreted as an aggregated dual variable that captures all gradient information up to iteration t.
To map back from the dual variable s_t to a primal iterate w_t, Nesterov introduces a proximal projection operator

$$\Pi_{\beta_t}(s) = \arg\min_{w \in \mathcal{F}_D} \Big\{ \langle -s, w\rangle + \beta_t \psi(w) \Big\},$$

where $\beta_t > 0$ is non decreasing. This gives us the following algorithm

Algorithm 11: Primal-dual Subgradient method

Data: convex closed set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, prox function ψ, step sizes $\{\lambda_k\}$
Initialize: $s_0 = 0 \in \mathcal{K}^*$, $w_0 \in \mathcal{K}$, $\beta_0 = 1$
for *t=0 to T* **do**
  Compute $g_t \in \partial f_t(w_t)$
  Update dual: $s_{t+1} = s_t + \lambda_t g_t$
  Update: $\beta_{t+1} = \beta_t + \frac{1}{\beta_t}$
  Update primal: $w_{t+1} = \Pi_{\beta_{t+1}}(-s_{t+1}) = \arg\min_{w \in \mathcal{K}} \Big\{ \langle s_{t+1}, w\rangle + \beta_{t+1}\psi(w) \Big\}$
end
return w_T

Remark: We start at $t = 0$, as Nesterov often does.

Before we move on to convergence analysis, let's take another look at the gap function. The following applies

$$\begin{aligned}
\delta_k(D) &\overset{(111)}{=} \sum_{i=0}^{k} \lambda_i \langle g_i, w_i\rangle + \max_{w \in \mathcal{F}_D} \Big\{ \sum_{i=0}^{k} \lambda_i \langle g_i, -w\rangle \Big\} \\
&= \sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w_0\rangle + \sum_{i=0}^{k} \lambda_i \langle g_i, w_0\rangle + \max_{w \in \mathcal{F}_D} \Big\{ \sum_{i=0}^{k} \lambda_i \langle g_i, -w\rangle \Big\} \\
&\overset{(112)}{=} \sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w_0\rangle + \langle s_{k+1}, w_0\rangle + \max_{w \in \mathcal{F}_D} \Big\{ \langle s_{k+1}, -w\rangle \Big\} \\
&= \sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w_0\rangle + \max_{w \in \mathcal{F}_D} \Big\{ \langle -s_{k+1}, w - w_0\rangle \Big\}
\end{aligned}$$

$$\overset{(110)}{=} \sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w_0 \rangle + \max_{w \in \mathcal{F}_D} \min_{\beta \geq 0} \Big\{ \langle -s_{k+1}, w - w_0 \rangle + \beta_{k+1}\big(D - \psi(w)\big) \Big\}$$

$$\leq \sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w_0 \rangle + \min_{\beta \geq 0} \max_{w \in \mathcal{F}_D} \Big\{ \langle -s_{k+1}, w - w_0 \rangle + \beta_{k+1}\big(D - \psi(w)\big) \Big\}$$

$$\leq \sum_{i=0}^{k} \lambda_i \langle g_i, w_i - w_0 \rangle + \beta_{k+1} D + \underbrace{\max_{w \in \mathcal{F}_D} \Big\{ \langle -s_{k+1}, w - w_0 \rangle - \beta_{k+1}\psi(w) \Big\}}_{:= V_{\beta_{k+1}}(-s_{k+1})} \tag{113}$$

We want to use the function

$$V_\beta(s) = \max_{w \in \mathcal{F}_D} \Big\{ \langle s, w - w_0 \rangle - \beta \psi(w) \Big\} \tag{114}$$

to establish a suitable estimate for the gap. It is easy to see that for $\beta_{t+1} > \beta_t$ for all $t \in \mathbb{N}$ the following applies

$$V_{\beta_{t+1}}(s) \leq V_{\beta_t}(s). \tag{115}$$

Before we get to a convergence analysis, we need to consider the following lemma.

Lemma 5.1

Let ψ be continous and strong convex with parameter σ. The function $V_\beta(\cdot)$ is convex and differentiable on $\mathcal{K}^*$. Moreover, its gradient is Lipschitz continous with constant $\frac{1}{\beta\sigma}$, i.e for all $s_1, s_2 \in \mathcal{K}^*$

$$\|\nabla V_\beta(s_1) - \nabla V_\beta(s_2)\| \leq \frac{1}{\beta\sigma} \|s_1 - s_2\|^*. \tag{116}$$

For any $s \in \mathcal{K}^*$, the vector $\nabla V_\beta(s)$ belongs to $\mathcal{K}$

$$\nabla V_\beta(s) = \Pi_\beta(s) - w_0 \tag{117}$$

Proof: The proof follows directly from Theorem 1 in [12].
Due to Lemma 3.3, the linearity in s and the continuity of ψ, we obtain convexity and continuity. Furthermore, due to the strict convexity and the linear argument, the maximum is unique, which is why the subgradient in Lemma 3.3 is a gradient. For this reason, we receive

$$\nabla V_\beta(s) = \nabla_s \Big(\max_{w \in \mathcal{F}_D} \Big\{ \langle s, w \rangle - \beta \psi(w) \Big\} \Big) - \nabla_s \langle s, w_0 \rangle = \Pi_\beta(s) - w_0,$$

what (117) shows. To show (116), we omit the terms that do not depend on w, as they cancel each other out. Consider two points s_1 and s_2 and let $\tilde{\psi}(s) \in \partial\psi(s)$. From the first-order optimality conditions we have

$$\langle w(s_1) - \beta\tilde{\psi}(w(s_1)),\, w(s_2) - w(s_1) \rangle \leq 0,$$
$$\langle w(s_2) - \beta\tilde{\psi}(w(s_2)),\, w(s_1) - w(s_2) \rangle \leq 0.$$

Adding these inequalities and using strong convexity of $\psi(\cdot)$, we continue as follows

$$\langle (s_1 - s_2), \big(w(s_1) - w(s_2)\big) \rangle \geq \beta \langle \tilde{\psi}\big(w(s_1)\big) - \tilde{\psi}\big(w(s_2)\big), w(s_1) - w(s_2) \rangle \overset{(5)}{\geq} \beta\sigma \|w(s_1) - w(s_2)\|_2^2. \tag{118}$$

Hence with $w(s) = \Pi_\beta(s)$

$$\|\nabla V_\beta(s_1) - \nabla V_\beta(s_2)\|^2 = \|w(s_1) - w(s_2)\|^2 \overset{(118)}{\leq} \frac{1}{\beta\sigma} \langle (s_1 - s_2), \big(w(s_1) - w(s_2)\big) \rangle$$
$$\overset{(10)}{\leq} \frac{1}{\beta\sigma} \|w(s_1) - w(s_2)\| \|s_1 - s_2\|^*.$$

Now divide by $\|w(s_1) - w(s_2)\|$ and obtain

$$\|w(s_1) - w(s_2)\| = \|\nabla V_\beta(s_1) - \nabla V_\beta(s_2)\| \leq \frac{1}{\beta\sigma}\|s_1 - s_2\|^*.$$

□

An important consequence of (116) and by the descent lemma for functions with L-Lipschitz gradient is

$$V_\beta(s+d) \leq V_\beta(s) + \langle d, \nabla V_\beta(s)\rangle + \frac{1}{2\sigma\beta}\|d\|_*^2, \quad \forall s, d \in K^*. \tag{119}$$

From our assumption (109), we obtain with $s = 0$

$$V_\beta(d) \leq \frac{1}{2\sigma\beta}\|d\|^{*2} \tag{120}$$

With these considerations, we obtain the following theorem to evaluate our solution

Theorem 5.1

Let $\{w_i\}$, $\{g_i\}$ and $\{\lambda_i\}$ generated by algorithm 11. Then for any $k \geq 0$ and $D \geq 0$ we have

$$\delta_k(D) \leq \beta_{k+1}D + \frac{1}{2\sigma}\sum_{i=0}^{k}\frac{\lambda_i^2}{\beta_i}\|g_i\|_2^{*2}. \tag{121}$$

Furthermore assume that a solution w^* in the sense (108) exists. Then

$$\frac{1}{2}\|w_{k+1} - w^*\|_2^2 \leq \psi(w^*) + \frac{1}{2\sigma\beta_{k+1}}\sum_{i=0}^{k}\frac{\lambda_i^2}{\beta_i}\|g_i\|_2^{*2}. \tag{122}$$

Proof: The proof follows directly from Theorem 1 in [10].
We obtain for all $i \geq 1$

$$\begin{aligned} V_{\beta_{i+1}}(-s_{i+1}) &\overset{(115)}{\leq} V_{\beta_i}(-s_{i+1}) \\ &\overset{(119)}{\leq} V_{\beta_i}(-s_i) - \lambda_i\langle g_i, \nabla V_{\beta_i}(-s_i)\rangle + \frac{\lambda_i^2}{2\sigma\beta_i}\|g_i\|_2^{*2} \\ &\overset{(117)}{=} V_{\beta_i}(-s_i) + \lambda_i\langle g_i, w_0 - w_i\rangle + \frac{\lambda_i^2}{2\sigma\beta_i}\|g_i\|_2^{*2}, \end{aligned}$$

hence for all $i \in \{1, ..., k\}$

$$\lambda_i\langle g_i, w_0 - w_i\rangle \leq V_{\beta_i}(-s_i) - V_{\beta_{i+1}}(-s_{i+1}) + \frac{\lambda_i^2}{2\sigma\beta_i}\|g_i\|_2^{*2}.$$

The summation of all these inequalities results in

$$\sum_{i=0}^{k}\lambda_i\langle g_i, w_0 - w_i\rangle \leq V_{\beta_1}(-s_1) - V_{\beta_{k+1}}(-s_{k+1}) + \frac{1}{2\sigma}\sum_{i=1}^{k}\frac{\lambda_i^2}{\beta_i}\|g_i\|_2^{*2}. \tag{123}$$

It holds

$$V_{\beta_1}(-s_1) = V_{\beta_1}(-\lambda_0 g_0) \overset{(120)}{\leq} \frac{\lambda_0^2}{2\sigma\beta_1}\|g_0\|_2^{*2} \leq \frac{\lambda_0^2}{2\sigma\beta_0}\|g_0\|_2^{*2},$$

plug this in (123) and obtain

$$\sum_{i=0}^{k}\lambda_i\langle g_i, w_0 - w_i\rangle \leq -V_{\beta_{k+1}}(-s_{k+1}) + \frac{1}{2\sigma}\sum_{i=0}^{k}\frac{\lambda_i^2}{\beta_i}\|g_i\|_2^{*2} \leq \beta_{k+1}D + \frac{1}{2\sigma}\sum_{i=0}^{k}\frac{\lambda_i^2}{\beta_i}\|g_i\|_2^{*2}. \tag{124}$$

This shows (121). To show (122), we consider that w^* exists. Then

$$
\begin{aligned}
0 &\overset{(108)}{\leq} \sum_{i=0}^{k} \lambda_i \langle g_i,\, w_i - w^* \rangle \\
&\overset{(124),(108)}{\leq} \langle s_{k+1},\, w_0 - w^* \rangle - V_{\beta_{k+1}}(-s_{k+1}) + \frac{1}{2\sigma} \sum_{i=0}^{k} \frac{\lambda_i^2}{\beta_i} \|g_i\|_2^{*2} \\
&\overset{(114)}{=} \langle s_{k+1},\, w_{k+1} - w^* \rangle + \beta_{k+1}\psi(w_{k+1}) + \frac{1}{2\sigma} \sum_{i=0}^{k} \frac{\lambda_i^2}{\beta_i} \|g_i\|_2^{*2} \\
&\leq \beta_{k+1}\psi(w^*) - \tfrac{1}{2}\beta_{k+1}\sigma \|w_{k+1} - w^*\|^2 + \frac{1}{2\sigma} \sum_{i=0}^{k} \frac{\lambda_i^2}{\beta_i} \|g_i\|_2^{*2},
\end{aligned}
$$

where we refer to Lemma 6 of [10] for the last inequality.

5.4 Regularized Dual Averaging Method

After looking at Nesterov's primal-dual method, we now return to the online setting. To add structure to the solution—such as sparsity or smoothing—the update step is extended by a regularizer. This leads us to Xiao's (2010) Regularized Dual Averaging (RDA) method, which transfers Nesterov's dual averaging approach to the online learning setting and explicitly takes regularization into account. The central update step is

$$
w_{t+1} = \arg\min_{w} \left\{ \frac{1}{t} \sum_{\tau=1}^{t} \langle g_\tau, w \rangle + \varphi(w) + \frac{\beta_t}{t} \psi(w) \right\}, \tag{125}
$$

where g_τ is the subgradient of the loss function observed in previous iterations, $\varphi(w)$ is the regularization term, and $\psi(w)$ is a strongly convex prox function.
In contrast to the classical primal-dual gradient method, the RDA approach thus takes the entire regularization term directly into account within the minimization, instead of merely using its subgradient. This integrates earlier gradient information in the form of its average $\frac{1}{t}\sum_{\tau=1}^{t} g_\tau$ and simultaneously stabilizes it through the regularizer. The additional term $(\beta_t/t)\,\psi(w)$ acts as an adaptive damping or smoothing parameter that controls the convergence and stability of the method. Below, we specify the RDA algorithm

Algorithm 12: Regularized dual averaging method (RDA)

Data: convex closed set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$, prox function ψ, regularizer φ
Initialize: $s_0 = 0 \in \mathcal{K}^*$, $\beta_1 = 1$
for *t=1 to T* **do**
  Compute $g_t \in \partial f_t(w_t)$
  Update dual average: $s_t = \frac{t-1}{t} s_{t-1} + \frac{1}{t} g_t$
  Update: $\beta_{t+1} = \beta_t + \frac{1}{\beta_t}$
  Update primal: $w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\{ \langle s_t, w \rangle + \varphi(w) + \frac{\beta_{t+1}}{t} \psi(w) \right\}$
end
return w_T

We now want to start a regret analysis.

Theorem 5.2

Let the sequences $\{w_t\}$ and $\{g_t\}$ be generated by Algorithm 12. Furthermore let G defined as (22) and ψ strong convex with parameter σ. Define

$$c_t = \beta_t D + \frac{G^2}{2} + \sum_{\tau=1}^{t-1} \frac{1}{\sigma\tau + \beta_\tau} \frac{G^2}{2}.$$

Then for any $w \in \mathcal{F}_D$(110), we have

- The regret is bounded by

$$R_t(w) \leq c_t.$$

- The primal variables are bounded as

$$\|w_{t+1} - w\|^2 \leq \frac{2}{\sigma t + \beta_t}(c_t - R_t(w)).$$

We refer to the proof in Theorem 1 and Appendix B in [11]. We will refrain from providing proof because the derivations are very closely based on those in Nesterov's paper [10].

5.5 Diagonal AdaGrad with RDA and Regret Analysis

We now combine the update step from AdaGrad with diagonal matrices with the update step from RDA and obtain the update step

$$w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\{ \eta \left\langle \frac{1}{t} \sum_{\tau=1}^{t} g_\tau, w \right\rangle + \eta\varphi(w) + \frac{1}{t}\psi_t(w) \right\}, \tag{126}$$

where $\psi_t(w) = \frac{1}{2}\langle w, h_t \odot w \rangle$. This gives us the following pseudocode

Algorithm 13: Diagonal AdaGrad with RDA

Data: convex closed set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$,step size $\eta > 0$,$\delta \geq 0$, regularizer φ
Initialize: $w_1 = 0 \in \mathbb{R}^n$ with $\varphi(w_1) = 0$, $\tilde{s}_0 = 0 \in \mathbb{R}^n$, $\bar{g} = 0 \in \mathbb{R}^n$
for *t=1 to T* **do**
  Observe convex loss f_t and compute subgradient $g_t \in \partial f_t(w_t)$
  Update: $\tilde{s}_t = \tilde{s}_{t-1} + g_t \odot g_t$
  Update: $\bar{g} = \bar{g} + g_t$
  $s_t = (\sqrt{(\tilde{s}_t)_1}, \ldots \sqrt{(\tilde{s}_t)_n})^\top$
  Set: $h_t = \delta 1_n + s_t$, $\psi_t(w) = \frac{1}{2}\langle w, h_t \odot w \rangle$
  Update: $w_{t+1} = \arg\min_{w \in \mathcal{K}} \left\{ \eta \left\langle \frac{1}{t}\bar{g}, w \right\rangle + \eta\varphi(w) + \frac{1}{t}\psi_t(w) \right\}$
end
return w_T

Remark:

- The remarks by Algorithm 9 applies here as well.

The following theorem gives an initial estimate of regret.

Theorem 5.3

Let the sequence $\{w_t\}_{t=1}^T$ be defined by the update in Algorithm 13 and let ψ_t strongly convex with respect to a norm $\|\cdot\|_{\psi_t}$. Furthermore let $\|\cdot\|_{\psi_t^*}$ be the dual norm of

$$\psi_t^*(g) = \sup_{w\in\mathcal{K}} \Big\{ \langle g, w\rangle - t\,\varphi(w) - \tfrac{1}{\eta}\psi_t(w) \Big\}. \tag{127}$$

For any $w^* \in \mathcal{K}$ holds

$$R_T^{\text{RDA Diag AdaGrad}} = \sum_{t=1}^T f_t(w_t) + \varphi(w_t) - f_t(w^*) - \varphi(w^*) \le \frac{1}{\eta}\psi_T(w^*) + \frac{\eta}{2}\sum_{t=1}^T \|g_t\|_{\psi_t^*}^2, \tag{128}$$

where $g_t \in \partial f_t(w_t)$.

Proof: Since ψ_t/η is $1/\eta$-strongly convex with respect to the norm $\|\cdot\|_{\psi_t}$, similar to the proof of Lemma 5.1, we obtain that the function ψ_t^* has η-Lipschitz continuous gradients with respect to $\|\cdot\|_{\psi_t^*}$

$$\|\nabla\psi_t^*(g_1) - \nabla\psi_t^*(g_2)\|_{\psi_t} \le \eta\|g_1 - g_2\|_{\psi_t^*} \tag{129}$$

for any g_1, g_2. Further,

$$\nabla\psi_t^*(g) = \arg\min_{w\in\mathcal{K}} \left\{ -\langle g, w\rangle + t\varphi(w) + \frac{1}{\eta}\psi_t(w) \right\}. \tag{130}$$

For a more detailed derivation and proof, we refer to Theorem 1 of [12]. Let $g_t \in \partial f_t(w_t)$ and defining $z_t = \sum_{\tau=1}^t g_\tau$, we have

$$\begin{aligned}
\sum_{t=1}^T f_t(w_t) + \varphi(w_t) - f_t(w^*) - \varphi(w^*) &\overset{(47)}{\le} \sum_{t=1}^T \langle g_t, w_t - w^*\rangle - \varphi(w^*) + \varphi(w_t) \\
&\le \sum_{t=1}^T \langle g_t, w_t\rangle + \varphi(w_t) + \sup_{w\in\mathcal{K}} \left\{ -\sum_{t=1}^T \langle g_t, w\rangle - T\varphi(w) - \frac{1}{\eta}\psi_T(w) \right\} + \frac{1}{\eta}\psi_T(w^*) \\
&\overset{(127)}{=} \frac{1}{\eta}\psi_T(w^*) + \sum_{t=1}^T \langle g_t, w_t\rangle + \varphi(w_t) + \psi_T^*(-z_T).
\end{aligned}$$

By defintion $\psi_{t+1} \ge \psi_t$, it is clear that

$$\begin{aligned}
\psi_T^*(-z_T) &\overset{(127)}{=} -\sum_{t=1}^T \langle g_t, w_{T+1}\rangle - T\varphi(w_{T+1}) - \frac{1}{\eta}\psi_T(w_{T+1}) \\
&\overset{(\psi_{t+1}\ge\psi_t)}{\le} -\sum_{t=1}^T \langle g_t, w_{T+1}\rangle - (T-1)\varphi(w_{T+1}) - \varphi(w_{T+1}) - \frac{1}{\eta}\psi_{T-1}(w_{T+1}) \\
&\le \sup_{w\in\mathcal{K}} \left(-\langle z_T, w\rangle - (T-1)\varphi(w) - \frac{1}{\eta}\psi_{T-1}(w) \right) - \varphi(w_{T+1}) \overset{(127)}{=} \psi_{T-1}^*(-z_T) - \varphi(w_{T+1}).
\end{aligned} \tag{131}$$

The Lipschitz continuity of $\nabla\psi_t^*$ and the fact that $z_T - z_{T-1} = -g_T$ give

$$\begin{aligned}
\sum_{t=1}^T f_t(w_t) + \varphi(w_{t+1}) - f_t(w^*) - \varphi(w^*) &\le \frac{1}{\eta}\psi_T(w^*) + \sum_{t=1}^T \langle g_t, w_t\rangle + \varphi(w_{t+1}) + \psi_{T-1}^*(-z_T) - \varphi(w_{T+1}) \\
&\overset{(5),(129)}{\le} \frac{1}{\eta}\psi_T(w^*) + \sum_{t=1}^T \langle g_t, w_t\rangle + \varphi(w_{t+1}) - \varphi(w_{T+1}) + \psi_{T-1}^*(-z_{T-1}) - \langle \nabla\psi_{T-1}^*(-z_{T-1}), g_T\rangle + \frac{\eta}{2}\|g_T\|_{\psi_{T-1}^*}^2 \\
&\overset{(w_T=\nabla\psi_{T-1}^*(-z_{T-1}))}{=} \frac{1}{\eta}\psi_T(w^*) + \sum_{t=1}^{T-1} \langle g_t, w_t\rangle + \varphi(w_{t+1}) + \psi_{T-1}^*(-z_{T-1}) + \frac{\eta}{2}\|g_T\|_{\psi_{T-1}^*}^2.
\end{aligned} \tag{132}$$

We can repeat the sequence of steps from (131) to (132) and get

$$\sum_{t=1}^{T} f_t(w_t) + \varphi(w_{t+1}) - f_t(w^*) - \varphi(w^*) \leq \frac{1}{\eta}\psi_T(w^*) + \frac{\eta}{2}\sum_{t=1}^{T} \|g_t\|^2_{\psi^*_{t-1}} + \psi_0^*(-z_0).$$

Recalling that $w_1 = \arg\min_{w\in\mathcal{K}}\{\varphi(w)\}$ and that $\psi_0^*(0) = 0$ completes the proof. □

Theorem 5.4

Let the sequences $\{w_t\}_{t=1}^T$ and $\{s_t\}_{t=1}^T$ be defined by the update in Algorithm 13. Then for any $w^* \in \mathcal{K}$

$$R_T^{\text{RDA Diag Adagrad}} \leq \frac{\delta}{2\eta}\|w^*\|_2^2 + \frac{1}{2\eta}\|w^*\|_\infty^2\|s_T\|_1 + \eta\|s_T\|_1. \tag{133}$$

Furthermore set $\gamma_T = \|s_T\|_1$ and $\eta = \frac{\|w^*\|_\infty}{\sqrt{2}}$, then we have for any $w^* \in \mathcal{K}$

$$R_T^{\text{RDA Diag Adagrad}} \leq \sqrt{2}\|w^*\|_\infty\gamma_T + \frac{\delta}{\sqrt{2}}\frac{\|w^*\|_2^2}{\|w^*\|_\infty} \leq \sqrt{2}\|w^*\|_\infty\gamma_T + \frac{\delta}{\sqrt{2}}\|w^*\|_1. \tag{134}$$

Proof: We reproduce the proof from Theorem 5 in [3]. Be careful, we have slightly other results because we take the factor $\frac{1}{2}$ in count.
It holds for any $w^* \in \mathcal{K}$

$$\begin{aligned}\psi_T(w^*) &= \frac{1}{2}\langle w^*, \big(\delta I_n + \text{Diag}(s_T)\big)w^*\rangle = \frac{\delta}{2}\|w^*\|_2^2 + \frac{1}{2}\langle w^*, \text{Diag}(s_T)w^*\rangle \\ &\overset{(13)}{\leq} \frac{\delta}{2}\|w^*\|_2^2 + \frac{1}{2}\|w^*\|_\infty^2\|s_T\|_1 \end{aligned} \tag{135}$$

Now plug (97) and (135) into (128) to obtain (133). Define $\gamma_T = \|s_T\|_1$ and $\eta = \frac{\|w^*\|_\infty}{\sqrt{2}}$ and plug this in (133). Use $\frac{\|w^*\|_2^2}{\|w^*\|_\infty} \leq \|w^*\|_1$ and obtain (134). This choice of η requires knowledge of w^* and is used only to establish the theoretical bound □

5.6 Full AdaGrad with RDA and Regret Analysis

We now combine the update step from AdaGrad with full matrices with the update step from RDA and obtain the update step

$$w_{t+1} = \arg\min_{w\in\mathcal{K}}\left\{\eta\Big\langle\frac{1}{t}\sum_{\tau=1}^{t} g_\tau, w\Big\rangle + \eta\varphi(w) + \frac{1}{t}\psi_t(w)\right\}, \tag{136}$$

where $\psi_t(w) = \frac{1}{2}\langle w, H_t w\rangle$. This gives us the following pseudocode

Algorithm 14: Full AdaGrad with RDA

Data: convex closed set $\mathcal{K}$, iteration number $0 < T \in \mathbb{N}$,step size $\eta > 0$, $\delta \geq 0$,regularizer φ
Initialize: $0 = w_1 \in \mathbb{R}^n$, $0 = G_0 \in \mathbb{R}^{n\times n}$, $0 = \bar{g} \in \mathbb{R}^n$
for *t=1 to T* **do**
  Observe convex loss f_t and compute subgradient $g_t \in \partial f_t(w_t)$
  Update: $G_t = G_{t-1} + g_t g_t^\top$, $S_t = G_t^{\frac{1}{2}}$
  Update: $\bar{g} = \bar{g} + g_t$
  Set: $H_t = \delta I_n + S_t$, $\psi_t(w) = \frac{1}{2}\langle w, H_t w\rangle$
  Update: $w_{t+1} = \arg\min_{w\in\mathcal{K}}\left\{\eta\Big\langle\frac{1}{t}\bar{g}, w\Big\rangle + \eta\varphi(w) + \frac{1}{t}\psi_t(w)\right\}$
end
return w_T

We would like to point out that (128) also applies to Algorithm 14. We use it to prove the following theorem.

Theorem 5.5

Let G_t be the outer product matrix from Algorithm 14 and the sequence $\{w_t\}_{t=1}^T$ produced by Algorithm 14. If $\delta \geq 0$, then for any $w^* \in \mathcal{K}$

$$R_T^{\text{RDA Full Adagrad}} \leq \frac{\delta}{2\eta}\|w^*\|_2^2 + \frac{1}{2\eta}\|w^*\|_2^2 \text{Tr}(G_T^{1/2}) + \eta \text{Tr}(G_T^{1/2}). \tag{137}$$

Furthermore set $\eta = \frac{\|w^*\|_2}{\sqrt{2}}$, we have

$$R_T^{\text{RDA Full Adagrad}} \leq \sqrt{2}\|w^*\|_2 \text{Tr}(G_T^{1/2}) + \frac{\delta}{\sqrt{2}}\|w^*\|_2. \tag{138}$$

Proof: We reproduce the proof from Theorem 7 in [3]. Be careful, we have slightly other results because we take the factor $\frac{1}{2}$ in count.
It holds for any $w^* \in \mathcal{K}$

$$\begin{aligned}
R_T &\overset{(128)}{\leq} \frac{1}{\eta}\psi_T(w^*) + \frac{\eta}{2}\sum_{t=1}^T \|g_t\|_{\psi_t^*}^2 = \frac{\delta}{2\eta}\|w^*\|_2^2 + \frac{1}{2\eta}\langle w^*, S_t w^*\rangle + \frac{\eta}{2}\sum_{t=1}^T \|g_t\|_{\psi_t^*}^2 \\
&\overset{(11),(12)}{\leq} \frac{\delta}{2\eta}\|w^*\|_2^2 + \frac{1}{2\eta}\|w^*\|_2^2 \text{Tr}(S_t) + \frac{\eta}{2}\sum_{t=1}^T \|g_t\|_{\psi_t^*}^2 \\
&\overset{(105)}{\leq} \frac{\delta}{2\eta}\|w^*\|_2^2 + \frac{1}{2\eta}\|w^*\|_2^2 \text{Tr}(G_t^{1/2}) + \eta \text{Tr}(G_T^{1/2})
\end{aligned}$$

This gives us (137). Set $\eta = \frac{\|w^*\|_2}{\sqrt{2}}$ and obtain (138). □

6 Derived Algorithms using concrete regularization

Chapters 4 and 5 developed the AdaGrad methods in the general framework of Composite Mirror Descent (COMID) and Regularized Dual Averaging (RDA). The resulting updates depend on three components

1. the sequence of loss functions f_t,
2. the *regularizer* φ,
3. the convex decision set $\mathcal{K} \subseteq \mathbb{R}^n$.

For all AdaGrad variants considered in this work, the proximal term is already fixed. Following [3], we use the time-dependent quadratic form

$$\psi_t(w) \;=\; \frac{1}{2}\,\langle w, H_t w\rangle,$$

where $H_t \succ 0$ is the diagonal AdaGrad matrix constructed from past gradients. The update rules in Chapters 4 and 5 are therefore fully determined once the choices of the regularizer φ and the feasible set $K \subset \mathbb{R}^n$ are specified.
In this chapter, we make these choices explicit and obtain several concrete AdaGrad variants that differ in their geometry and in the resulting proximal or projection operators.

6.1 Choice of the Regularizer and Feasible Set

We consider four types of regularizers that commonly arise in practical applications

1. **No explicit regularization**
$$\varphi(w) \equiv 0,$$
which yields "pure" AdaGrad over the convex set $\mathcal{K}$.

2. **ℓ_1-regularization**
$$\varphi_1(w) \;=\; \lambda\,\|w\|_1,$$
which results to LASSO (least absolute shrinkage and selection operator).

3. **ℓ_2-regularization**
$$\varphi_2(w) \;=\; \lambda\,\|w\|_2.$$

For the feasible region $\mathcal{K}$, we consider the following convex sets

1. **Unconstrained domain**
$$\mathcal{K} = \mathbb{R}^n.$$

2. **Box constraints**
$$\mathcal{K} \;=\; \{\, w \in \mathbb{R}^n : l_i \le w_i \le r_i \,\},$$

3. **ℓ_2-ball**
$$\mathcal{K} \;=\; \{\, w \in \mathbb{R}^n : \|w\|_2 \le R \,\},$$

4. **ℓ_1-ball**
$$\mathcal{K} \;=\; \{\, w \in \mathbb{R}^n : \|w\|_1 \le R \,\},$$

Substituting these choices of φ and K into the general AdaGrad updates from Chapters 4 and 5 leads to optimization problems of the form

$$w_{t+1} \;=\; \arg\min_{w\in K}\Big\{\langle u_t, w\rangle + \eta\varphi(w) + \tfrac{1}{2}\,\langle w, H_t^x w\rangle\Big\},$$

where the vector u_t and the matrix H_t^x depends on whether the update is based on COMID or RDA. Using some simple calculations, we conclude that for COMID

$$u_t = \eta g_t - H_t^x w_t \qquad \text{and} \qquad H_t^x = H_t \tag{139}$$

and for the RDA update, we get

$$u_t = \frac{\eta}{t}\sum_{i=1}^{t} g_i \qquad \text{and} \qquad H_t^x = \frac{1}{t}H_t. \tag{140}$$

We would like to note that the scientific paper [3] did not take step sizes (η and t) into account, and we have decided to incorporate this more precisely. For this reason our update formula slightly differ from [3]. For the following we set

$$f_\varphi(w) := \langle u_t, w\rangle + \eta\varphi(w) + \tfrac{1}{2}\langle w, H_t^x w\rangle \qquad \text{and} \qquad h_t^x = \operatorname{diag}(H_t^x).$$

6.2 Formula for $\mathcal{K} = \mathbb{R}^n$

We now consider the problem

$$w_{t+1} = \arg\min_{w\in\mathbb{R}^n}\left\{\langle u_t, w\rangle + \eta\varphi(w) + \tfrac{1}{2}\langle w, H_t^x w\rangle\right\}$$

and receive the following

1. **No explicit regularization** $\varphi(w) \equiv 0$: This simplifies to the normal AdaGrad update without regularizer with update formula

$$w_{t+1} = -(H_t^x)^{-1}u_t.$$

2. **ℓ_1-regularization** $\varphi_1(w) = \lambda\|w\|_1$: Here we give the solution for diagonal H_t. In the full-matrix case, there is no simple closed form. We obtain

$$\partial f_{\varphi_1}(w) = u_t + \eta\lambda\partial||w||_1 + H_t^x w,$$

where

$$\partial|w_i| = \begin{cases} 1, & \text{if } w_i > 0,\\ -1, & \text{if } w_i < 0,\\ [-1,1] & \text{if } w_i = 0.\end{cases}$$

We will examine the problem component by component.
Case $w_i > 0$: It follows

$$(\partial f_{\varphi_1}(w))_i = u_i + \eta\lambda + (h_t^x)_i w_i,$$

what leads to

$$(w_{t+1})_i = -\frac{u_i + \eta\lambda}{(h_t^x)_i}.$$

Since $(h_t^x)_i > 0$, $\eta > 0$ it follows that $u_i + \eta\lambda < 0$, so $u_i < -\eta\lambda$.
Case $w_i < 0$: It follows

$$(\partial f_{\varphi_1}(w))_i = u_i - \eta\lambda + (h_t^x)_i w_i,$$

what leads to

$$(w_{t+1})_i = -\frac{u_i - \eta\lambda}{(h_t^x)_i}.$$

Since $(h_t^x)_i > 0$, $\eta > 0$ it follows that $u_i - \eta\lambda > 0$, so $u_i > \eta\lambda$.
Case $w_i = 0$: It follows

$$(\partial f_{\varphi_1}(w))_i \in u_i + \eta\lambda[-1,1],$$

For optimality, the following must hold

$$-u_i \in \eta\lambda[-1,1]$$

Hence $w_i = 0$ is optimal for $|u_i| \le \eta\lambda$.
With these case distinctions, we obtain

$$(w_{t+1})_i = \frac{1}{(h_t^x)_i}\operatorname{sign}(-u_i)\max(|u_i| - \eta\lambda, 0) \tag{141}$$

3. ℓ_2**-regularization** $\varphi_2(w) = \lambda \|w\|_2$: Define

$$f(w) = u_t^\top w + \eta\lambda\|w\|_2 + \frac{1}{2} w^\top H_t^x w.$$

The first-order optimality (subgradient) condition is

$$0 \in u_t + H_t^x w + \eta\lambda\, \partial\|w\|_2.$$

Case $w_{t+1} = 0$: It holds $\partial\|0\|_2 = \{s \in \mathbb{R}^n : \|s\|_2 \leq 1\}$. Thus $w = 0$ is optimal if and only if

$$\exists s,\ \|s\|_2 \leq 1 : \quad 0 = u + \eta\lambda s \quad \Longleftrightarrow \quad u = -\eta\lambda s \quad \Longleftrightarrow \quad \|u\|_2 \leq \eta\lambda.$$

Hence

$$\|u\|_2 \leq \eta\lambda \implies w_{t+1} = 0.$$

Case $w_{t+1} \neq 0$: For $w \neq 0$ we have $\partial\|w\|_2 = \left\{\frac{w}{\|w\|_2}\right\}$, so the optimality condition becomes

$$H_t^x w + u + \eta\lambda \frac{w}{\|w\|_2} = 0.$$

Introduce the scalar

$$\beta := \frac{\eta\lambda}{\|w\|_2} > 0.$$

Rearranging yields

$$\Big(H_t^x + \beta I\Big) w = -u_t.$$

Since $\beta > 0$ is $H_t^x + \beta I$ not singular, this gives a closed form:

$$w = -\Big(H_t^x + \beta I\Big)^{-1} u_t.$$

The remaining unknown $\beta > 0$ is determined by consistency with its definition. Plugging into $\beta = \frac{\eta\lambda}{\|w\|_2}$ gives

$$\beta = \frac{\eta\lambda}{\|(H_t^x + \beta I)^{-1} u_t\|_2}$$

In the regime $\|u_t\|_2 > \eta\lambda$, this scalar equation has a unique solution $\beta > 0$ (and thus $w_{t+1} \neq 0$). In summary, we obtain

$$w_{t+1} = \begin{cases} 0, & \|u_t\|_2 \leq \eta\lambda, \\ -\Big(H_t^x + \beta I\Big)^{-1} u_t, & \|u\|_2 > \eta\lambda, \end{cases}$$

where $\beta > 0$ is the unique solution of

$$0 = \frac{\eta\lambda}{\beta} - \left\| \Big(H_t^x + \beta I\Big)^{-1} u_t \right\|_2.$$

To find the optimal $\beta > 0$, a bisection method is proposed in [3]. We try a different approach and determine $\beta > 0$ using the Newton method, similar to how one would solve a trust region problem. We refer to Moré und Sorensen in [6]. Define

$$\theta(\beta) = \frac{\beta}{\eta\lambda} - \frac{1}{\|(H_t^x + \beta I)^{-1} u_t\|_2} \tag{142}$$

which has the following derivative

$$\theta'(\beta) = \frac{1}{\lambda\eta} - \frac{\|(H_t^x + \beta I)^{-3/2} u_t\|_2^2}{\|(H_t^x + \beta I)^{-1} u_t\|_2^3}.$$

The function $-\theta$ is asymptotically linear, concave, and grows strictly monotonically, so the Newton method should converge quickly from a valid starting point. In practice, three to four Newton steps are sufficient to approximate β sufficiently well. If the function value increases during the course of the Newton step, this means that the previous Newton step was too long and must be discarded and repeated with a step size damping $c \in (0, 1)$.

6.3 Box constraints

We now consider the problem

$$w_{t+1} = \arg\min_{w\in\mathbb{R}^n : l_i \le w_i \le r_i} \Big\{ \langle u_t, w\rangle + \eta\varphi(w) + \tfrac{1}{2}\langle w, H_t^x w\rangle \Big\}.$$

Fortunately, we can use the previous results and we can apply the unconstrained solution followed by component-wise clipping. To do this, define

$$\Pi_{[l_i,r_i]}(w_i) = \min\big(\max(w_i, l_i), r_i\big),$$

in vector form, taking the maximum and minimum entry by entry

$$\Pi_{[l,r]}(w) = \min\big(\max(w, l), r\big),$$

where $l, r, w \in \mathbb{R}^n$ and $l_i < r_i$ for all i. We obtain

1. **No explicit regularization** $\varphi(w) \equiv 0$: This simplifies to
$$w_{t+1} = \Pi_{[l,r]}(-(H_t^x)^{-1}u_t).$$
2. ℓ_1**-regularization** $\varphi_1(w) = \lambda\|w\|_1$: Here, too, we can cut off the unbounded case
$$(w_{t+1})_i = \Pi_{[l_i,r_i]}\Big(\frac{1}{(h_t^x)_i}\mathrm{sign}(-u_i)\max(|u_i| - \eta\lambda, 0)\Big)$$
for all $i \in \{1, 2, ..., n\}$.
3. ℓ_2**-regularization**: $\varphi_2(w) = \lambda\|w\|_2$: Here, too, we can cut off the unbounded case
$$w_{t+1} = \begin{cases} 0, & \|u_t\|_2 \le \eta\lambda, \\ \Pi_{[l,r]}\Big(-(H_t^x + \beta I)^{-1}u_t\Big), & \|u_t\|_2 > \eta\lambda, \end{cases}$$
where $\beta > 0$ is the unique solution of
$$0 = \frac{\eta\lambda}{\beta} - \Big\|\Big(H_t^x + \beta I\Big)^{-1}u_t\Big\|_2.$$

6.4 ℓ_2-ball

We now consider the problem

$$w_{t+1} = \arg\min_{w\in\mathbb{R}^n : \|w\|_2 \le R} \Big\{ \langle u_t, w\rangle + \eta\varphi(w) + \tfrac{1}{2}\langle w, H_t^x w\rangle \Big\}.$$

Fortunately, we can use the previous results and we can apply the unconstrained solution followed by component-wise clipping. To do this, define

$$\Pi_{\ell_2,R}(w) = \begin{cases} w, & \text{if} \quad \|w\|_2 \le R \\ R\frac{w}{\|w\|_2}, & \text{if} \quad \|w\|_2 > R \end{cases}.$$

We obtain

1. **No explicit regularization** $\varphi(w) \equiv 0$: This simplifies to
$$w_{t+1} = \Pi_{\ell_2,R}(-(H_t^x)^{-1}u_t).$$
2. ℓ_1**-regularization** $\varphi_1(w) = \lambda\|w\|_1$: Calculate
$$(w_{t+1})_i = \frac{1}{(h_t^x)_i}\mathrm{sign}(-u_i)\max(|u_i| - \lambda\eta, 0)$$
for all $i \in \{1, 2, ..., n\}$ and create the resulting vector $w = ((w_{t+1})_1, (w_{t+1})_2, ..., (w_{t+1})_n)^\top$. We obtain then $w_{t+1} = \Pi_{\ell_2,R}(w)$.
3. ℓ_2**-regularization**: $\varphi_2(w) = \lambda\|w\|_2$: Here, too, we can cut off the unbounded case
$$w_{t+1} = \begin{cases} 0, & \|u_t\|_2 \le \eta\lambda, \\ \Pi_{\ell_2,R}\Big(-(H_t^x + \beta I)^{-1}u\Big), & \|u_t\|_2 > \eta\lambda, \end{cases}$$
where $\beta > 0$ is the unique solution of
$$0 = \frac{\eta\lambda}{\beta} - \Big\|\Big(H_t^x + \beta I\Big)^{-1}u_t\Big\|_2.$$

6.5 l_1-ball

We consider the problem

$$w_{t+1} = \arg\min_{\|w\|_1 \le R} \Big\{ \langle u_t, w\rangle + \eta\lambda\|w\|_1 + \frac{1}{2}\langle w, H_t^x w\rangle \Big\},$$

where $H_t = \mathrm{diag}(h_1, \dots, h_n) \succ 0$ is diagonal. There is no simple projection as for the ℓ_2-ball. For the ℓ_2-ball, the Euclidean projection has a closed form because the ℓ_2-ball is rotationally symmetric and the Euclidean distance is induced by the same geometry. In contrast, the ℓ_1-ball is a polytope with corners (not rotationally symmetric), hence the Euclidean projection onto $\{w : \|w\|_1 \le R\}$ is not obtained by uniform scaling, instead it is given by a (global) soft-thresholding rule with a threshold chosen to satisfy the ℓ_1-constraint.

Introduce a multiplier $\nu \ge 0$ for the constraint $\|w\|_1 \le R$ and consider

$$\min_w \langle u_t, w\rangle + (\eta\lambda + \nu)\|w\|_1 + \frac{1}{2}\langle w, H_t^x w\rangle.$$

At the optimum $(w^\star, \nu^\star)$, the KKT conditions hold:

$$\|w^\star\|_1 \le R, \qquad \nu^\star \ge 0, \qquad \nu^\star(\|w^\star\|_1 - R) = 0.$$

Since H_t^x is diagonal, the problem separates over coordinates (similar to the calculation steps in (141)). For fixed ν,

$$w_i(\nu) = -\frac{1}{h_{ii}} \,\mathrm{sign}(u_i)\, \max\big(|u_i| - (\eta\lambda + \nu),\, 0\big), \qquad i = 1, \dots, n.$$

Define

$$g(\nu) := \|w(\nu)\|_1 = \sum_{i=1}^{d} \frac{1}{(h_t^x)_i} \max\big(|u_i| - (\lambda\eta + \nu), 0\big).$$

and the abbreviations

$$a_i := \frac{1}{(h_t^x)_i} > 0, \qquad b_i := |u_i| - \eta\lambda.$$

Then

- If $g(0) \le R$, the constraint is inactive and $\nu^\star = 0$, hence $w^\star = w(0)$.
- If $g(0) > R$, the constraint is active and $\nu^\star > 0$ is the unique value such that $g(\nu^\star) = R$.

The function $g(\nu)$ is continuous and piecewise linear, since it is a finite sum of hinge functions $\max(b_i - \nu, 0)$. Moreover, $g(\nu)$ is monotone non-increasing in ν, for any $\nu_1 \le \nu_2$ and any i we have

$$\max(b_i - \nu_1, 0) \ge \max(b_i - \nu_2, 0),$$

and thus $g(\nu_1) \ge g(\nu_2)$. Hence, whenever a solution exists, it is unique on any interval where g is strictly decreasing. This monotonicity and continuity make bisection a robust choice for computing ν.

Assuming we search for $\nu \ge 0$, a natural lower bound is

$$\nu_{\mathrm{lo}} := 0.$$

An upper bound can be chosen such that $g(\nu_{\mathrm{hi}}) = 0$. Since $\max(b_i - \nu, 0) = 0$ for all i whenever $\nu \ge \max_i b_i$, we may set

$$\nu_{\mathrm{hi}} := \max_{1 \le i \le n} b_i.$$

Then $g(\nu_{\mathrm{hi}}) = 0$ and the solution (if it exists) must lie in the interval $[\nu_{\mathrm{lo}}, \nu_{\mathrm{hi}}]$.

Note that $g(\nu) \ge 0$ for all ν and that $g(\nu) \to 0$ as $\nu \to +\infty$. Therefore:

- If $R > g(0)$, the constraint is inactive and $\nu^* = 0$.
- If $R = 0$, then any $\nu \ge \nu_{\mathrm{hi}}$ is $\nu^\star$.
- If $0 < R \le g(0)$, then there exists at least one solution in $[0, \nu_{\mathrm{hi}}]$.

7 Numerical Experiments

In this chapter, we apply the AdaGrad variants developed in Chapters 4 and 5 to a real-world online prediction problem, click-through rate (CTR) estimation in digital advertising. We begin by motivating the application domain and describing the dataset. We then introduce the prediction model, the loss function, and the evaluation metrics used. Finally, we specify the experimental protocol and the configurations under comparison.

7.1 Application Context: Click-Through Rate Prediction

In the digital advertising industry, a central problem is to predict whether a user will click on a displayed advertisement. Each time a webpage is loaded, an ad impression is generated, and the system must decide in real time which advertisement to display. The probability that a user clicks on a given ad, the *click-through rate* (CTR), directly determines the economic value of the impression to the advertiser.

The prediction of CTR is a binary classification problem, given a set of contextual features (the website, the user's device, the time of day, etc.), the model outputs a probability $p \in [0, 1]$ that a click will occur. This prediction must be made *before* the outcome is observed, and the model is updated *after* the label (click or no click) is revealed, precisely the online learning setting formalized in Chapter 2.

Automated CTR prediction systems are of significant practical importance. Advertising platforms process billions of impressions per day, making manual analysis infeasible. The ability to learn and adapt in real time, from a sequential stream of data, aligns naturally with the Online Convex Optimization framework and the adaptive subgradient methods studied in this thesis.

It may seem that a stream of billions of impressions, spread over many websites and many users, could equally well be collected and processed offline. Three properties of the application rule this out. First, decision and label are separated in time, the system must output a probability before the impression is served, while the label becomes available only seconds to minutes later. Second, the data are non-stationary, that means campaigns, ad inventory and user populations change continuously and so a model fitted on a fixed historical window begins to degrade as soon as it is deployed. Third, the sheer volume makes repeated retraining from scratch on the accumulated history prohibitively expensive, whereas an online method processes each sample once and with constant memory.

7.2 The Avazu Dataset

We use the Avazu CTR dataset, originally published as part of a prediction competition on the Kaggle platform[1]. The dataset contains approximately 40 million ad impressions collected over 10 consecutive days in October 2014. Each record contains a binary label (`click` $\in \{0, 1\}$) and a set of categorical features describing the context of the impression.

The features used in our experiments are:

Feature	Description	Unique values
`banner_pos`	Position of the ad on the page	7
`site_domain`	Domain of the website	$\sim$2 000
`device_model`	Model of the user's device	$\sim$4 600
`device_type`	Device category (mobile, desktop, ...)	4
`device_conn_type`	Connection type (WiFi, cellular, ...)	4
`C14, C17, C18, C19, C21`	Anonymized categorical features	4–600
`hour_of_day`	Hour extracted from timestamp	24

All features are treated as categorical variables and encoded via the *feature hashing trick*, each (field, value) pair is mapped to an index in $\{0, \ldots, B-1\}$ using a deterministic hash function, where B is the hash table size per field. This representation is standard in large-scale prediction [14] and avoids the need for explicit vocabulary construction.

In our experiments we use $B = 5000$ hash buckets per feature field, production systems typically use tables that are several orders of magnitude larger. Note that $B <$ (number of distinct

[1] `https://www.kaggle.com/c/avazu-ctr-prediction`

values) is not an error but the intended trade-off. For m distinct values the expected number of occupied rows is

$$B(1 - (1 - 1/B)^m),$$

which for device_model ($m \approx 4600, B = 5000$) is about 3000, that means collisions are frequent by construction, and this is the reason why the active-row counts in Table 2 do not exactly reproduce the true cardinalities.

Class imbalance: The dataset exhibits a pronounced class imbalance, only approximately 17% of impressions result in a click. This means the majority of observed gradients correspond to non-click events. The adaptive per-coordinate learning rates of AdaGrad are particularly well-suited for this setting, as they automatically adjust to the different gradient magnitudes across features. Features that are rarely associated with clicks receive larger effective step sizes when they do appear.

Data splitting and chronological order. To reflect the online setting, we split the data chronologically, days 1-9 are used for sequential training (one day at a time, in order), and day 10 serves as the held-out test set. Within each day, the data is processed in its original temporal order, preserving the sequential structure that is fundamental to the online learning framework.

7.3 Prediction Model

Since the theoretical analysis in Chapters 4 and 5 is developed for convex loss functions, we briefly discuss the model architecture and note where the non-convex setting departs from the theory.

Embedding layer: Each of the d feature fields maps its hashed index to a dense vector $e_i \in \mathbb{R}^k$ via an embedding lookup

$$e_i = E_i[\, h(x_i)\,], \quad i = 1, \ldots, d,$$

where $E_i \in \mathbb{R}^{B \times k}$ is a learnable embedding matrix for field i, the map $h : \mathcal{V} \to \{0, \ldots, B-1\}$ is a hash function with $\mathcal{V}$ the set of possible values of a field, and k is the embedding dimension. Note that each field owns a separate table of B rows, so that collisions occur within a field but never across fields.

Multi-layer perceptron: The concatenated embedding vector $z = (e_1, \ldots, e_d) \in \mathbb{R}^{dk}$ is passed through a multi-layer perceptron (MLP):

$$f_w(x) = W_L\, \sigma(\cdots \sigma(W_1 z + b_1) \cdots) + b_L,$$

where σ denotes the ReLU activation function $\sigma(t) = \max(t, 0)$, and $w = (E_1, \ldots, E_d, W_1, b_1, \ldots, W_L, b_L)$ collects all learnable parameters.
The output $f_w(x) \in \mathbb{R}$ is a scalar logit, the predicted click probability is obtained via the sigmoid function

$$\hat{p}(x) = \frac{1}{1 + e^{-f_w(x)}}.$$

In our experiments $d = 11$ feature fields, $k = 4$, and the MLP has two hidden layers of widths $(32, 16)$, i.e. $L = 3$, so that $z \in \mathbb{R}^{44}$ and $n \approx 2.2 \cdot 10^5$, of which more than 99 % of the parameters belong to the embedding matrices. Typical values in the CTR literature are larger: $d \approx 20$–40 fields, $k \approx 8$–64, and two to four hidden layers of width 200–1024. We deliberately work at the lower end of this range, since the object of study is the behaviour of the optimizers and not the predictive ceiling of the architecture.

Remark on non-differentiability: The activation $\sigma(t) = \max(t, 0)$ is not differentiable at $t = 0$, so $f_w(x)$ is not a differentiable function of w, not even locally. The vectors g_t used by the algorithms of Chapters 4 and 5 are therefore subgradients rather than gradients, which is precisely the setting for which those methods were developed. In the implementation, automatic differentiation resolves the ambiguity at the kink by the convention $\sigma'(0) := 0$.

Remark on convexity: The composition of embeddings, ReLU activations, and linear layers makes $f_w(x)$ a non-convex function of the parameters w. The regret bounds established in Theorems 4.3

and 5.4 therefore do not directly apply. However, the application of adaptive subgradient methods to non-convex objectives is widespread in practice and consistently yields strong empirical results. Our experiments serve to evaluate the *relative* performance of the AdaGrad with COMID and RDA variants, rather than to verify the theoretical bounds.

7.4 Loss Function

We use the *binary cross-entropy loss* (also known as logistic loss), which for a single sample (x, y) with label $y \in \{0, 1\}$ and logit $f_w(x) \in \mathbb{R}$ is defined as:

$$\ell(w; x, y) = -\big[\, y \log \hat{p}(x) + (1 - y) \log(1 - \hat{p}(x)) \,\big] = \log\big(1 + e^{-f_w(x)}\big) + (1 - y)\, f_w(x). \tag{143}$$

This loss is convex in the logit $f_w(x)$ (though not jointly convex in w due to the non-linear model). It is bounded below by zero and achieves its minimum when $\hat{p}(x) = y$, with the usual convention $0 * \log(0) := 0$. For finite logits $\hat{p}(x) \in (0, 1)$ strictly, so this value is approached but never attained. In the online learning setting, at each round t, the learner predicts $\hat{p}_t = \hat{p}(x_t)$, observes the label y_t, and incurs loss $\ell_t(w_t) = \ell(w_t; x_t, y_t)$. The gradient $g_t = \nabla_w \ell_t(w_t)$ is then used by the optimizer to update the parameters.

7.5 Evaluation Metrics

We use two standard metrics for binary classification.

LogLoss (cross-entropy): The average binary cross-entropy over N test samples

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^{N} \big[\, y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \,\big].$$

LogLoss measures the quality of the predicted probabilities. A perfect model achieves LogLoss $= 0$, a model that always predicts the base rate achieves LogLoss $= -[r \log r + (1 - r) \log(1 - r)]$ where r is the click rate.

AUC (Area Under the ROC Curve): The AUC measures the model's ability to *rank* positive examples (clicks) above negative examples (non-clicks). Formally, for a set of predictions $\{(\hat{p}_i, y_i)\}_{i=1}^{N}$, the AUC equals the probability that a randomly chosen positive example receives a higher predicted probability than a randomly chosen negative example:

$$\text{AUC} = \mathbb{P}\big[\hat{p}_i > \hat{p}_j \mid y_i = 1,\, y_j = 0\big].$$

An AUC of 0.5 corresponds to random guessing, an AUC of 1.0 indicates perfect ranking. The AUC is threshold-independent and robust to class imbalance, making it the preferred metric for CTR prediction.

7.6 Experimental Protocol

Prequential evaluation: To track the model's generalization over time, we employ a *prequential* (predictive sequential) evaluation scheme. After training on day i, we evaluate the model on day $i + 1$ *before* training on it. This produces a sequence of AUC and LogLoss values that reflects how well the model generalizes to unseen future data at each point in time.

Baseline: We additionally evaluate the untrained model (randomly initialized weights) on the first day to establish a baseline. The initial AUC should be close to 0.5 (random ranking).

Algorithms under comparison:

1. *Diagonal AdaGrad with COMID* (Algorithm 9): The update follows the Composite Mirror Descent framework, where the regularizer φ is incorporated into the proximal step. The method performs a gradient step followed by a proximal operation, making it analogous to proximal gradient descent with adaptive step sizes.

2. *Diagonal AdaGrad with RDA* (Algorithm 13): The Regularized Dual Averaging variant accumulates the gradient sum $\bar{g}_t$ and recomputes the solution from scratch at each step. This "global" view of the gradient history tends to produce *sparser* solutions under ℓ_1-regularization than COMID Diagonal AdaGrad, because the thresholding operates on the accumulated average gradient rather than on a single proximal step.

3. *PyTorch AdaGrad* (`torch.optim.Adagrad`): The standard implementation, serving as a baseline to verify that our custom implementations produce comparable results in the unregularized case.

Configurations: Each optimizer is tested with the following regularization and constraint options (corresponding to the derived algorithms in Chapter 6)

Regularizer φ	Feasible set $\mathcal{K}$	Thesis reference
$\varphi \equiv 0$ (none)	$\mathbb{R}^n$ (unconstrained)	§6.2 case 1
$\lambda\|w\|_1$	$\mathbb{R}^n$	§6.2 case 2, eq. (141)
$\lambda\|w\|_2$	$\mathbb{R}^n$	§6.2 case 3, eq. (142)
$\lambda\|w\|_1$	$\{w : \|w\|_1 \leq R\}$	§6.5
$\varphi \equiv 0$	$\{w : \|w\|_2 \leq R\}$	§6.4

Empirical regret. To connect the experimental results to the theoretical analysis, we compute the empirical regret

$$R_T = \sum_{t=1}^{T} \ell_t(w_t) - \sum_{t=1}^{T} \ell_t(w_T),$$

where w_T denotes the final model after all T training steps. The quantity w_T serves as an approximation of the best fixed comparator $w^* = \arg\min_w \sum_{t=1}^{T} \ell_t(w)$. If the regret is sublinear, i.e., $R_T/T \to 0$ as $T \to \infty$, the online algorithm converges to the performance of the best fixed model in hindsight. We verify this empirically for both COMID and RDA.

The CTR prediction setting combines several properties that are favorable for adaptive subgradient methods":

- *Sparse, heterogeneous features:* High-cardinality categorical features (such as `site_domain` with $\sim$2 000 unique values) produce highly sparse gradient vectors. Most embedding rows receive zero gradients at each step. AdaGrad's per-coordinate learning rates ensure that rarely updated coordinates take larger steps when they do receive a gradient signal.

- *Class imbalance:* With only 17% positive labels, the gradient landscape is dominated by the non-click class. The adaptive scaling of AdaGrad automatically adjusts for this asymmetry.

- *Non-stationary data:* User behavior and ad inventory change over time. The online learning framework, processing data sequentially without revisiting past examples, naturally handles this temporal structure.

The complete source code for all experiments, including the custom optimizer implementations and evaluation scripts, is publicly available at
https://github.com/Eddie1337-code/adaptive-subgradient-methods.

7.7 Hyperparameter Selection

Before presenting the experimental results, we describe how the free parameters of the algorithms were chosen. The model architecture, embedding dimension $k = 4$, two hidden layers of sizes $(32, 16)$, and hash table size $B = 5\,000$ per feature field, was fixed for all experiments, yielding approximately 222 000 trainable parameters. We deliberately chose a compact architecture, since the focus of this thesis is the comparison of optimization algorithms rather than model design, a smaller model reduces computational cost and allows for a more systematic exploration of the algorithmic parameters.
The remaining degrees of freedom are the learning rate η, the regularization strength λ, and (where applicable) the constraint parameter R or c. These were selected through a structured grid search, using exclusively the first three days of data. Days 1 and 2 for training, day 3 for evaluation. This

separation ensures that no information from the test period (days 4–10) influences the parameter choice, a requirement that is particularly important in the online setting, where the temporal ordering of data carries meaning.

Step 1: Learning rate: The learning rate is the most influential hyperparameter, as it scales the entire update. We searched a logarithmic grid

$$\eta \in \{10^{-3}, 5 \cdot 10^{-3}, 10^{-2}, 5 \cdot 10^{-2}, 10^{-1}, 5 \cdot 10^{-1}\},$$

independently for COMID, RDA, and PyTorch AdaGrad, without regularization ($\varphi \equiv 0$, $\mathcal{K} = \mathbb{R}^n$). All three methods achieved their best AUC at $\eta = 0.1$. Notably, the RDA version diverged entirely at $\eta = 0.5$ (AUC ≈ 0.5, equivalent to random guessing), while the COMID version remained stable. This sensitivity can be understood from the RDA update, the direction vector $u_t = (\eta/t)\bar{g}_t$ scales linearly with η, and an excessively large η causes the solution $w = -(H_t^x)^{-1}u_t$ to overshoot, particularly in the early iterations when t is small and H_t^x has not yet accumulated sufficient curvature information.

Step 2: Regularization strength: With $\eta = 0.1$ fixed, we tuned λ over the grid

$$\lambda \in \{10^{-9}, 10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$$

for both ℓ_1- and ℓ_2-regularization. The results revealed a striking sensitivity of ℓ_1-regularization. At $\lambda = 10^{-2}$, the soft-thresholding operator $\max(|u_i| - \eta\lambda, 0)$ zeroed out all coordinates (since $\eta\lambda = 10^{-3}$ exceeded the typical gradient magnitudes), producing a trivial model with 100% sparsity and no predictive power. The optimal values were:

	$\lambda^*_{\ell_1}$	$\lambda^*_{\ell_2}$
COMID	10^{-8}	10^{-4}
RDA	10^{-6}	10^{-5}

The fact that RDA requires a smaller $\lambda^*_{\ell_2}$ than COMID Diag AdaGrad is consistent with the structure of the respective updates. In RDA Diag AdaGrad, the regularizer $\eta\varphi(w)$ is applied to a solution that is recomputed from the full gradient history, amplifying the effect of λ compared to COMID Diag AdaGrad incremental proximal step.

Step 3: Constraint parameters: Box bounds and ball radius were tuned with $\eta = 0.1$ and $\varphi \equiv 0$. The optimal box bound was $c = 2.0$ for both methods. For the ℓ_2-ball, COMID Diag AdaGrad required a large radius $R = 100$ (the unconstrained weight norms remained well below this threshold, rendering the constraint effectively inactive), while RDA Diag AdaGrad performed best at $R = 10$, reflecting its naturally smaller weight magnitudes due to the $1/t$ scaling in $H_t^x = H_t/t$.

7.8 Results-Overall Comparison

All experiments reported in this section use the full dataset without subsampling, mini-batch size 128, a single chronological pass over the data, and the hyperparameters selected in §7.7. Each configuration was trained on days 1–9 (approximately 36 million samples) and evaluated on day 10 (approximately 4.2 million samples).

Table 1 presents the test-day AUC for all 13 configurations, along with the wall-clock training time.

Computational cost: The training times in Table 1 reveal a clear pattern. Methods without regularization or with ℓ_1-regularization require 230–320 seconds, as their updates involve only element-wise operations (soft-thresholding for ℓ_1, division for the unregularized case). The ℓ_2-regularized variants, by contrast, require 740–900 seconds — roughly three times longer. This overhead is due to the Newton iteration required to solve $\theta(\beta) = 0$ (142) at each optimizer step, even though convergence is typically achieved in 3–4 iterations, each iteration involves computing $\|(H_t^x + \beta I)^{-1}u_t\|_2$ and its derivative, which requires a full pass over all n parameters. For $n \approx 222\,000$, this accumulates to a significant overhead across 280 000 mini-batch updates. The PyTorch baseline is fastest (142 s) because it uses a highly optimized C++ implementation without the Python-level overhead of our custom optimizers.

Table 1: Test results on day 10, sorted by AUC. Training time includes prequential evaluation after each day. All experiments ran on a single CPU core with mini-batch size 128.

Rank	Configuration	φ	$\mathcal{K}$	AUC	Time
1	RDA + ℓ_1 + ℓ_2-ball	$10^{-6}\|w\|_1$	$\|w\|_2 \le 10$	0.7163	288 s
2	RDA + ℓ_2	$10^{-5}\|w\|_2$	$\mathbb{R}^n$	0.7137	897 s
3	RDA + ℓ_1	$10^{-6}\|w\|_1$	$\mathbb{R}^n$	0.7099	308 s
4	COMID + ℓ_1	$10^{-8}\|w\|_1$	$\mathbb{R}^n$	0.7087	322 s
5	RDA	0	$\mathbb{R}^n$	0.7060	232 s
6	RDA + box	0	$\|w_i\| \le 2$	0.7060	228 s
7	COMID + ℓ_1 + box	$10^{-8}\|w\|_1$	$\|w_i\| \le 2$	0.7007	293 s
8	COMID + box	0	$\|w_i\| \le 2$	0.6971	233 s
9	PyTorch AdaGrad	0	$\mathbb{R}^n$	0.6970	142 s
10	RDA + ℓ_2-ball	0	$\|w\|_2 \le 10$	0.6958	228 s
11	COMID + ℓ_2	$10^{-4}\|w\|_2$	$\mathbb{R}^n$	0.6934	739 s
12	COMID	0	$\mathbb{R}^n$	0.6838	277 s
13	COMID + ℓ_2-ball	0	$\|w\|_2 \le 100$	0.6838	233 s

All results refer to a single dataset, a single model architecture and one run per configuration; no confidence intervals were computed, so small differences in AUC should not be over-interpreted.

7.9 COMID vs. RDA Diag AdaGrad

In the configurations tested, the RDA variant attained a higher AUC than the COMID variant. Without regularization, the gap is substantial. RDA Diag AdaGrad achieves AUC = 0.7060 versus COMID Diag AdaGrad 0.6838, a difference of over two percentage points.

Since an AUC of 0.5 corresponds to random ranking, absolute differences in AUC understate the size of an improvement, and it is instructive to normalise them. Two normalisations are common. Measuring the remaining gap to perfect ranking, ℓ_1-regularization reduces $1 - \text{AUC}$ for COMID from 0.3162 to 0.2914, i.e.

$$\frac{1 - 0.6838}{1 - 0.7087} \;=\; 1.085,$$

so the residual gap shrinks by 7.8%. Measuring instead the informative range above chance, the Gini coefficient $G = 2\,\text{AUC} - 1$ rises from 0.368 to 0.417, a relative gain of 13.5% (Table 2).We report absolute AUC throughout, since in the CTR literature differences of 0.001 are already regarded as practically relevant.

Configuration	AUC	$1 - \text{AUC}$	$G = 2\,\text{AUC} - 1$
COMID	0.6838	0.3162	0.3676
COMID + ℓ_1	0.7087	0.2913	0.4174
RDA	0.7060	0.2940	0.4120
RDA + ℓ_1 + ℓ_2-ball	0.7163	0.2837	0.4326

Table 2: Normalised view of the test-day results. Relative to the unregularized COMID baseline, the Gini coefficient increases by 13.5%, 12.1% and 17.7%, respectively.

ℓ_1-regularization: Adding $\varphi(w) = \lambda\|w\|_1$ improves both methods: RDA Diag AdaGrad from 0.7060 to 0.7099 and COMID Diag AdaGrad from 0.6838 to 0.7087. The improvement is particularly striking for COMID Diag AdaGrad (+2.5 points), suggesting that COMID Diag AdaGrad benefits strongly from the structural bias that ℓ_1-regularization provides. This is consistent with the observation that COMID Diag AdaGrad, unlike RDA Diag AdaGrad, does not inherently produce sparse solutions (see §7.11 below), and therefore relies on explicit regularization to eliminate irrelevant parameters.

ℓ_2-regularization: The regularizer $\varphi(w) = \lambda\|w\|_2$ yields the second-best result overall (RDA + ℓ_2, AUC = 0.7137). Unlike ℓ_1, the ℓ_2-regularizer does not produce exact zeros but penalizes the

overall magnitude of w, encouraging the optimizer to distribute its capacity more evenly across coordinates. Interestingly, COMID + ℓ_2 (rank 11) performs *worse* than the unregularized COMID (rank 12). A possible explanation is that the Newton-based proximal operator for ℓ_2 (142) introduces a global coupling between coordinates (through the scalar β that satisfies $\theta(\beta) = 0$), which may interact unfavorably with COMID's local, coordinate-wise gradient step. RDA, whose update is already "global" in nature, integrates the ℓ_2-proximal more naturally.

7.10 Effect of Constraints

Constraints from §6.3–6.5 alone (without regularization) have a negligible effect on predictive performance. Comparing the unconstrained baselines (ranks 5, 12) with their constrained counterparts (ranks 6, 8, 10, 13), we observe that the AUC values are nearly identical. For the box constraint with $c = 2.0$, this is easily explained. The unconstrained weights already satisfy $|w_i| < 2.0$ for most coordinates, so the projection $\Pi_{[-c,c]}(w)$ acts as the identity in practice. For the ℓ_2-ball with $R = 100$ (COMID Diag AdaGrad), the same reasoning applies.

More fundamentally, AdaGrad provides a form of *implicit regularization* through its adaptive step sizes. The effective learning rate $\eta/(h_t)_i = \eta/(\delta + \sqrt{\tilde{s}_{t,i}})$ decays as the cumulative squared gradient $\tilde{s}_{t,i}$ grows, automatically limiting the magnitude of each coordinate. This mechanism serves a similar purpose as explicit norm constraints, explaining why the latter provide little additional benefit.

The exception is the combination of ℓ_1-regularization with an ℓ_2-ball constraint (rank 1, AUC = 0.7163). Here, the two mechanisms are complementary. ℓ_1 promotes sparsity (setting irrelevant coordinates to zero), while the ℓ_2-ball controls the overall energy of the solution (preventing the remaining non-zero coordinates from growing too large). This result demonstrates the practical value of the combined regularizer-constraint framework developed in Chapter 6.

7.11 Sparsity

One difference between the two updates concerns their sparsity behavior under ℓ_1-regularization. The COMID Diag AdaGrad update (141) applies soft-thresholding with threshold $\eta\lambda$ to each iterate

$$(w_{t+1})_i = \frac{1}{(h_t^x)_i} \operatorname{sign}(-u_i) \max(|u_i| - \eta\lambda, 0).$$

The threshold $\eta\lambda$ is *constant* across iterations. A coordinate that is zeroed out at step t may well become non-zero at step $t + 1$ if the new gradient g_{t+1} is sufficiently large.

The RDA Diag AdaGrad update, by contrast, applies thresholding to the *cumulative* average gradient. When expanded using $u_t = (\eta/t)\bar{g}_t$ and $H_t^x = H_t/t$, the ℓ_1 solution becomes

$$(w_{t+1})_i = \frac{\eta}{(h_t)_i} \operatorname{sign}(-\bar{g}_i) \max(|\bar{g}_i| - \lambda t, 0).$$

Here the threshold λt *grows with* t. As training progresses, an increasing proportion of coordinates satisfy $|\bar{g}_i| \leq \lambda t$ and are set exactly to zero. Once zeroed out, a coordinate can only return if its cumulative gradient magnitude grows faster than λt which is an increasingly unlikely event.

This theoretical prediction is consistent with the experiments. Table 3 compares the number of active (non-zero) embedding rows per feature field for COMID and RDA AdaGrad versions, both *without* explicit ℓ_1-regularization.

Table 3: Active embedding rows per feature field (out of $B = 5\,000$), without regularization. The "True unique" column shows the actual number of distinct values observed in the training data.

Feature	True unique	COMID active	RDA active
`site_domain`	∼2 000	5 000 (100%)	3 871 (77%)
`device_model`	∼4 600	5 000 (100%)	4 017 (80%)
`C14`	∼600	5 000 (100%)	1 932 (39%)
`C17`	∼160	5 000 (100%)	388 (8%)
`C19`	∼40	5 000 (100%)	66 (1.3%)
`banner_pos`	7	5 000 (100%)	7 (0.1%)
`hour_of_day`	24	5 000 (100%)	24 (0.5%)

The observed pattern matches the prediction of Section 6.2. RDA Diag AdaGrad correctly identifies the intrinsic cardinality of each feature. `banner_pos` has exactly 7 unique values in the data, and RDA Diag AdaGrad activates exactly 7 out of 5 000 embedding rows. Similarly, `hour_of_day` has 24 possible values, and RDA Diag AdaGrad retains exactly 24 rows. For higher-cardinality features, RDA Diag AdaGrad activates a number of rows that correlates with, but does not exactly match, the true cardinality. This is expected due to hash collisions that map multiple feature values to the same embedding row.

COMID Diag AdaGrad, by contrast, maintains non-zero entries across all $B = 5\,000$ rows for every feature field, regardless of the actual number of unique values. This is a direct consequence of the constant threshold in COMID's soft-thresholding. Even embedding rows that have been accessed only once retain a non-zero value indefinitely.

This structural sparsity of RDA has practical implications beyond predictive accuracy. A model with 84% fewer active parameters requires proportionally less memory for storage and inference, which is relevant in deployment scenarios where the model must be served at scale.

7.12 Feature Importance

The per-field embedding norms $\|E_i\|_2$ provide a natural measure of feature importance. Fields whose embeddings carry larger norms exert a stronger influence on the model's output, since the logit $f_w(x)$ depends linearly on the concatenated embedding vector.

Table 4: Feature importance by embedding norm (consistent across all optimizer configurations).

Rank	Feature	$\|E_i\|_2$	Share
1	`site_domain`	46.74	27.5%
2	`device_model`	45.91	27.0%
3	`C14` (anonymized)	36.36	21.4%
4	`C17` (anonymized)	18.13	10.7%
5	`C19` (anonymized)	6.01	3.5%
6	`C21` (anonymized)	5.44	3.2%
7	`banner_pos`	3.31	2.0%
8	`device_type`	2.36	1.4%
9	`device_conn_type`	2.08	1.2%
10	`hour_of_day`	1.73	1.0%
11	`C18` (anonymized)	1.68	1.0%

Two features dominate, the website domain and the device model together account for 54.5% of the total embedding norm. This is plausible from a domain perspective, the identity of the website determines the type of content and audience, while the device model correlates with user demographics and browsing behavior. The anonymized features `C14` and `C17` contribute an additional 32%, suggesting that they encode information complementary to the named features.

Importantly, this ranking is *stable across all 13 configurations*. The same features appear in the same order regardless of whether COMID or RDA AdaGrad versions is used, and regardless of regularization or constraints. This consistency indicates that the ranking reflects genuine properties of the data rather than artifacts of a particular optimization method.

7.13 Adaptive Learning Rates

The per-coordinate learning rate $\eta/(h_T)_i$, where $(h_T)_i = \delta + \sqrt{\sum_{t=1}^{T}(g_t)_i^2}$, is the central mechanism of AdaGrad. After training on 36 million samples, the ratio $\max_i(\eta/(h_T)_i)/\min_i(\eta/(h_T)_i)$ exceeds 10^4 for the embedding parameters. This means that the most frequently updated coordinate receives an effective step size more than 10 000 times smaller than the least frequently updated one.

To understand why this is necessary, consider the structure of the gradient. At each step, only the embedding rows corresponding to the current input's feature values receive a non-zero gradient. A popular website that appears in 30% of impressions contributes to the gradient in roughly one-third of all updates, while a rare device model appearing in 0.01% of impressions contributes only once every 10 000 updates. Without per-coordinate adaptation, a uniform step size would

either be too large for the frequent features (causing oscillation) or too small for the rare features (preventing learning). AdaGrad resolves this dilemma automatically through the accumulated squared gradients $\tilde{s}_{t,i}$.

7.14 Empirical Regret

Theorems 4.3 and 5.4 establish that the regret is sublinear for convex losses, implying $R_T/T \to 0$. To investigate whether this behaviour extends to our non-convex setting, we compute the empirical proxy

$$\hat{R}_T \;=\; \sum_{t=1}^{T} \ell_t(w_t) \;-\; \sum_{t=1}^{T} \ell_t(w_T),$$

where the final iterate w_T replaces the best fixed comparator. Note that $\hat{R}_T \le R_T$, since $w_T \in K$. Moreover, the regret itself is not sign-constrained, the comparator ranges over *fixed* points, while the algorithm plays a time-varying sequence, so negative values are possible, unlikely in a stationary stream, but not excluded. The bounds of Chapters 4 and 5 are upper bounds and remain valid either way.
In all configurations, $\hat{R}_T$ is slightly negative (with $\hat{R}_T/T$ in the range $[-0.004, 0]$). A negative value means that the sequence of online predictions $\{w_t\}$ achieved a *lower* total loss than the fixed final model w_T replayed over the same data. It does, however, illustrate a practical point, that in non-stationary environments the adaptive nature of online learning can be an advantage rather than a limitation, since the iterates track an evolving data distribution that no fixed comparator can match.

7.15 Summary

The numerical experiments validate the algorithms developed in Chapters 4–6 and reveal several findings of both theoretical and practical interest. The following observations refer to this dataset, model architecture and hyperparameter grid, with a single run per configuration. They are not general statements about COMID versus RDA.

1. *RDA outperformed COMID in the configurations tested here:* The Regularized Dual Averaging method (Algorithm 13) chieved a higher AUC in every configuration we ran, with differences of up to 2.2 percentage points. A plausible explanation is RDA's gradient averaging, which produces more stable updates as t grows.

2. *Regularization is effective:* Both ℓ_1- and ℓ_2-regularization improve generalization. The best overall result (AUC = 0.7163) is achieved by combining ℓ_1-regularization with an ℓ_2-ball constraint, a configuration that exploits the full generality of the framework from Chapter 6.

3. *Constraints require regularization to be effective:* Feasible-set constraints (§6.3–6.5) alone do not improve performance, as AdaGrad's adaptive step sizes provide implicit norm control. However, when paired with regularization, constraints offer a complementary benefit.

4. *RDA induced structural sparsity in our runs:* Even without explicit ℓ_1-regularization, RDA left only about 16% of the embedding rows non-zero and identifies the intrinsic cardinality of each feature field. This is consistent with the theoretical prediction that RDA's growing threshold λt produces monotonically increasing sparsity.

5. *ℓ_2-regularization is computationally expensive:* The Newton iteration for the ℓ_2-proximal (142) increases training time by a factor of 3 compared to ℓ_1 or unregularized training, due to the per-step evaluation of $\|(H_t^x + \beta I)^{-1} u_t\|_2$ and its derivative.

6. *Adaptivity mattered for this problem:* Effective learning rates vary by a factor exceeding 10^4 across coordinates

7. *Implementation correctness:* The PyTorch AdaGrad baseline (rank 9) and our COMID implementation (rank 12) produce comparable results in the unregularized case, validating the correctness of the custom implementations. All derived projection operators (box, ℓ_2-ball, weighted ℓ_1-ball bisection) function as specified in Chapter 6.

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Erklärung zur KI-Nutzung

Ich, Eduard Bopp, erkläre hiermit, dass ich im Rahmen meiner Masterarbeit mit dem Titel Adaptive Subgradient Methods for Online Deep Learning bei Prof. Dr. Jarre KI-gestützte Werkzeuge verwendet habe. Die folgende Aufstellung führt sämtliche eingesetzten Werkzeuge auf und beschreibt, in welcher Weise und zu welchem Zweck diese verwendet wurden.

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- **Sprachliche Überarbeitung:** Kontrolle hinsichtlich Orthographie, Grammatik, Syntax und Satzbau; stilistische Überarbeitung hinsichtlich Semantik sowie Formulierungshilfen. *(Claude, Anthropic)*
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Sämtliche mithilfe der genannten Werkzeuge erzeugten Inhalte und Vorschläge habe ich eigenständig überprüft, kritisch bewertet und gegebenenfalls überarbeitet. Die inhaltliche und wissenschaftliche Verantwortung für die vorliegende Arbeit liegt vollständig bei mir.

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Hiermit versichere ich, dass ich die Masterarbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Düsseldorf, den 11 08 2026



(Eduard Bopp)